Master thesis

Master's program in Energy Smart Innovation in the Built Environment, 120 credits

Design and Simulation of Renewable Energy Systems to Meet Industrial and Service Sector Electricity Demand in Swedish Regions

Thesis in Construction Engineering with Specialization in Renewable Energy, 30 credits

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Abstract

The shift towards renewable energy systems poses technological opportunities and structural issues, especially in areas with high variability and diverse demand. This thesis explores the design and performance of renewable energy systems aimed at supplying the industrial and service sector demand for electricity on various scales and geographic locations in Sweden. A data-oriented simulation approach has been employed in this study to assess Solar Photovoltaic systems, Wind energy, and Battery energy storage systems at various scales.

The building-scale case studies address data that was collected from municipal energy monitoring in Halmstad and provide a real-world assessment of integrating renewables. The city-scale studies focus on a generic industrial load curve and climate data from a particular region, that allow for the separation of the effect of the availability of renewables. In all simulation scenarios, there is an hourly resolution and an entire year span.

The analysis indicates that building-level systems can considerably lower dependence on the grid during a high availability period of renewables, but it adds less on an annual basis. Also, city-level systems can increase penetration levels with mutual support between wind and solar outputs, but there needs to be a large oversize margin for low unserved demand. Battery storage can enhance short-term flexibility but will be ineffective for seasonal mismatch.

The results show that improving one aspect of the system, such as reliability, efficiency, or cost performance, often comes at the expense of another. This highlights the importance of designing renewable energy systems based on regional conditions and evaluating them through multiple scenarios rather than relying on a single optimal solution.

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List of Abbreviations and Symbols

Abbreviations

PV – Photovoltaic
WT – Wind Turbine
BESS – Battery Energy Storage System
Batt – Battery

DSM – Demand Side Management

FTX / MHRV – Mechanical Ventilation with Heat Recovery

CAPEX – Capital Expenditure
OPEX – Operation and Maintenance Cost
TAC – Total Annualized Cost
CRF – Capital Recovery Factor

PVGIS – Photovoltaic Geographical Information System
SAM – System Advisor Model

SE1 – Swedish electricity price zone 1 (Northern Sweden)
SE2 – Swedish electricity price zone 2 (Central-Northern Sweden)
SE3 – Swedish electricity price zone 3 (Central-Southern Sweden)
SE4 – Swedish electricity price zone 4 (Southern Sweden)

EMU1 – Energy Monitoring Unit 1
EMU2 – Energy Monitoring Unit 2
EMU3 – Energy Monitoring Unit 3
EMU4 – Energy Monitoring Unit 4

6WT – Configuration with six wind turbines
3WT – Configuration with three wind turbines

kWp – Kilowatt-peak
kWh – Kilowatt-hour
MWh – Megawatt-hour
GWh – Gigawatt-hour

Symbols

Time and indices

t – Time index (hour)
Δt – Time step (hour)

Solar photovoltaic system

$P_{PV}(t)$ – PV power output at time *t*
$P_{PV,nom}$ – Nominal installed PV capacity
$G(t)$ – Global solar irradiance at time *t*
G_{ref} – Reference irradiance (1000 W/m²)
$\eta_{PV}(t)$ – PV efficiency at time *t*
η_{ref} – Reference PV efficiency
β – PV temperature coefficient
$T_c(t)$ – PV cell temperature at time *t*
T_{ref} – Reference temperature

Wind energy system

$P_{wg}(V)$ – wind generator (turbine) electrical power output as a function of wind speed V.
V – wind speed (at the turbine hub height after any height adjustment, if applied).
P_r – rated power (maximum continuous electrical output under normal operation).
V_{cin} – cut-in wind speed (minimum wind speed at which the turbine begins generating).
V_{rat} – rated wind speed (wind speed at which the turbine first reaches P_r).
V_{cou} – cut-out wind speed (wind speed above which the turbine is shut down for protection).
P_r – Rated wind turbine power
v_{ref} – Reference wind speed
H – Wind turbine hub height
H_{ref} – Reference height
α – Wind shear exponent

Battery energy storage system

SOC(t) – Battery state of charge at time *t*
SOC(t−1) – Battery state of charge at previous time step
SOC_{min}– Minimum allowable state of charge
SOC_{max}– Maximum allowable state of charge
η_{ch}– Battery charging efficiency
η_{dis}– Battery discharging efficiency
P_{ch} (t) – Battery charging power at time *t*
$P_{dis}(t)$ – Battery discharging power at time *t*

System-level power balance

$P_{gen}(t)$ – Total renewable power generation at time *t*
$P_{bat,ch}(t)$ – Battery charging power
$P_{bat,dis}$ (t) – Battery discharging power
P_{load} (t) – Electricity demand
P_{waste} (t) – Curtailed (wasted) power
P_{res} (t) – Residual load after wind generation

Economic parameters

C_i – Unit investment cost of component *i*
S_i – Installed size of component *i*
O_i – Annual operating cost of component *i*

Performance indicators

E_{load} – Total electrical demand
E_{unmet} – Unserved electrical energy
E_{waste} – Wasted renewable energy
$E_{generated}$ – Total renewable energy generation
$E_{renewable}$ – Renewable energy supplied to the load

Shortage – Unmet load ratio
Waste – Wasted energy ratio
Coverage – Renewable energy coverage ratio

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1 Introduction

The industrial and service sectors are among the largest consumers of electricity worldwide and play a key role in both national and regional energy systems. In recent years, industries have increasingly faced rising electricity prices, stricter climate targets, and growing pressure to reduce greenhouse gas emissions. As a result, understanding and managing industrial electricity demand has become a significant challenge rather than a purely technical issue.

At the same time, the ongoing transition towards renewable energy systems adds further complexity. Renewable electricity generation is inherently variable and strongly dependent on local conditions, which makes system planning and demand matching more challenging for industrial and service sector applications.

Though the role of industry is of prime relevance to energy transition models, accurate and publicly available data regarding industrial electricity consumption do not exist to any great extent at present. This is because previous research regarding industrial electricity use is normally conducted by means of approximate figures and industrial electricity load models that do not represent and reflect actual industrial electricity loads to any great extent.

In recent years, simulation-based methods have played an increasingly important role in the assessment of renewable energy systems. These methods enable the evaluation of how technologies such as solar photovoltaics, wind energy, and energy storage can be integrated to supply electricity demand in the industrial and service sectors. Through scenario-based simulations, different system designs can be examined under diverse climatic, geographical, and demand-related conditions, while maintaining clarity and reproducibility in the analysis.

Such an analytical approach is particularly relevant for countries like Sweden, where variations in climate, availability of renewable resources, and electricity market conditions differ significantly between regions and can strongly influence system performance.

In response to these conditions, this thesis applies a systematic simulation framework to investigate renewable energy system solutions for meeting electricity demand in industrial and service-oriented contexts across multiple Swedish regions. By integrating analyses at both the building and city scales, the study aims to capture local characteristics as well as broader regional trends. The results are intended to contribute to more practical and informed decision-making in future energy system planning.

1.1 Background

The electrification of industrial processes is widely seen as an important pathway toward decarbonization, particularly in regions with ambitious climate goals such as the European Union. In this context, Sweden has set long-term targets aimed at achieving a fossil-free energy system, with a growing reliance on renewable electricity generation.

As a result, technologies such as solar photovoltaic systems, wind turbines, and battery energy storage systems are expected to play an increasingly significant role in supplying electricity to both industrial and urban consumers.

However, integrating renewable energy into industrial electricity supply is not a straightforward task. Renewable electricity generation is inherently variable and does not always align with industrial load profiles, which are often marked by high peak demands and relatively stable baseloads during operating hours.

If this mismatch is not carefully considered during the design phase, it can result in challenges such as energy shortages, excess electricity generation, or inefficient system sizing.

Previous research has examined individual aspects of renewable energy integration, including technology performance, economic feasibility, and environmental impacts. However, many studies tend to focus either on national-scale energy systems or on isolated case studies, without offering a coherent comparison across different spatial scales and regional conditions.

In addition, the reliance on assumed or generic system sizes and load profiles can reduce the practical relevance of the results, particularly when applied to real-world industrial contexts.

To address these limitations, there is an increasing need for transparent, scenario-based simulation frameworks that rely on realistic input data, clearly stated assumptions, and consistent evaluation criteria. Such frameworks make it possible to systematically compare alternative system configurations across different regions and spatial scales, while reducing the risk of overgeneralization.

This approach provides the methodological foundation for the present thesis.

1.2 Aim and Research Objectives

The overall aim of this thesis is to design and simulate renewable energy systems capable of meeting industrial and service sector electricity demand under realistic regional conditions in Sweden.

To achieve this aim, the thesis addresses the following specific objectives:

- To model industrial electricity and service sector demand using representative load profiles at both building and city scales
- To simulate electricity generation from solar photovoltaic and wind energy systems under region-specific climatic conditions
- To integrate battery energy storage systems and evaluate their role in reducing energy shortages and wasted electricity
- To apply a structured, scenario-based approach for comparing different system configurations

- To assess overall system performance using technical, economic, and energy-related indicators

Through these objectives, the thesis seeks to provide a consistent and transparent basis for evaluating renewable energy system designs tailored to industrial and service sector electricity demand.

1.3 Delimitations

This thesis is subject to several delimitations that define the scope of the analysis. First, the study focuses exclusively on electricity demand and electricity-based renewable energy systems. Thermal energy demands and sector coupling options are therefore not considered.

Second, the analysis is limited to selected case studies at the building and city levels within Sweden. These cases are chosen to represent different regional and climatic conditions, rather than to provide a nationwide optimization of energy systems.

Furthermore, the system configurations examined in this thesis are based on predefined and non-random scenarios. The aim is not to identify globally optimal solutions through automated optimization, but instead to enable transparent comparisons between realistic and interpretable system designs.

Finally, the analysis relies on simulation inputs and assumptions derived from established datasets and technology catalogues. Future policy changes or speculative technology developments beyond the defined scope are not included.

2 Literature Review

Industrial and service sector electricity consumption represents a substantial share of total electricity demand and shows complex temporal characteristics that distinguish it from residential and commercial sectors. Unlike other types of electricity use, industrial demand is strongly linked to production schedules, operational constraints, and the continuous operation of auxiliary systems (Dehning et al., 2019; Herrmann and Thiede, 2009; Henriksson et al., 2014)

Several studies indicate that industrial facilities maintain significant levels of electricity consumption even during non-production periods. This is mainly due to standby equipment, ventilation systems, safety requirements, and other process-related auxiliary systems (Dehning et al., 2023; Herrmann and Thiede, 2009; Gontarz et al., 2015; Henriksson et al., 2014). As a result, a persistent base load is often present, highlighting the importance of considering both production and non-production electricity use when assessing industrial energy systems (Scaramuzzino et al., 2019).

A recurring challenge highlighted in the literature is the limited availability of high-resolution and publicly accessible industrial load data. As a result, many studies rely on aggregated statistics or simplified assumptions, which do not fully capture short-term fluctuations and operational dynamics that are critical for system design and demand-side management (Karlsson, 2011; Koivisto et al., 2019; Dehning et al., 2019; Kan, 2023; Schelin, 2019)

Consequently, comparisons between different studies can be difficult, and the practical applicability of reported findings is often constrained by methodological inconsistencies and data limitations (Kan, 2023).

Empirical investigations into industrial load behavior reveal distinct patterns associated with production and non-production periods. Dehning et al. (2019) demonstrate that electricity demand rises sharply during production shifts and remains relatively stable throughout operation, while dropping significantly during nighttime and non-working days. However, even during non-productive periods, electricity consumption does not approach zero, indicating the presence of essential systems that operate continuously (Herrmann and Thiede, 2009; Henriksson et al., 2014)

Figure 1 presents a typical weekly industrial load profile, illustrating the contrast between production and non-production periods and highlighting the persistence of base loads outside active operation. This type of representation serves as a useful reference for identifying potential opportunities to reduce unnecessary electricity consumption, while maintaining operational readiness and system reliability (Dehning et al., 2019; Herrmann and Thiede, 2009; Gontarz et al., 2015)

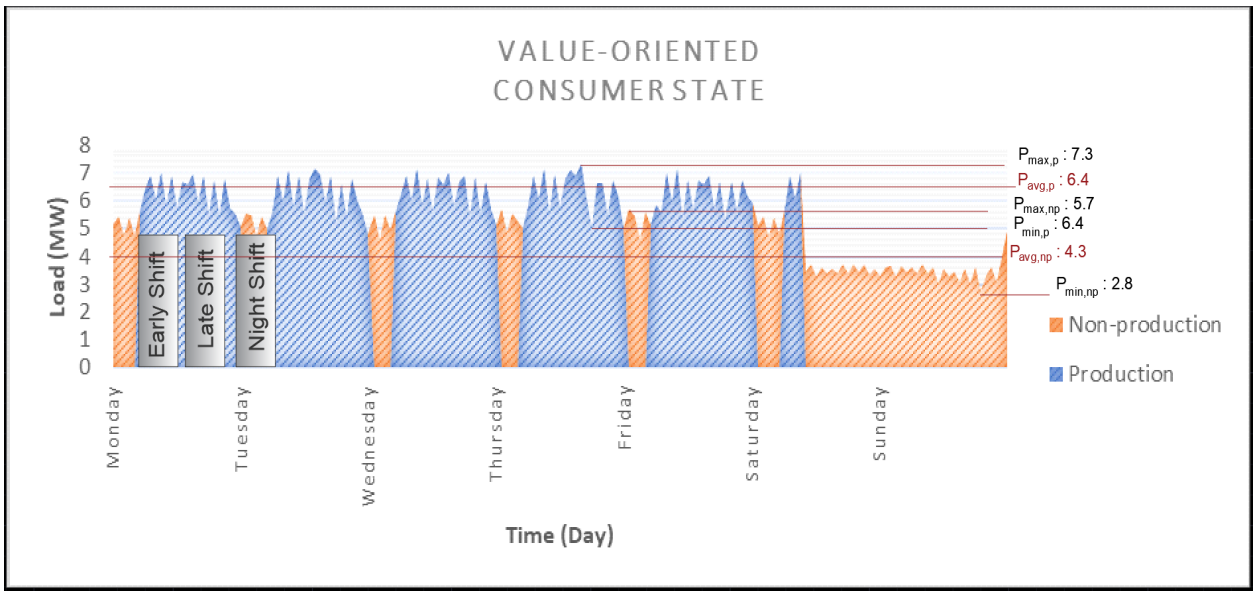


Figure 1. Typical weekly industrial electricity load profile illustrating production and non-production periods, reconstructed based on the load profile concept presented by Dehning et al. (2019).

Demand-side management (DSM) has therefore emerged as an important strategy for improving energy efficiency and better aligning industrial electricity consumption with overall system and grid requirements. Common DSM measures, such as peak shaving, load shifting, and demand response, aim to modify electricity consumption patterns while preserving production reliability.

Previous studies highlight that industrial environments often offer substantial DSM potential, mainly due to the presence of large controllable loads and relatively predictable operating schedules (Salam et al., 2024; Karlsson, 2011). At the same time, the feasibility of DSM implementation differs across industrial sectors and is strongly shaped by process criticality and the degree of operational flexibility (Scaramuzzino et al., 2019).

To support DSM strategies, several authors have proposed classifications of industrial loads based on their flexibility and operational characteristics. Salam et al. (2024) classify industrial electricity demand into categories such as induction motor loads, lighting loads, thermostatically controlled loads, and flexible loads that can be shifted without affecting production output.

This type of classification supports more targeted DSM interventions by helping to identify which load types are suitable for rescheduling or temporary reduction during peak demand periods (Karlsson, 2011).

Figure 2 presents a categorized industrial demand curve, illustrating how different load types contribute to overall electricity consumption throughout the day and highlighting their respective flexibility potentials. Such representations offer a useful conceptual basis for understanding how DSM measures can redistribute electricity demand and contribute to peak load reduction. The demand curve is decomposed into different load categories, including full

load (FL), induction motor load characterized by the voltage imbalance factor (IM VIF), thermostatically controlled loads (TCL), and lighting loads (LL) (Salam et al., 2024).

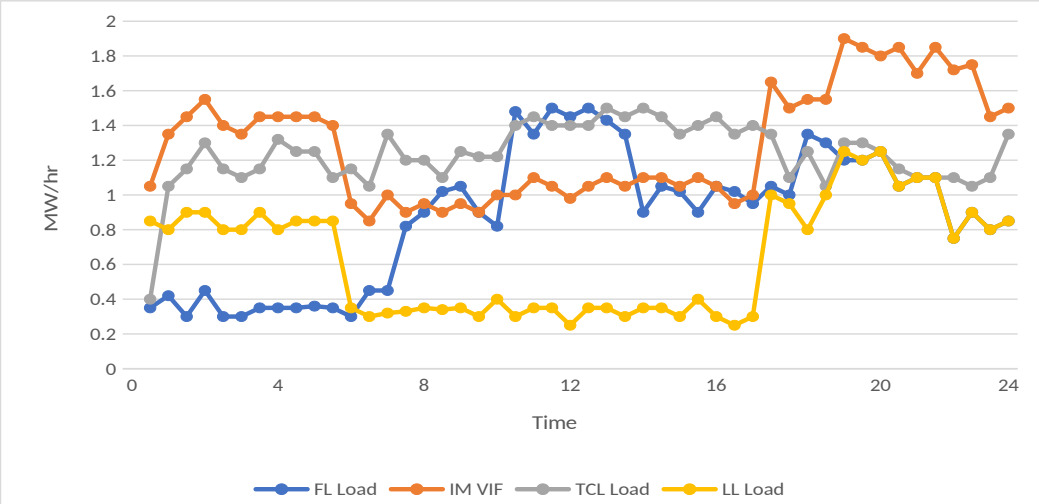


Figure 2. Representative hourly industrial electricity demand profile with classified load components, reconstructed based on the demand-side load classification presented by Salam et al. (2024).

Induction motor loads, which are common in many industrial processes, are particularly sensitive to operational conditions such as voltage imbalance and scheduling. Optimizing the operation of these loads can lead to notable energy savings, especially when combined with off-peak operation strategies (Salam et al., 2024; Herrmann and Thiede, 2009).

Lighting loads, in contrast, are strongly influenced by occupancy patterns and daylight availability. This makes them suitable candidates for demand reduction through measures such as daylight utilization and automated control systems (Gontarz et al., 2015).

Flexible loads, including pumps and auxiliary equipment, further increase DSM potential by allowing electricity consumption to be shifted to periods of lower system demand or higher renewable electricity generation (Karlsson, 2011).

Accurate characterization of industrial and service sector electricity demand depends strongly on data resolution and monitoring capabilities. High-resolution measurements and real-time monitoring systems make it possible to identify inefficiencies and consumption patterns that often remain hidden in aggregated datasets (Gontarz et al., 2015; Koivisto et al., 2019)

More recent research also points to the increasing use of advanced analytics, artificial intelligence, and machine learning techniques for forecasting electricity demand and supporting DSM-related decision-making (Koivisto et al., 2018). Nevertheless, the adoption of these technologies is still uneven and is frequently constrained by implementation costs as well as limited data availability.

While demand-side management strategies play an important role in improving energy efficiency and operational flexibility, the primary focus of this thesis is not on modifying or controlling industrial loads. Instead, DSM-related studies are reviewed here to provide context

for the characteristics and structure of industrial electricity demand. The main contribution of this work lies in the supply-side design and simulation of renewable energy systems capable of meeting such demand under different regional and system-level conditions (Potrč et al., 2021).

Despite extensive research on industrial energy consumption and demand-side management, several limitations remain in the existing literature. Many studies rely on assumed or generic load profiles that fail to capture regional differences or sector-specific operational characteristics, which can limit the transferability of the results (Dehning et al., 2019; Salam et al., 2024; Henriksson et al., 2014; Guena and Leblanc, 2006).

In addition, analyses often focus on a single spatial scale, such as individual facilities or national energy systems, without systematically comparing performance across building-level and city-level contexts (Karlsson, 2011; Koivisto et al., 2019). While optimization-based approaches offer powerful analytical tools, they may also emphasize mathematical optimality at the expense of transparency and practical interpretability (Herrmann and Thiede, 2009; Zhong et al., 2021; Sjölander and Espling, 2023; Drgham, 2023)

These gaps point to the need for simulation-based, scenario-driven analyses that combine realistic load profiles, region-specific renewable resource data, and transparent evaluation criteria. In particular, existing research offers a limited systematic comparison of renewable energy system performance across multiple spatial scales within a single country, especially when consistent methodological assumptions are maintained (Kan, 2023; Koivisto et al., 2019; Schelin, 2019; Sjölander and Espling, 2023)

The present thesis responds to this gap by analyzing renewable energy systems designed to meet industrial electricity demand at both building and city scales across different regions in Sweden (Kan 2023). In doing so, the study builds on existing literature while extending its applicability to more practical and regionally grounded energy system design.

3 Materials and methods

3.1 Overall Methodological Framework

This thesis applies a data-driven, scenario-based simulation framework to evaluate the performance of renewable electricity systems at both building and city scales. The methodology combines realistic and literature-consistent electricity demand profiles with physics-based models of solar photovoltaic (PV) systems, wind turbines, and battery energy storage systems (BESS). Rather than pursuing automated optimization, the focus is placed on assessing system behavior under predefined, non-random scenarios, with particular attention to reliability, surplus electricity generation, and economic implications.

All simulations are carried out with an hourly temporal resolution over a full annual period. This time-resolved approach allows for a realistic representation of load variability, renewable resource intermittency, and storage dynamics. To ensure methodological consistency and comparability, the same modeling logic and evaluation indicators are applied across all case studies and cities.

In this study, the AI-based language assistance tools, specifically ChatGPT, were used in a limited and supportive manner during the writing process. Their use was restricted to improving language clarity, correcting grammatical and typographical errors, and refining sentence structure to enhance readability and academic presentation.

All the methodological decisions, data selection, system modeling, scenario definition, simulation processes, and result interpretations were designed, performed, and validated by the author. AI tools did not contribute to analytical reasoning, numerical calculations, model formulation, or the drawing of scientific conclusions.

The use of these tools is considered comparable to advanced proofreading or language-editing software, and it has been used only to support clear and effective scientific communication.

The overall methodological workflow includes the following steps: (i) data acquisition and preprocessing of building-scale, city-scale, and climate-related input data; (ii) component-level modeling of solar photovoltaic systems, wind energy systems, and battery energy storage systems; (iii) scenario definition based on predefined and realistic combinations of generation and storage capacities; (iv) hourly energy balance simulation to capture the interaction between electricity production, storage operation, and demand; (v) economic assessment based on investment and operational cost parameters; and (vi) performance evaluation using standardized technical, energy-related, and economic indicators.

3.2 Data Sources

3.2.1 Building-Scale Data

Four building-scale case studies located in Halmstad municipality form the empirical basis of the building-level analysis. Official authorization was granted to access municipal energy

monitoring systems, providing real measured datasets for each building. The collected data include hourly electricity consumption over a full year and, for buildings equipped with on-site PV systems, hourly electricity generation and technical specifications of the installed systems.

All datasets were quality-checked for completeness and temporal consistency prior to use. Measured load and generation profiles were used directly in the simulations to preserve realistic operational characteristics.

3.2.2 City-Scale Load Profile

For city-scale analyses, a representative industrial electricity demand profile was derived from the literature based on a typical factory load pattern. This profile was expanded to an annual hourly time series by accounting for weekends, public holidays, and non-production periods. The resulting synthetic load profile is consistent with reported industrial consumption characteristics and was applied uniformly across all cities.

By keeping the demand profile identical for all city-scale cases, the analysis isolates the influence of regional climate conditions and renewable resource availability on system performance, rather than differences in consumption behavior.

3.2.3 Climate and Renewable Resource Data

Solar and wind resource data were obtained from established datasets commonly used in energy system analysis. Hourly global horizontal irradiance and ambient temperature data were extracted from the PVGIS database for PV simulations. Wind energy simulations were based on hourly wind speed data, adjusted to turbine hub height using standard vertical wind profile formulations. These climate datasets served as direct inputs to the hourly simulation model, where solar and wind generation were calculated and compared against the corresponding industrial electricity demand to assess supply–demand matching under each scenario.

3.3 Renewable Energy System Context in Sweden

Sweden is recognized as one of the leading European countries in the deployment and integration of renewable energy technologies. The national target of achieving net-zero greenhouse gas emissions by 2045 has led to a strong expansion of electricity generation from renewable sources, particularly wind power, solar photovoltaics, and complementary energy storage systems (Swedish Energy Agency, 2023).

The Swedish electricity market is divided into four bidding zones (SE1–SE4), reflecting regional differences in electricity demand, renewable resource availability, and transmission grid capacity. These regional variations significantly influence the suitability and performance

of renewable energy technologies across the country (Holmberg and Tangerås, 2023). Northern regions (SE1 and SE2) benefit from high average wind speeds and large available land areas, making wind power the dominant renewable resource, especially during winter months when solar availability is limited. In contrast, southern regions (SE3 and SE4) experience higher solar irradiation and longer daylight periods during spring and summer, which enhances the performance of photovoltaic systems in cities such as Stockholm, Malmö, and Halmstad (Nedstrand, 2024).

Hybrid renewable energy systems combining wind power, solar PV, and battery energy storage have increasingly been identified as an effective approach to improving system reliability and reducing electricity deficits. Wind and solar resources in Sweden exhibit complementary seasonal patterns, where wind generation tends to peak during winter while solar PV generation dominates during summer months (Roy et al., 2022). Battery energy storage systems support these hybrid configurations by increasing self-consumption of renewable electricity, mitigating short-term imbalances between generation and demand, and enhancing system flexibility (IEA-PVPS, 2022).

Technological developments have further strengthened the feasibility of integrated renewable systems. High-efficiency photovoltaic modules, advanced wind turbine designs, and improved battery performance have contributed to increased energy yields and reduced costs over time (IEA-PVPS, 2022; Lindahl et al., 2022). At the same time, economic and regulatory factors—such as grid limitations, permitting procedures, and regional planning constraints—continue to shape renewable energy deployment, particularly in southern Sweden (MDU University, 2022).

Within this national and regional context, the present thesis adopts a scenario-based modeling approach to investigate how combinations of solar PV, wind power, and battery energy storage can be configured to meet industrial and service sector electricity demand under Swedish conditions. By explicitly accounting for regional climate characteristics and applying consistent modeling assumptions across building-scale and city-scale analyses, the study aims to provide realistic and comparable insights into renewable energy system performance across different Swedish regions.

3.4 System Modeling Approach

3.4.1 Solar Photovoltaic (PV) Modeling

Electricity generation from solar photovoltaic (PV) systems was modeled using an hourly, physics-based approach that relates solar irradiance, installed system capacity, and temperature-dependent efficiency to electrical power output. This modeling framework follows established formulations widely applied in renewable energy system simulations and is suitable for both building-scale and city-scale analyses.

At each hourly time step *t*, the electrical power output of the PV system is calculated as a function of the incident solar irradiance and the nominal installed PV capacity, adjusted by an overall system efficiency factor.

$$P_{PV(t)} = P_{PV},nom * \left(\frac{G(t)}{G_{ref}}\right) * \eta_{PV(t)} \qquad (1)$$

In this formulation, the PV power output ($P_{PV(t)}$) at each time step is determined by three main components: the installed PV capacity, the available solar irradiance, and the effective system efficiency. The term representing the nominal installed PV capacity (P_{PV},nom) defines the maximum possible electrical output of the system under standard conditions. The irradiance-dependent term reflects the direct influence of solar resource availability on PV electricity generation, where the global solar irradiance G(t) represents the amount of solar energy incident on the PV surface at hour *t*.

The PV power output at time *t* is expressed in a normalized form based on irradiance, using the simplified PV generator model reported by Diaf et al. (2008) (based on Markvart, 2000). The original relationship links PV output to incident irradiance and an effective conversion efficiency. In this thesis, the formulation is rewritten in terms of the nominal PV power $P_{PV,nom}$defined at a reference irradiance G_{ref}, which removes the explicit area term by absorbing it into $P_{PV,nom}$. The ratio $G(t)/G_{ref}$scales the output from reference to operating irradiance, while $\eta_{PV}(t)$represents the time-varying overall efficiency (e.g., temperature and system effects), ensuring consistency with the simplified physical model while matching the available simulation inputs.

By normalizing the hourly irradiance with respect to the reference irradiance (G_{ref}) under standard test conditions, the formulation ensures consistency with conventional PV performance assessment methods. The efficiency term ($\eta_{PV(t)}$) aggregates the combined effects of module characteristics, temperature-related losses, and balance-of-system components, thereby providing a realistic representation of PV system behavior under varying operating conditions (Technical Solar PV – MDU, n.d.).

The temperature dependence of PV efficiency is explicitly considered, as higher operating temperatures reduce the electrical efficiency of photovoltaic modules. The effective PV efficiency at each time step is therefore adjusted based on the deviation of the operating cell temperature from reference conditions, following established empirical relationships reported in the literature.

The PV efficiency is modeled as a linear function of cell temperature. As adopted in Diaf et al. (2008), the efficiency at time *t* is expressed relative to a reference efficiency at standard test conditions. The coefficient βrepresents the temperature sensitivity of the PV module, while $T_c(t)$and T_{ref}denote the operating and reference cell temperatures, respectively. This formulation is consistent with established PV performance models and is used here to account for temperature-induced efficiency variations in the MATLAB-based simulation. (Habib et al., 1999, as cited in Diaf et al., 2008).

$$\eta_{PV(t)} = \eta_{ref} * \left(1 - \beta * \left(T_{c(t)} - T_{ref}\right)\right) \quad (2)$$

Here, the temperature coefficient (β) represents the sensitivity of PV module efficiency to temperature variations, while the reference efficiency (η_{ref}) corresponds to standard operating conditions. The effective PV system efficiency ($\eta_{PV(t)}$) therefore reflects the actual system efficiency at time *t*, accounting for temperature-related losses as well as balance-of-system effects. The operating cell temperature ($T_{c(t)}$) is estimated using ambient temperature (T_{ref}) and irradiance data, ensuring consistency with commonly applied PV performance models.

For building-scale case studies with existing PV installations, the modeled PV generation was validated against measured hourly production data obtained from municipal records. This validation step ensured that the selected formulation realistically reproduces observed system behavior before being applied to scenario-based capacity variations.

For city-scale analyses, the same PV modeling approach was applied consistently across all locations. While the installed PV capacity varies across scenarios, the underlying performance model remains unchanged, allowing differences in PV generation to be attributed solely to climatic conditions rather than methodological inconsistencies.

Solar resource data used in the PV simulations were obtained from the PVGIS database, which provides location-specific hourly irradiance and temperature data and is widely used in European solar energy assessments.

3.4.2 Wind Energy Modeling

Wind electricity generation was modeled using a simplified, piecewise representation of the wind turbine power curve, which relates electrical power output to wind speed at turbine hub height. This modeling approach is widely used in hybrid renewable energy system studies, as it captures the essential non-linear behavior of wind turbines while maintaining computational simplicity and transparency (Sohoni et al., 2016).

The wind turbine's power output $P_{wg}(V)$ is modeled using a simplified piecewise power-curve formulation that represents the turbine's operational behavior across different wind speed ranges. At wind speeds below the cut-in value V_{cin}, the turbine does not generate electrical power and the output is assumed to be zero. When the wind speed lies between the cut-in wind speed $V_{cin}$and the rated wind speed V_{rat}, the electrical power output increases linearly with wind speed, reaching the rated power P_r at V_{rat}.

$$P_{wg}(V) = \begin{cases} P_r \dfrac{V - V_{cin}}{V_{rat} - V_{cin}} & V_{cin} \le V \le V_{rat} \\ P_r & V_{rat} \le V \le V_{cou} \\ 0 & V \le V_{cin} \ \wedge \ V \ge V_{cou} \end{cases} \quad (3)$$

For wind speeds between the rated wind speed and the cut-out wind speed V_{cou}, the turbine operates at constant rated power (Hongxing et al., 2007; Lu et al., 2002a,b; Diaf et al., 2008). When the wind speed exceeds the cut-out threshold, the turbine is shut down for safety reasons, and the power output is again set to zero. Using this formulation, the power output of the wind turbine can be determined provided that the cut-in, rated, and cut-out wind speeds, as well as the rated power, are known (Danish Energy Agency, 2016).

Wind speed data were adjusted to the turbine hub height (H) before power output calculation to account for vertical wind shear effects (*alpha*). The adjustment was performed using a power-law relationship between wind speed and height, which is commonly applied in wind energy assessments. (Ilinka et al., 2003; Johnson, 1985, cited in Diaf et al., 2008)

$$V = V_{ref} * \left(\frac{H}{H_{ref}} \right)^{alpha} \quad (4)$$

In this formulation, the wind speed at hub height V, is estimated based on a reference wind speed (V_{ref}) measured at a reference height (H_{ref}), while the shear exponent represents surface roughness and atmospheric stability conditions. This approach is consistent with standard wind engineering practices and is widely adopted in wind resource assessment studies.

This wind energy modeling framework provides a robust and literature-supported basis for city-scale scenario analysis, enabling consistent comparison of wind power performance across different regions and system configurations.

3.4.3 Battery Energy Storage Modeling

Battery behavior was modeled using an energy balance formulation based on the state of charge (SOC), which represents the amount of energy stored in the battery at a given time. This modeling approach is widely used in energy system simulations to capture the dynamic interaction between charging, discharging, and storage capacity in a transparent and physically consistent manner (Diaf et al., 2008).

At each hourly time step *t*, the state of charge ($SOC(t)$) is updated as a function of the charging power ($P_{ch(t)}$) and discharging power ($P_{dis(t)}$), while accounting for the respective charging (η_{ch}) and discharging efficiencies (η_{dis}) of the battery system. (Diaf *et al.*, 2008; Bin *et al.*, 2003; Bogdan and Salameh, 1996).

$$SOC(t) = SOC(t-1) + \frac{1}{\eta_{ch}} \left(P_{ch(t)} - \left(\frac{P_{load}(t) - P_{dis(t)}}{\eta_{dis}} \right) \right) \Delta t \quad (5)$$

In this formulation, the charging and discharging processes are treated asymmetrically to reflect real battery behavior, where efficiency losses occur during both energy input and output. The time-step duration (Δt) ensures consistency with the hourly resolution adopted throughout the simulation framework. This SOC balance formulation has been extensively applied in energy

system modeling studies to ensure transparent and reproducible representation of storage dynamics.

Operational constraints were imposed to ensure realistic battery operation and to prevent overcharging or deep discharging. These constraints were implemented by defining a minimum (SOC_{min}) and maximum allowable SOC levels (SOC_{max}), which represent the usable energy capacity of the battery system. (Diaf *et al.*, 2008; Bin *et al.*, 2003; Bogdan and Salameh, 1996).

$$SOC_{min} \leq SOC(t) \leq SOC_{max} \quad (6)$$

In addition to energy constraints, maximum charging and discharging power limits were applied based on technology-specific specifications. These limits ensure that the battery operates within safe and realistic power ranges and reflect inverter and system design limitations reported in technology catalogues. These parameters ensure that the battery operates within safe and realistic power ranges consistent with commercial battery system designs.

Although the simulation is implemented using a time-stepping loop in MATLAB, the underlying battery behavior is governed by this SOC balance formulation. This approach ensures transparent, reproducible, and physically meaningful representation of battery storage dynamics across all scenarios and spatial scales.

The battery modeling framework follows established formulations reported in hybrid renewable energy system studies and is consistent with technology performance parameters provided in official energy storage catalogues (Danish Energy Agency, 2023).

3.4.4 System-Level Energy Balance and Scenario Definition

At each hourly time step, the system operation is governed by an electricity balance, in which renewable electricity generation, battery charging and discharging, load demand, and curtailed energy are explicitly accounted for. This type of energy balance formulation is commonly used in hybrid renewable energy system simulations to ensure physical consistency between supply, storage, and demand (Diaf et al., 2008; Connolly et al., 2010).

$$P_{gen(t)} + P_{bat}, dis(t) = P_{load(t)} + P_{bat}, ch(t) + P_{waste(t)} \quad (7)$$

In this formulation, renewable electricity generation from solar PV and wind turbines is first allocated to meet the instantaneous electricity demand ($P_{load(t)}$). When generation exceeds demand, the surplus energy is directed to battery charging, subject to storage capacity and power constraints. Any remaining excess electricity that cannot be stored is classified as wasted (curtailed) electricity ($P_{waste(t)}$). Conversely, when renewable generation and available battery discharge are insufficient to meet demand, the remaining deficit is recorded as unmet demand. This explicit accounting ensures a transparent and physically consistent representation of system operation, which is widely adopted in hybrid renewable energy system modeling.

Scenario design in this study follows a purpose-driven structure, in which system capacities are systematically varied to explore the interaction between renewable electricity generation, storage, and industrial electricity demand. To support this process, a residual electricity demand term is defined to represent the remaining demand after wind power generation has been accounted for. This residual demand, denoted as $P_{res(t)}$, is calculated as the difference between the total electricity demand $P_{load(t)}$and wind power generation $P_{WT(t)}$.

$$P_{res(t)} = P_{load(t)} - P_{WT(t)} \quad (8)$$

This formulation allows solar PV and wind power to be treated as complementary contributors within the same system configuration and avoids non-physical exclusion of solar generation under high wind penetration conditions. Rather than suppressing PV output when wind generation exceeds demand, the formulation preserves simultaneous multi-source generation, enabling a transparent assessment of surplus energy, storage utilisation, and unmet load. The same formulation framework is applied consistently at both building scale and urban scale, while capacity definitions and scenario objectives differ according to the level of analysis.

At the building level, a case-study-based approach is applied to capture realistic operational behavior under actual consumption and production conditions. For each case study, the real hourly electricity load profile of the building is used directly as input to the simulation model, without any scaling, normalization, or smoothing. In buildings with existing photovoltaic systems, the installed PV capacity (City of Halmstad, Technical and Property Management Department) and the corresponding measured production data are treated as fixed system inputs throughout the analysis.

Following the simulation of a base case without energy storage, the temporal behavior of each system is examined in detail, including the timing and magnitude of PV surplus, daily mismatch between generation and demand, and periods of unmet load. Based on these observations, battery storage is introduced as a short-term balancing mechanism.

Battery capacities are deliberately limited to represent only a few effective hours of operation within the daily cycle. The batteries are not designed to cover total daily demand or provide long-term or seasonal storage. This design choice avoids unrealistic over-dimensioning and preserves the practical and implementable nature of the case-study scenarios.

To evaluate system sensitivity, stepwise variations in PV and battery capacities are defined. PV capacities are varied within a narrow and realistic range, while battery capacities represent small, medium, and relatively larger short-term storage levels. These combinations enable transparent comparison of how incremental changes in generation and storage affect key performance indicators such as self-consumption, surplus electricity, and unmet demand. As a result, the case-study scenarios are not based on abstract optimization, but on observed system behavior and traceable design logic.

Urban-scale scenarios are defined to investigate the operational behavior of hybrid renewable energy systems at the city level, rather than to identify a single numerically optimal configuration. The objective is to understand how solar PV, wind power, and battery storage

interact under different climatic conditions and load characteristics across Swedish electricity regions.

All urban scenarios are referenced to the peak demand of a typical industrial load profile, which is applied consistently to all cities. Solar PV capacity is defined using predefined PV factors relative to this peak load from a typical industrial load profile in Figure 30 (Dehning et al., 2019), ensuring that system sizing remains directly linked to the maximum demand that must be met. Battery capacity is then defined as a function of the installed PV capacity (Battery factor × PV capacity), allowing storage requirements to scale proportionally with renewable generation rather than being selected arbitrarily (Danish Energy Agency, 2023).

Two structurally distinct scenario groups are evaluated: three-wind-turbine configurations and six-wind-turbine configurations. This separation reflects an intentional investigation of the role of wind power in system reliability. In the three-turbine group, wind contributes a limited share of annual and winter electricity supply, forcing the system to rely heavily on increased PV capacity and large batteries to compensate for seasonal deficits. In contrast, the six-turbine group represents systems with a stronger wind backbone, where higher winter wind availability partially mitigates seasonal supply gaps, although temporal mismatches and energy surplus remain significant.

Across all urban scenarios, solar PV is treated as a complementary resource rather than being excluded under high wind capacity conditions. Even when wind generation is substantial, PV remains part of the system to capture seasonal and diurnal solar availability. This modelling choice enables a transparent assessment of multi-source system behavior, including surplus generation, unmet load, and battery utilisation patterns, which would not be observable if one technology were artificially suppressed.

3.4.5 Economic Assessment Method

The economic assessment of the proposed renewable energy systems is based on three main cost components: total investment cost (CAPEX), fixed operation and maintenance cost (OPEX), and total annualized cost (TAC). This framework enables a consistent and transparent comparison of different system configurations and scenarios from an economic perspective and is widely adopted in energy system analysis studies (Diaf et al., 2008; Danish Energy Agency, 2018).

Technology-specific investment costs, fixed operation and maintenance costs, and component lifetimes were obtained from the Danish Energy Agency Technology Data Catalogue. These parameters reflect representative costs for commercially available technologies and ensure that the economic evaluation is grounded in realistic and up-to-date data.

3.4.6 Total Investment Cost (CAPEX)

The total investment cost ($CAPEX$) represents the upfront capital required to install the renewable energy system components considered in each scenario. This includes the investment costs associated with solar PV systems, wind turbines, and battery energy storage systems (i). The total investment cost is calculated as the sum of the individual component costs, expressed as the product of installed capacity (S_i) and unit investment cost (C_i) for each technology (Stock et al., 2024).

$$CAPEX = \sum_{i=1}^{n} C_i * S_i \quad (9)$$

In this formulation, the investment cost for each system component is determined by multiplying the installed capacity of that component by its corresponding unit capital cost. The summation accounts for all technologies included in the scenario. This approach ensures that differences in system size and technology mix are directly reflected in the total investment cost.

In this study, the capital expenditure model is simplified to include only the investment costs of the renewable electricity generation and storage components under consideration, namely photovoltaic systems, wind turbines, and battery energy storage. Other infrastructure-related cost components are excluded, as they fall outside the scope of the present system configuration.

3.4.7 Operation and Maintenance Cost (OPEX)

Fixed operation and maintenance costs ($OPEX$) represent the recurring annual expenses required to operate and maintain the installed energy system components. These costs include routine maintenance, inspection, and system management activities and are assumed to be independent of actual energy production (Stock et al., 2024).

$$OPEX = \sum_{i} O_i \quad (10)$$

Here, the total annual OPEX is calculated as the sum of fixed operation and maintenance costs associated with each installed technology, where the term (O_i) denotes the annual fixed operation and maintenance cost of technology i. The use of fixed OPEX values is consistent with the cost representation provided in the Danish Energy Agency Technology Data Catalogue and is commonly applied in comparative energy system assessments.

The operating expenditure is limited to fixed operation and maintenance costs associated with the considered renewable energy components. Variable fuel costs and network-related operating expenses are not included, as the system under study is fully based on renewable electricity generation.

3.4.8 Total Annualized Cost (TAC)

To enable economic comparison between scenarios with different investment sizes and component lifetimes, investment costs were converted into equivalent annual costs using an annualization approach. The total annualized cost (TAC) represents the combined annual burden of capital investment and fixed operation and maintenance costs ($OPEX$).

$$TAC = CAPEX * CRF + OPEX \quad (11)$$

In this formulation, the capital recovery factor (CRF) is used to distribute the upfront investment cost over the lifetime of the system, accounting for the time value of money. By combining the annualized investment cost with annual OPEX, the TAC provides a single, comparable economic indicator across all scenarios (Stock et al., 2024).

The total annualized cost is calculated by annualizing the total investment cost and adding the fixed operation and maintenance costs. This simplified formulation is adopted to ensure consistent economic comparison between renewable energy system configurations within the defined study scope.

This economic assessment framework ensures that scenario comparisons reflect not only technical performance but also the economic implications of different system configurations, supporting robust evaluation of cost–performance trade-offs.

3.4.9 Performance Indicators

System performance was evaluated using a consistent set of quantitative indicators applied uniformly across all building-scale case studies and city-scale scenarios. These indicators were selected to capture key aspects of system behavior, including reliability of supply, surplus generation resulting from oversizing, and the contribution of renewable electricity to overall demand. The use of standardized performance indicators enables transparent comparison between scenarios and supports robust interpretation of results (Diaf et al., 2008; Connolly et al., 2010; Rashid and Safari, 2021).

3.4.10 Shortage Ratio

The shortage ratio (Sh) quantifies the proportion of electricity demand that cannot be met by renewable generation (E_{unmet}) and available battery discharge. This indicator is used to assess system reliability and reflects the extent to which the proposed energy system fails to satisfy demand under given scenario conditions.

$$Sh = \frac{\sum E_{unmet}}{\sum E_{load}} \quad (12)$$

In this formulation, unmet electricity demand (E_{unmet}) represents the cumulative energy deficit over the simulation period, while total electricity demand (E_{load}) corresponds to the aggregate

load. A lower shortage ratio indicates higher system reliability and improved adequacy of renewable generation and storage capacity. This indicator is commonly used in simulation-based assessments of renewable energy systems to evaluate supply adequacy and system reliability.

3.4.11 Wasted Energy Ratio

The wasted energy ratio (WR) represents the proportion of renewable electricity generation that cannot be utilized due to insufficient demand or limited storage capacity. This indicator is used to evaluate oversizing effects and system inefficiencies associated with excessive renewable generation.

$$WR = \frac{\sum E_{waste}}{\sum E_{generated}} \quad (13)$$

Here, wasted electricity (E_{waste}) corresponds to renewable energy that is curtailed after battery charging constraints are reached, while total renewable generation ($E_{generated}$) represents the aggregate electricity produced by solar PV and wind systems. A high wasted energy ratio indicates that system capacities exceed the effective utilization potential under the given demand profile. This indicator is used in simulation-based renewable energy system studies to assess surplus generation and oversizing effects under different demand and storage conditions.

3.4.12 Renewable Coverage Ratio

The renewable coverage ratio (Cov) measures the fraction of total electricity demand supplied by renewable energy sources. This indicator provides an overall assessment of the system's ability to meet demand using renewable electricity and is particularly relevant for evaluating decarbonization potential.

$$Cov = \frac{\sum E_{renewable}}{\sum E_{load}} \quad (14)$$

In this formulation, renewable electricity (E_renewable) refers to the share of solar PV and wind generation that is either directly consumed or delivered through battery discharge, while total electricity demand (E_load) represents the aggregated load over the simulation period. A higher coverage ratio, therefore, reflects a greater reliance on renewable energy sources and a reduced dependence on external electricity supply. This indicator is commonly used to quantify renewable energy penetration and to assess the extent to which electricity demand is supplied by renewable sources within a given system configuration.

Taken together, these performance indicators offer complementary insights into system reliability, efficiency, and renewable integration. Their combined application allows for a

comprehensive evaluation of scenario outcomes and supports meaningful comparisons across different system configurations and spatial scales.

4 Urban Case Studies (Building Scale)

4.1 Kärlekens Skola

4.1.1 Case study overview and building characteristics

Kärlekens Skola is a modern educational building located at Stenstorpsvägen 2 in Halmstad, Sweden. The school was constructed in 2020 and has a total heated floor area (Atemp) of 6,998 m². It accommodates students from Grade 4 to Grade 9 and consists of two floors. The building includes typical school facilities such as classrooms, teachers' offices, a full-scale kitchen, sports areas, and conference rooms.

The school operates on weekdays between 06:00 and 18:00. Regular teaching activities take place until 16:00, after which selected areas remain in use for after-school activities. This extended operational schedule makes the building a suitable case study for analyzing the relationship between occupancy patterns and electricity demand. All electricity consumption and system operation data used in this case study were obtained from official municipal datasets provided by the Municipality of Halmstad and represent measured operational performance.

An aerial view of the building is provided in Figure 3 to illustrate the building context and surrounding area based on municipal geospatial data from the City of Halmstad (Halmstad Municipality).



Figure 3. Figure 3. Aerial view of Kärlekens Skola and its surrounding area, based on municipal geospatial data provided by the City of Halmstad.

4.1.2 Energy systems and available municipal data

Kärlekens Skola is connected to the district heating network and does not employ active air conditioning systems. Ventilation is provided by five mechanical ventilation units equipped with heat recovery (FTX/MHRV). Four of these units use rotary heat exchangers, while one unit employs a plate heat exchanger.

The building also hosts an on-site solar photovoltaic system. Detailed technical specifications of the installed PV system, including system capacity and configuration, were obtained from municipal records and supplementary documentation provided by the municipality.

A key strength of this case study lies in the availability of high-resolution, real operational data obtained directly from the municipality. The dataset includes:

- Hourly electricity consumption data covering a full calendar year
- Disaggregated electricity consumption for major end-use categories (ventilation, kitchen, refrigeration)
- Hourly PV electricity production data for the installed system

These datasets form the basis for both baseline performance assessment and subsequent scenario-based analyses.

4.1.3 Baseline electricity consumption characteristics

The annual electricity consumption profile of Kärlekens Skola exhibits strong dependence on occupancy patterns and seasonal operation. Hourly electricity demand data reveal pronounced daily and weekly variations associated with school opening hours, kitchen operation, and ventilation requirements.

To illustrate typical operational behavior, a representative weekly electricity consumption profile is presented in Figure 4, covering the period from 7 October to 13 October, from 00:00 on Monday to 23:00 on Sunday. This figure highlights the clear distinction between weekday and weekend operations, with peak electricity demand occurring during school hours and significantly reduced consumption during evenings and weekends.

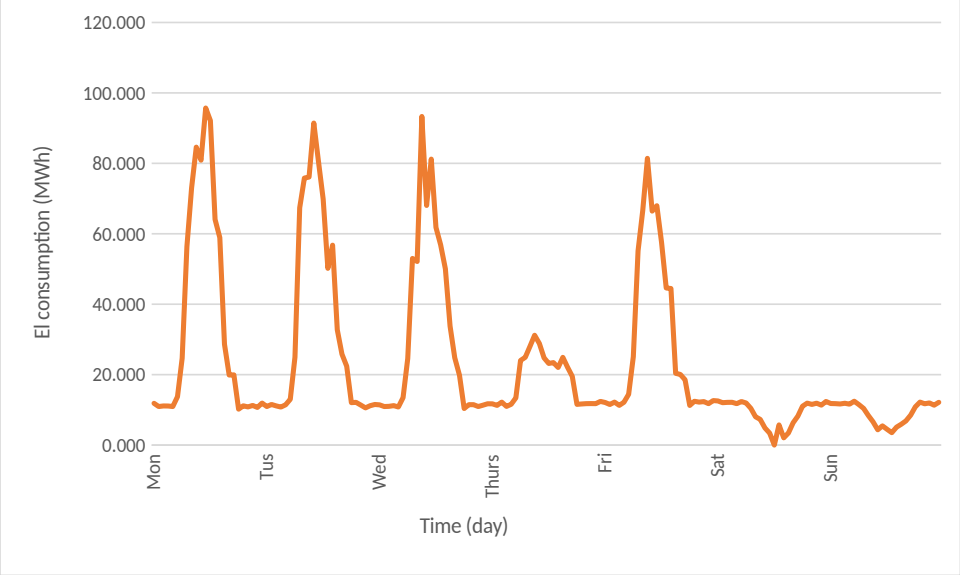


Figure 4. Weekly electricity consumption profile for Kärlekens Skola, reconstructed from hourly municipal energy monitoring data provided by the City of Halmstad (representative week).

The full annual hourly electricity load profile (8,760 hours) and detailed monthly electricity consumption breakdowns by end-use category are provided in Appendix A, as these datasets support model input validation but are not required for direct interpretation of scenario results.

Seasonal trends in electricity demand further reflect the operational characteristics of the school. Electricity consumption is highest during winter months, driven primarily by increased ventilation requirements for indoor air quality control under cold outdoor conditions. In contrast, electricity demand decreases substantially during summer vacation periods, although it does not fall to zero due to continuous operation of essential systems such as refrigeration, security lighting, and minimum ventilation.

4.1.4 Case-specific insights

The baseline analysis of Kärlekens Skola demonstrates that electricity demand is predominantly governed by occupancy-driven ventilation and kitchen operation, rather than by static base loads. This behavior distinguishes the building from residential or continuously operated facilities and underscores the importance of using real operational data when evaluating renewable energy integration strategies.

The availability of detailed municipal datasets enables accurate representation of both temporal demand variability and system operation, providing a robust foundation for scenario-based assessment of renewable energy systems at the building scale.

4.2 Nymansgatan 23

4.2.1 Case study overview and building characteristics

Nymansgatan 23 is a mixed-use building located in Halmstad, Sweden, constructed in 1979, with a total heated floor area of 9,415 m². The building combines two distinct functions: a school located on the ground floor and office spaces on the upper floor. This dual-use configuration results in differentiated occupancy patterns and electricity demand characteristics within the same building envelope. An aerial view of the building location and surrounding urban context is provided in Figure 5 based on municipal geospatial data from the City of Halmstad.

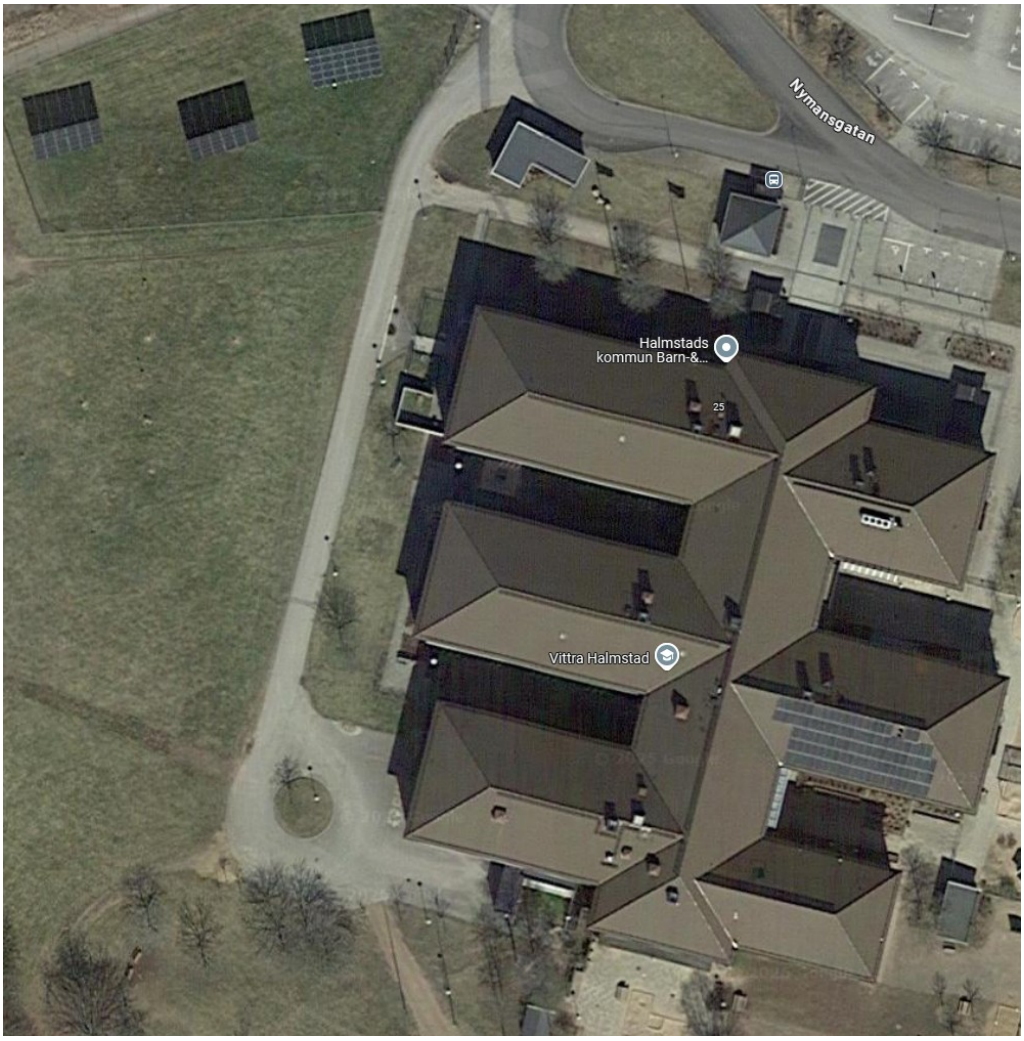


Figure 5. Aerial view of Nymansgatan 23 and its surrounding area, based on municipal geospatial data provided by the City of Halmstad.

The building is supplied with district heating (*fjärrvärme*) and does not employ any mechanical cooling system. Ventilation is provided through a centralized system consisting of eleven mechanical ventilation units with heat recovery (FTX). Among these units, three are equipped with plate heat exchangers and eight utilize rotary heat exchangers. In addition, the building hosts a full-scale kitchen serving the school operations.

A rooftop solar photovoltaic (PV) system is installed on the building and contributes to on-site electricity generation. Detailed operational electricity consumption data and PV production data were obtained directly from municipal records, providing a high-resolution and reliable basis for analysis. All electricity consumption and system operation data used in this case study were obtained from official municipal datasets provided by the Municipality of Halmstad and represent measured operational performance.

4.2.2 Operational schedules and electricity demand characteristics

The operational schedules of the two building functions differ notably. The school operates between 06:00 and 18:00 on weekdays, while the office spaces are typically occupied between 07:00 and 16:00. These staggered schedules lead to a composite electricity demand profile characterized by multiple daily peaks.

A representative weekly electricity consumption profile is shown in Figure 6, based on hourly municipal energy monitoring data from the City of Halmstad. This illustrates covering the period from 7 October to 13 October, from 00:00 on Monday to 23:00 on Sunday, illustrating pronounced weekday peaks associated with simultaneous operation of lighting, ventilation systems, office equipment, and kitchen activities. Electricity demand decreases substantially during evenings and weekends, reflecting reduced occupancy and limited operation of non-essential systems.

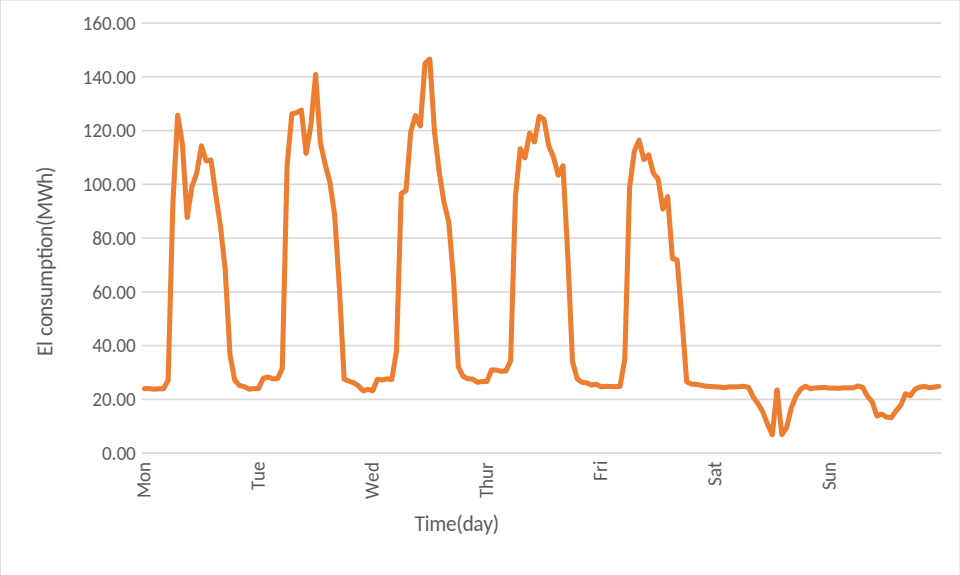


Figure 6. Weekly electricity consumption profile for Nymansgatan 23, reconstructed from hourly municipal energy monitoring data provided by the City of Halmstad (representative week).

4.2.3 Seasonal and system-specific electricity use

Figure 7 presents the monthly electricity consumption of the central ventilation system for the years 2023 and 2024. In both years, consumption levels vary noticeably from month to month,

without exhibiting a clear or consistent seasonal pattern. While some of the higher values occur during late winter and early spring, similar consumption levels are also observed during parts of the summer and autumn periods. Overall, the ventilation-related electricity use in 2024 is generally lower than in 2023 across several months, although the differences are not uniform throughout the year. These results indicate that the observed variations are primarily driven by operational or usage-related factors rather than by a distinct seasonal effect.

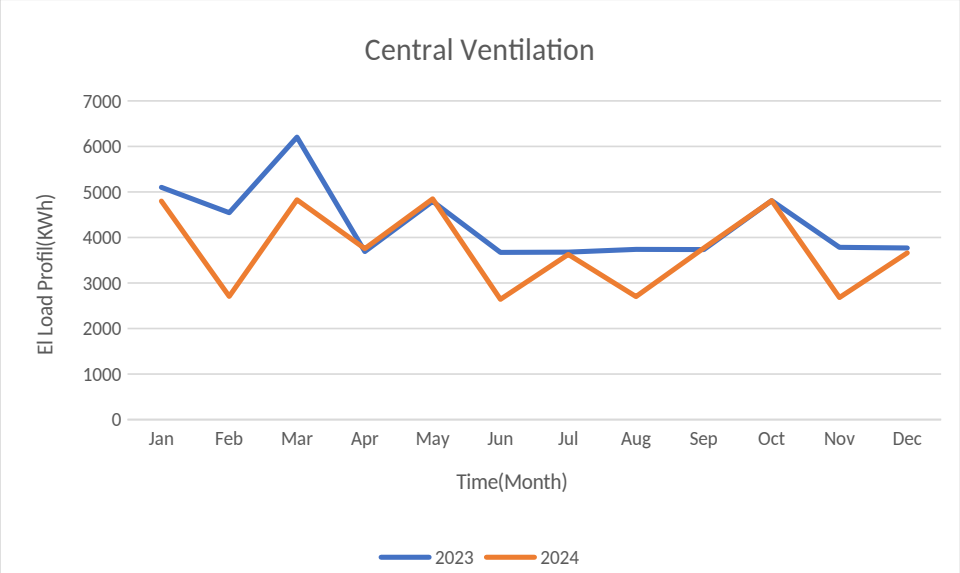


Figure 7. Monthly electricity consumption of the central ventilation system for Nymansgatan 23 for the years 2023 and 2024, reconstructed from hourly municipal energy monitoring data provided by the City of Halmstad.

Detailed hourly electricity load profiles and end-use electricity breakdowns for both years are provided in Appendix B to support model validation and detailed system understanding.

4.2.4 Case-specific insights

The Nymansgatan 23 case study highlights the impact of mixed-use occupancy on electricity demand profiles. Unlike single-function educational buildings, electricity consumption reflects overlapping operational schedules and heterogeneous system requirements. Ventilation demand emerges as a dominant driver of electricity use, particularly during winter months, while the on-site PV system contributes to reducing reliance on external electricity supply.

The availability of multi-year, high-resolution municipal data allows for robust characterization of both baseline operation and system behavior, providing a strong foundation for subsequent scenario-based renewable energy analyses.

4.3 Sofiebergs äldreboende

4.3.1 Case study overview and building characteristics

Sofiebergs äldreboende is an elderly care facility located at Sofiebergsvägen 34 in Halmstad, Sweden. The building was constructed in 2014 and consists of three floors. As a healthcare-related facility, it operates continuously, 24 hours a day, throughout the year, imposing strict requirements on indoor environmental quality, ventilation, and thermal comfort.

An aerial view of the building location and surrounding area is provided in Figure 8 based on municipal geospatial data from the City of Halmstad.



Figure 8. Aerial view of Sofiebergs äldreboende and its surrounding area, based on municipal geospatial data provided by the City of Halmstad.

The building is well insulated compared to older structures, which reduces transmission losses; however, continuous operation and healthcare-specific requirements result in a relatively high and stable electricity demand. Electricity consumption data used in this study were obtained directly from municipal operational records and represented real measured performance. All electricity consumption and system operation data used in this case study were obtained from official municipal datasets provided by the Municipality of Halmstad and represent measured operational performance.

4.3.2 Operational characteristics and electricity demand

Unlike educational or office buildings, Sofiebergs äldreboende exhibits a persistent electricity demand throughout both weekdays and weekends. Daily variations are driven primarily by staff activities, resident routines, and system operation rather than occupancy schedules.

A representative weekly electricity consumption profile is shown in Figure 9, based on municipal geospatial data from the City of Halmstad. The figure illustrates sustained electricity demand across the entire week, covering the period from 7 October to 13 October, from 00:00 on Monday to 23:00 on Sunday (the same time as the other case studies), with moderate peaks occurring during morning routines, meal preparation periods, and evening activities. Night-time electricity consumption remains significant due to the continuous operation of ventilation systems, medical equipment, and essential services.

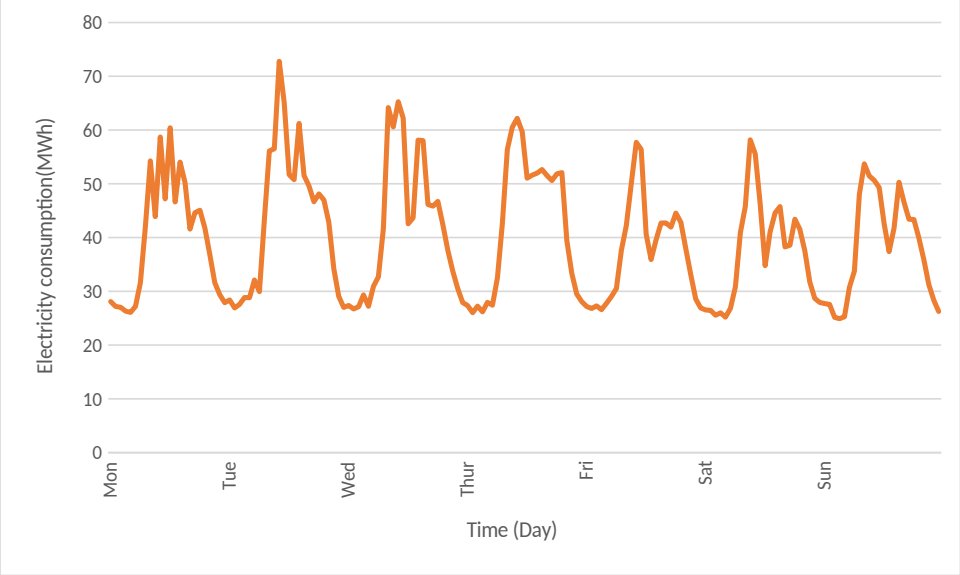


Figure 9. Weekly electricity consumption profile for Sofiebergs äldreboende, reconstructed from hourly municipal energy monitoring data provided by the City of Halmstad (representative week).

4.3.3 Seasonal ventilation and cooling behavior

Seasonal variations in electricity consumption are strongly linked to ventilation and cooling requirements. During cold periods, ventilation systems operate intensively to maintain indoor air quality and thermal comfort, while during warm periods, additional cooling-related ventilation becomes necessary.

Figures 10 and 11 present the monthly electricity consumption associated with cooling ventilation and cooling equipment for the years 2023 and 2024 (based on municipal geospatial data from the City of Halmstad). In 2023, cooling-related electricity use was largely confined to the period from May to September, with a pronounced peak in June. In contrast, cooling demand in 2024 started earlier in the year and reached its maximum in late summer, indicating increased cooling requirements likely associated with warmer outdoor conditions or altered operational practices.

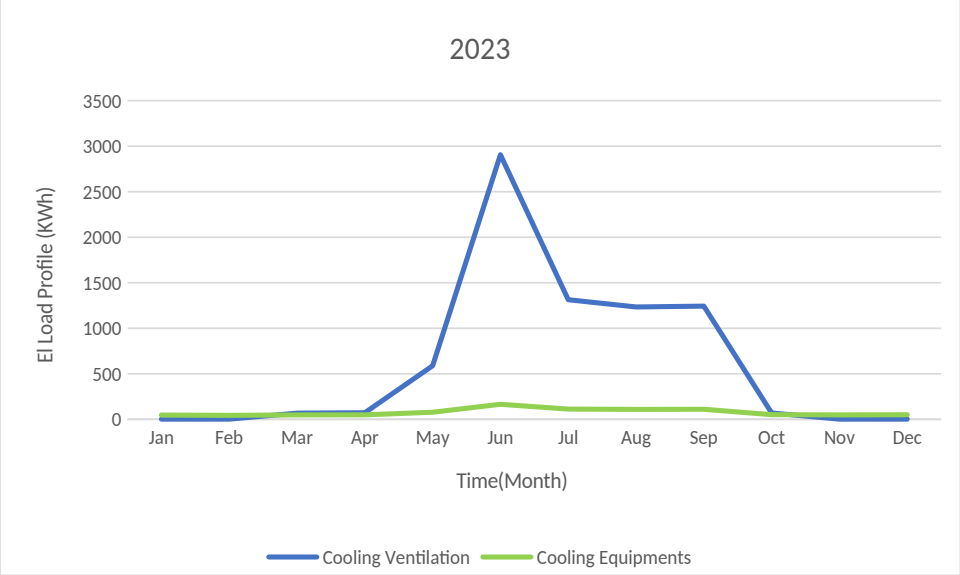


Figure 10. Monthly cooling ventilation electricity consumption Sofiebergs äldreboende. (2023) reconstructed from hourly municipal energy monitoring data provided by the City of Halmstad (representative week).

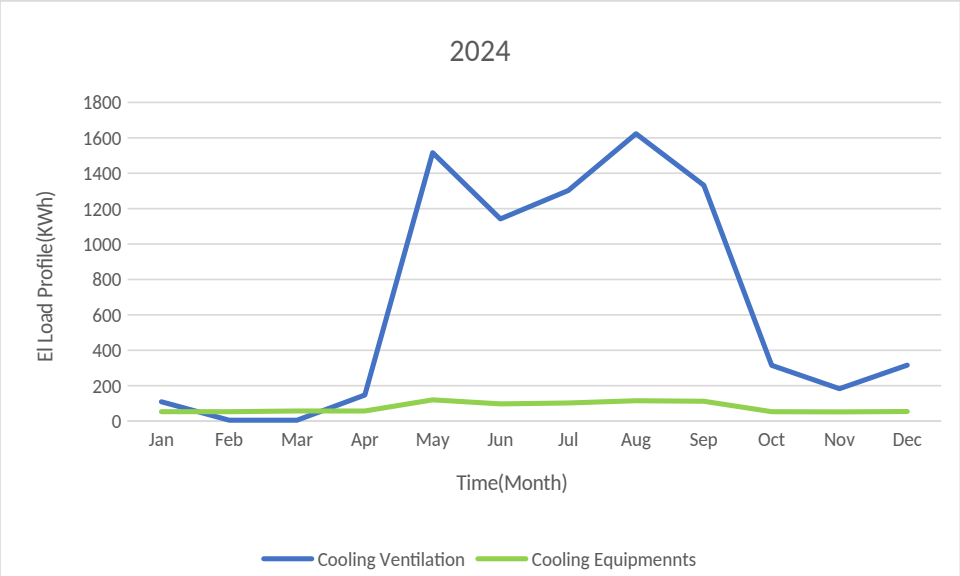


Figure 11. Monthly cooling ventilation electricity consumption Sofiebergs äldreboende. (2024) reconstructed from hourly municipal energy monitoring data provided by the City of Halmstad (representative week).

Cooling electricity consumption associated with equipment remained relatively stable across both years, reflecting the continuous operation of medical and essential devices independent of seasonal conditions.

Detailed hourly load profiles and end-use electricity breakdowns for ventilation, kitchen, and equipment systems are provided in Appendix C.

4.3.4 Case-specific insights

The Sofiebergs äldreboende case study highlights the fundamentally different electricity demand characteristics of healthcare facilities compared to educational or office buildings. Continuous operation, stringent indoor environmental requirements, and reliance on ventilation and medical equipment result in a high and stable baseline electricity demand throughout the year.

Seasonal variations are primarily driven by thermal and ventilation requirements rather than occupancy changes. These characteristics underline the importance of tailored energy efficiency and renewable integration strategies for elderly care facilities, where system reliability and indoor comfort are critical constraints.

4.4 Trönningeskolan

4.4.1 Case study overview and building characteristics

Trönningeskolan is located at Laxviksvägen 30 in Halmstad, Sweden, and was constructed in 1994. The school has a total heated floor area of 5,775 m², complemented by a separate sports hall with an additional heated area of 1,220 m². The building is supplied with a biogas-based district heating system and does not include any air conditioning or on-site solar photovoltaic installation.

An aerial view of the building location and surrounding area is provided in Figure 12, based on municipal geospatial data from the City of Halmstad.



Figure 12. Aerial view of Trönningeskolan and its surrounding urban context based on municipal geospatial data provided by the City of Halmstad.

Ventilation is provided through older mechanical ventilation systems with heat recovery (FTX). The main school building is served by three FTX units, two equipped with counterflow plate heat exchangers and one with a coil heat exchanger. The sports hall is ventilated by two additional FTX units, one using a rotary heat exchanger and the other a plate heat exchanger. A full-scale kitchen is also present within the facility.

All electricity consumption and system operation data used in this case study were obtained from official municipal datasets provided by the Municipality of Halmstad and represent measured operational performance.

4.4.2 Operational schedules and electricity demand characteristics

Trönningeskolan operates primarily on weekdays between 06:00 and 18:00. Outside these hours, the building exhibits a pronounced shutdown pattern, distinguishing it from continuously operated facilities such as healthcare buildings.

A representative weekly electricity consumption profile, covering the period from 7 October to 13 October, from 00:00 on Monday to 23:00 on Sunday, is shown in Figure 13 based on municipal geospatial data from the City of Halmstad. The profile demonstrates a clear reduction in electricity demand during nighttime hours, indicating systematic shutdown of non-essential

systems such as lighting, classroom equipment, and most ventilation units. Residual electricity consumption during nights and weekends is attributed to essential services, including security systems, refrigeration equipment, and limited background ventilation.

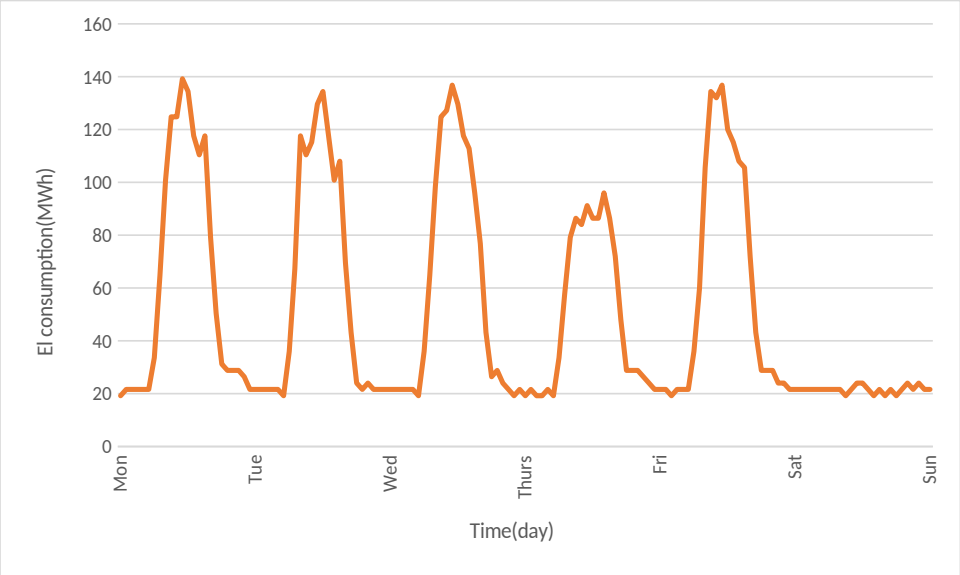


Figure 13. Weekly electricity consumption profile of Trönningeskolan based on municipal geospatial data provided by the City of Halmstad.

During operational hours, electricity demand increases sharply in the early morning as ventilation systems, lighting, and kitchen equipment are activated. A secondary peak is typically observed around midday, corresponding to meal preparation and increased activity in common areas. Electricity demand gradually decreases in the afternoon as teaching activities conclude.

4.4.3 Seasonal and operational characteristics (text-based analysis)

Seasonal variations in electricity demand are primarily driven by changes in ventilation intensity and heating-related operation rather than continuous occupancy. During the winter months, electricity demand increases due to higher ventilation and heating requirements under cold outdoor conditions. In contrast, during school holiday periods and milder seasons, electricity consumption decreases significantly as occupancy levels drop, and ventilation systems operate at reduced capacity.

Short periods of increased electricity demand are also observed during late spring and early summer, associated with cooling-related ventilation needs, particularly in the sports hall. These variations reflect adaptive system operation in response to both outdoor climate conditions and building use patterns.

Detailed hourly electricity load profiles and ventilation system-specific electricity consumption data for both 2023 and 2024 are provided in Appendix D to support model validation and further analysis.

4.4.4 Case-specific insights

Trönningeskolan represents a baseline educational building characterized by clear operational schedules, pronounced night-time shutdown, and the absence of on-site renewable electricity generation. Compared to newer school buildings, the electricity demand profile exhibits stronger sensitivity to seasonal conditions, reflecting older insulation standards and less advanced ventilation control strategies.

The observed operational patterns highlight the potential for targeted energy efficiency improvements and provide a robust reference case for evaluating renewable energy integration and storage strategies in subsequent scenario-based analyses.

A comparative analysis of electricity load profiles for all building-scale case studies during selected representative months (March, May, July, and October) is provided in Appendix E.

4.5 Baseline Solar Energy Performance

4.5.1 Existing Solar Energy Systems – Kärlekens Skola

Kärlekens Skola is equipped with an on-site rooftop photovoltaic (PV) system designed to partially offset the building's electricity demand through local renewable energy generation. The system represents a typical PV configuration for educational buildings and provides a suitable baseline for assessing the interaction between on-site solar production and electricity demand.

Monthly solar electricity generation and total building electricity demand for the years 2023 and 2024 are summarized in Table 1 and illustrated in Figure 14 based on municipal geospatial data from the City of Halmstad. Solar generation exhibits a clear seasonal pattern, with peak production occurring between May and July, while generation remains minimal during the winter months due to limited solar irradiance in southern Sweden.

Table 1. Monthly solar generation and electricity demand at Kärlekens Skola based on municipal geospatial data provided by the City of Halmstad.

Month	PV Generation 2023 (kWh)	PV Generation 2024 (kWh)	Total El. Load 2023 (kWh)	Total El. Load 2024 (kWh)
Jan	276	336	17,379	17,812
Feb	980	577	15,530	17,430
Mar	1,983	2,120	18,484	16,244
Apr	3,973	3,047	12,078	14,672

Month	PV Generation 2023 (kWh)	PV Generation 2024 (kWh)	Total El. Load 2023 (kWh)	Total El. Load 2024 (kWh)
May	4,762	4,653	13,231	14,417
Jun	4,608	4,311	8,162	8,744
Jul	4,762	4,083	6,435	8,074
Aug	3,881	3,992	13,079	14,190
Sep	2,660	2,971	16,190	17,396
Oct	1,252	1,444	17,541	17,204
Nov	433	437	17,605	19,537
Dec	155	168	16,033	17,164
Total	**29,725**	**28,139**	**171,747**	**182,884**

A comparison between PV generation and total electricity demand indicates that the solar system contributes a meaningful share of electricity during high-irradiance months. However, even during peak production periods, solar generation does not fully cover the building's electricity demand, highlighting the continued dependence on grid electricity.

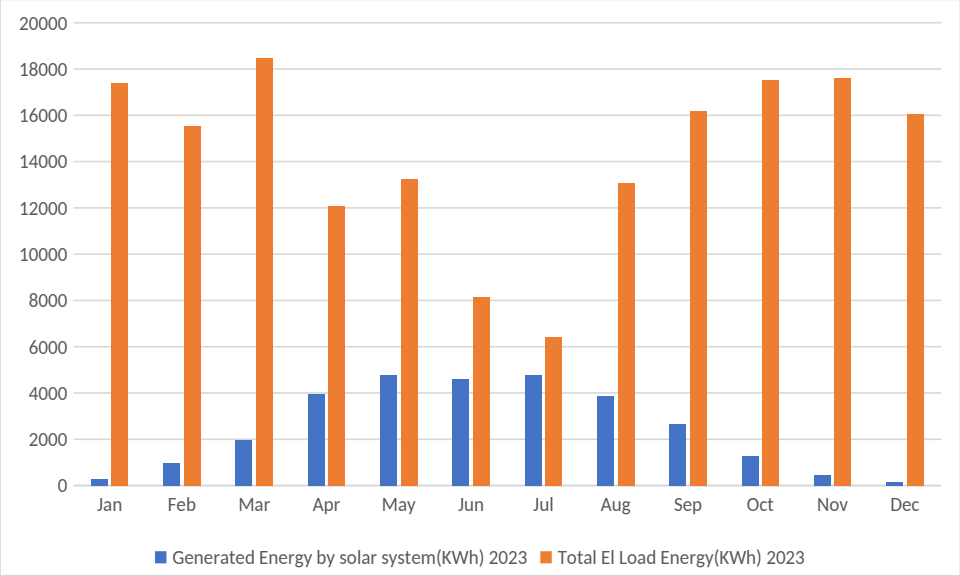


Figure 14. Monthly comparison of total solar PV generation and total electricity demand at Kärlekens Skola for 2023 based on municipal geospatial data provided by the City of Halmstad..

Interannual variations in PV output between 2023 and 2024 are relatively small and are primarily attributed to differences in weather conditions rather than changes in system performance. Overall, the PV system demonstrates stable and predictable operation across both years.

All solar production and electricity demand data presented in this section were obtained from official municipal monitoring systems provided by the Municipality of Halmstad and represent measured operational performance.

Figure 15 presents selected photographs of the photovoltaic installation at Kärlekens skola, illustrating the rooftop-mounted modules, inverter units, and associated electrical components. The images provide visual context for the solar energy system analyzed in this case study and support the interpretation of the measured production data and system performance discussed above. Additional visual documentation is provided in Appendix A. Source: City of Halmstad, Technical and Property Management Department.



Figure 15.Photographs of the installed photovoltaic system at Kärlekens skola, including rooftop PV modules and inverter equipment, City of Halmstad, Technical and Property Management Department (2025).

4.5.2 Existing Solar Energy Systems – Nymansgatan 23

Nymansgatan 23 is equipped with an on-site photovoltaic (PV) system that contributes to the building's electricity supply and provides an opportunity to evaluate the baseline solar performance under real operating conditions. The PV installation is designed primarily to support on-site electricity demand, while surplus electricity is exported to the grid during periods when instantaneous generation exceeds consumption. Given the mixed-use nature of the building, combining school and office functions, electricity demand is concentrated during daytime hours, resulting in partial temporal alignment with solar generation.

Figure 16 presents a comparison between total monthly PV electricity generation and total monthly electricity demand for the year 2023, based on municipal geospatial data from the City of Halmstad. This system-level representation enables a direct assessment of the extent to which solar energy contributes to meeting the building's electricity needs throughout the year. The results reveal a pronounced seasonal dependency, with PV generation increasing steadily from early spring, peaking during late spring and summer months, and declining significantly during autumn and winter. This pattern reflects variations in regional solar irradiance rather than limitations in system operation.

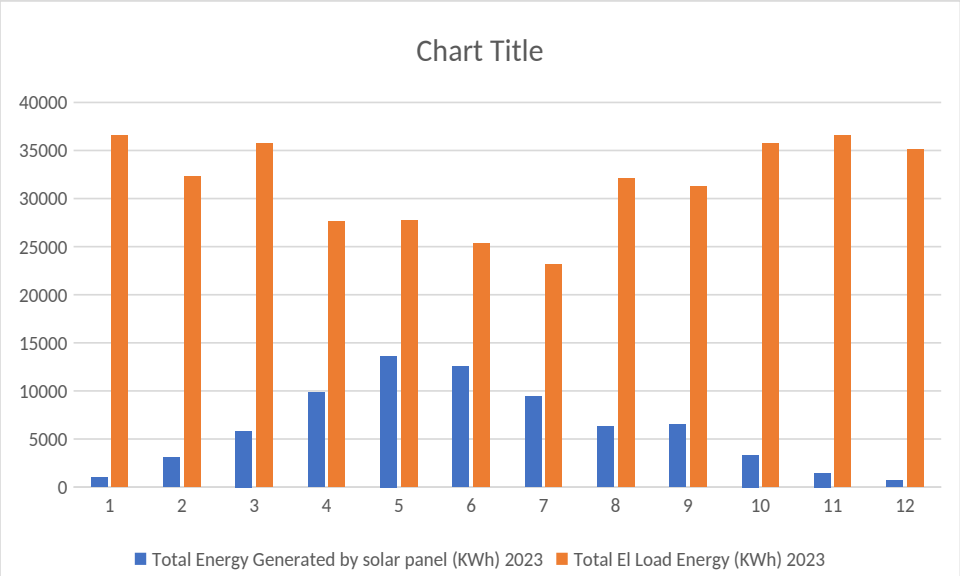


Figure 16. Comparison between total monthly photovoltaic electricity generation and total monthly electricity load at Nymansgatan 23 (2023), based on municipal geospatial data from the City of Halmstad.

During summer months, solar electricity covers a substantial share of the building's total demand, thereby reducing reliance on externally supplied electricity. In contrast, during winter months, PV generation remains limited and accounts for only a small fraction of the total load, highlighting a seasonal mismatch between renewable generation and electricity demand. This mismatch is characteristic of solar-based systems in Nordic climates and underscores the importance of complementary measures such as energy storage or grid interaction.

To support the graphical analysis, Table 2 summarizes monthly PV generation, total electricity demand, and net exported electricity for both 2023 and 2024, City of Halmstad, Technical and Property Management Department (2025). The table provides a quantitative reference for evaluating interannual variations and confirms that differences in annual solar production are primarily driven by weather-related factors rather than changes in system performance.

Table 2. Monthly photovoltaic electricity generation, total electricity demand, and exported electricity at Nymansgatan 23 (2023–2024) based on municipal geospatial data from the City of Halmstad.

Month	Total PV Generation 2023 (kWh)	Total PV Generation 2024 (kWh)	Total Electricity Load 2023 (kWh)	Total Electricity Load 2024 (kWh)
Jan	993	1745	36,518	35,796

Month	Total PV Generation 2023 (kWh)	Total PV Generation 2024 (kWh)	Total Electricity Load 2023 (kWh)	Total Electricity Load 2024 (kWh)
Feb	3,104	1,684	32,331	35,472
Mar	5,830	5,438	35,708	33,311
Apr	9,852	7,441	27,587	31,472
May	13,632	11,300	27,701	29,332
Jun	12,571	9,463	25,365	24,468
Jul	9,453	9,140	23,131	23,366
Aug	6,320	9,236	32,117	30,285
Sep	6,503	7,187	31,248	33,296
Oct	3,323	3,514	35,696	38,301
Nov	1,457	1,131	36,601	38,296
Dec	666	408	35,083	36,743
Total	**73,704**	**67,687**	**379,086**	**390,138**

The solar installation at Nymansgatan 23 demonstrates stable and consistent operational behavior across the analyzed years. The baseline solar performance established in this section forms a reference for subsequent scenario-based analyses, where the role of solar energy is further evaluated in combination with additional renewable technologies and energy storage systems.

Figure 17 presents selected photographs of the rooftop-mounted photovoltaic installation at Nymansgatan 23 provided by the City of Halmstad, Technical and Property Management Department, illustrating the panel layout, inverter equipment, and on-site monitoring interface. The images provide visual support for the building-scale PV system analyzed in this case study and complement the interpretation of the measured production data and scenario results discussed above. Additional visual material is included in Appendix B. Source: Municipality of Halmstad, Technical and Property Management Department.



Figure 17. Photographs of the rooftop photovoltaic system at Nymansgatan 23, including PV modules, inverter units, and monitoring equipment (City of Halmstad, Technical and Property Management Department).

4.5.3 Existing Solar Energy Systems – Sofieberg Elderly Care Home

Sofieberg Elderly Care Home is equipped with a hybrid solar photovoltaic system combining rooftop-mounted panels and façade-integrated solar shading installations. Unlike conventional rooftop-only systems, the façade-mounted panels were primarily introduced to improve indoor thermal comfort while simultaneously contributing to on-site electricity generation.

The rooftop PV system was installed in 2021 and consists of 100 monocrystalline modules with a total peak capacity of 45.5 kWp. In addition, façade-integrated PV systems were installed in stages between 2020 and mid-2024. Two additional façade systems (EMU2 and EMU3) became operational in June and July 2024, significantly expanding the overall solar capacity of the facility (City of Halmstad, Technical and Property Management Department).

Figure 18 presents the monthly total solar electricity generation for 2023 and 2024, based on municipal geospatial data from the City of Halmstad. A clear increase in electricity production is observed in 2024, particularly during summer months, reflecting the commissioning of additional façade-mounted PV systems. For example, solar production in June increased from approximately 2.9 MWh in 2023 to over 4.2 MWh in 2024, while August production rose by more than 50% compared to the previous year.

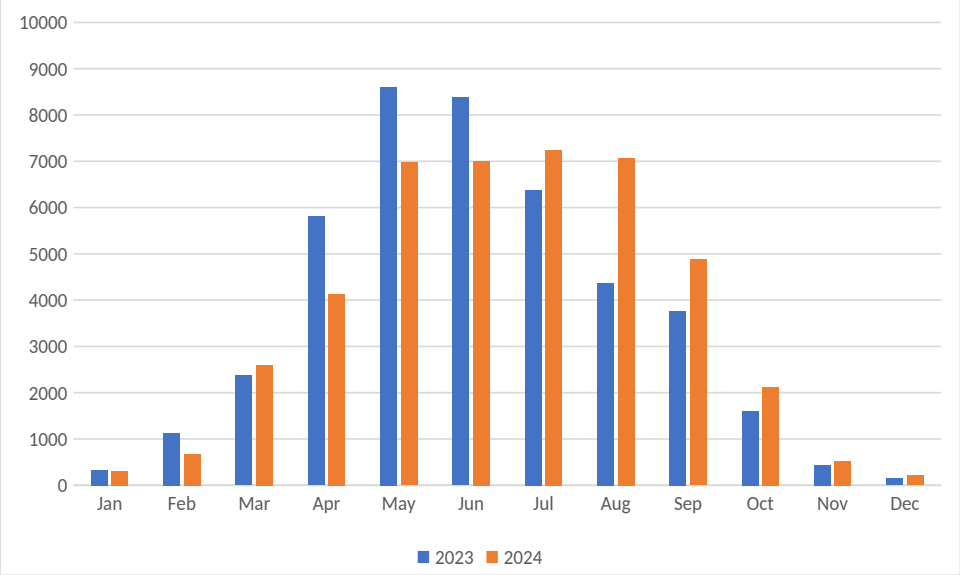


Figure 18: Monthly total solar electricity generation at Sofieberg Elderly Care Home (2023–2024) based on municipal geospatial data from the City of Halmstad.

To assess the role of solar energy within the building's overall electricity balance, Figure 19, based on municipal geospatial data from the City of Halmstad, compares total monthly PV generation with total building electricity demand. Although the facility operates continuously throughout the year and maintains a relatively high baseline demand, solar generation provides a substantial contribution during high-irradiance months. During summer periods, approximately 70–75% of generated solar electricity is consumed directly on-site, with the remaining share exported to the grid.

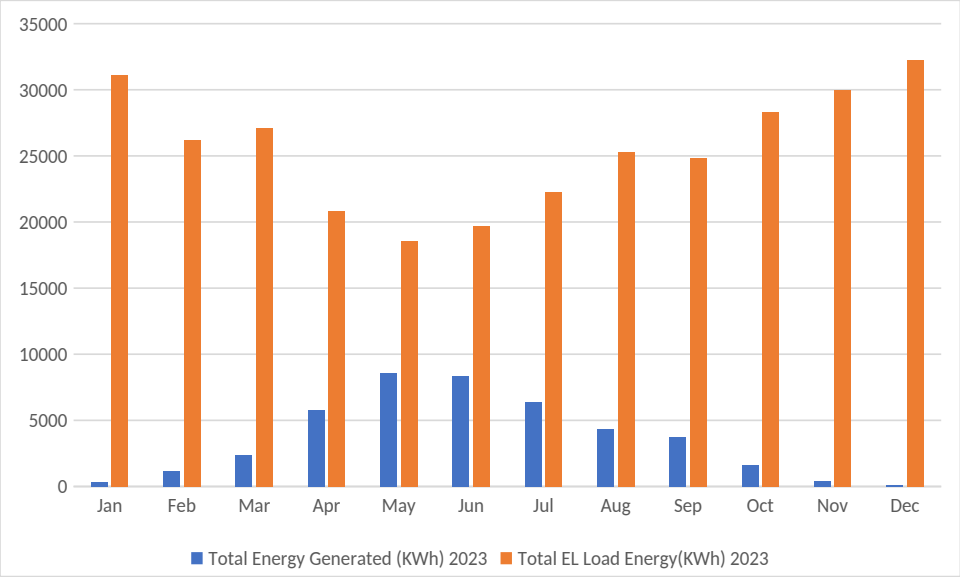


Figure 19. Monthly comparison of total PV generation and total electricity demand at Sofieberg Elderly Care Home, based on municipal geospatial data from the City of Halmstad.

Table 3, based on municipal geospatial data from the City of Halmstad, summarizes the annual electricity balance for the building. Despite the increase in solar production in 2024, total

electricity demand also rose slightly due to operational conditions, resulting in a comparable relative contribution of solar energy across both years.

Table 3. Annual solar electricity production and total electricity demand at Sofieberg Elderly Care Home, based on municipal geospatial data from the City of Halmstad.

Year	Total PV generation (kWh)	Total electricity load (kWh)	Share of PV in total demand (%)
2023	43,343	306,564	14.1
2024	43,752	315,845	13.9

Detailed monthly production data, subsystem-level monitoring (EMU-based breakdown), and electricity export figures are provided in Appendix C. All solar energy data presented in this section are based on measured operational records obtained from official municipal monitoring systems.

Figure 20 presents selected photographs of the façade-integrated photovoltaic system at Sofiebergs äldreboende, illustrating the architectural integration of semi-transparent PV elements, the building context, and the associated inverter and monitoring equipment.



Figure 20. Photographs of the façade-mounted photovoltaic installation at Sofiebergs äldreboende, including building integration, PV elements, and monitoring equipment, City of Halmstad, Technical and Property Management Department.

The images support the interpretation of the measured production data and scenario-based analysis by providing visual context for the building-scale renewable energy system. Additional visual material is provided in Appendix C. Source: City of Halmstad, Technical and Property Management Department.

5 Urban Case Studies (City Scale)

Urban-scale case studies are introduced to complement the building-scale analyses by extending the assessment to a broader system perspective. While the building-level case studies rely on measured municipal electricity data and site-specific characteristics, the city-scale investigations aim to evaluate how renewable energy system performance varies across different Swedish regions under controlled and comparable demand assumptions.

Four cities were selected to represent Sweden's major electricity market regions (SE1–SE4), each characterized by distinct climatic conditions and renewable resource availability. These cities also function as representative industrial and service sectors or urban centers within their respective regions, making them suitable for comparative system-level analysis. Halmstad is included in the city-scale assessment to maintain continuity with the building-scale case studies, enabling cross-scale comparison between measured building-level behavior and aggregated urban-level system performance within the same regional context.

To support clarity and continuity in the presentation of the city-scale analysis, selected data descriptions and scenario-related inputs are introduced alongside the corresponding simulation outputs. Although these elements form part of the analytical setup, they are presented together with the results to allow a more direct and transparent interpretation of system behavior. This structure enables the reader to follow the progression from input assumptions to observed outcomes without unnecessary back-and-forth between sections, thereby improving readability and supporting a coherent comparison across urban scenarios.

6 Results

This section presents the quantitative outcomes of the renewable energy system simulations conducted at both building and city scales. Results are reported in a structured manner, beginning with building-scale case studies based on measured operational data, followed by city-scale scenario analyses using standardized industrial and service sector load profiles. All results are derived from hourly simulations over a full annual period and are evaluated using consistent performance indicators, including unmet load, wasted energy, renewable coverage, and storage utilization.

6.1 Kärlekens Skola — Urban Case Study

6.1.1 Baseline System Context and Load Characteristics

Kärlekens Skola is a municipal school building located in an urban area of Halmstad. As an educational facility, its electricity demand is strongly time-dependent, with high consumption during school hours and substantially lower loads during evenings, nights, and weekends. The building is fully grid-connected and operates under typical urban constraints related to space availability and planning regulations.

The site is equipped with an existing rooftop PV system, installed before this study as part of the municipality's renewable energy program. No on-site energy storage was installed in the baseline configuration. Due to the dense urban context and proximity to residential areas, wind energy systems were excluded, and all scenarios focus exclusively on PV–battery integration.

The objective of this case study is to assess how battery storage sizing affects operational performance when added to an existing PV system, with particular emphasis on unmet load reduction and surplus (wasted) energy behavior.

Figure 21 presents the annual electricity load profile of Kärlekens Skola, based on municipal geospatial data from the City of Halmstad. The profile clearly reflects a weekday-dominated consumption pattern, with peak demand aligned with teaching hours and reduced demand during nights and weekends. Seasonal variation in electricity demand is moderate, indicating that system performance is primarily constrained by temporal mismatch rather than strong seasonal demand fluctuations.

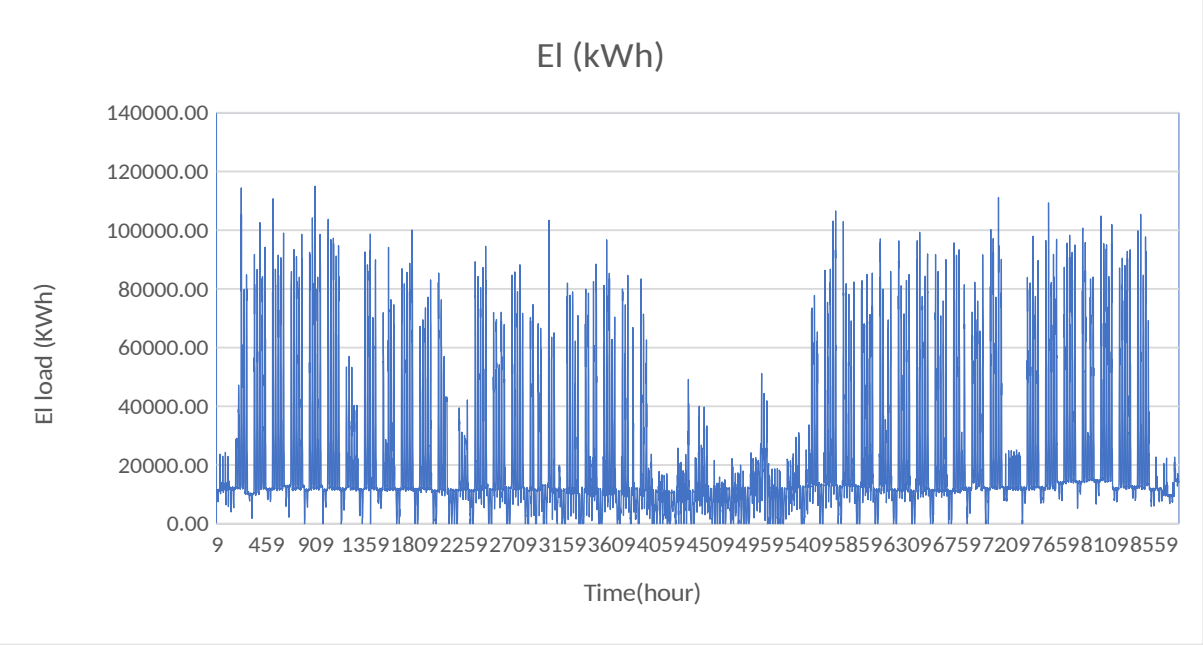


Figure 21. Annual electricity load profile of Kärlekens Skola (baseline operation) (Halmstad Municipality, 2023) based on municipal geospatial data from the City of Halmstad.

These load characteristics establish a realistic baseline against which PV generation and battery operation are evaluated.

6.1.2 Reference Scenario — System Interaction Overview

The reference scenario corresponds to a predefined system configuration with a PV factor of 12 and a battery factor of 3, denoted as configuration 12–3. This scenario serves as the baseline against which subsequent PV and battery scaling scenarios are evaluated.

Figure 22 compares PV generation with the building load for the reference scenario. PV production is highly concentrated around midday hours and during summer months, whereas electricity demand remains distributed across daytime hours throughout the year. As a result, surplus PV generation occurs frequently during midday, while grid imports persist during early mornings, late afternoons, and winter periods.

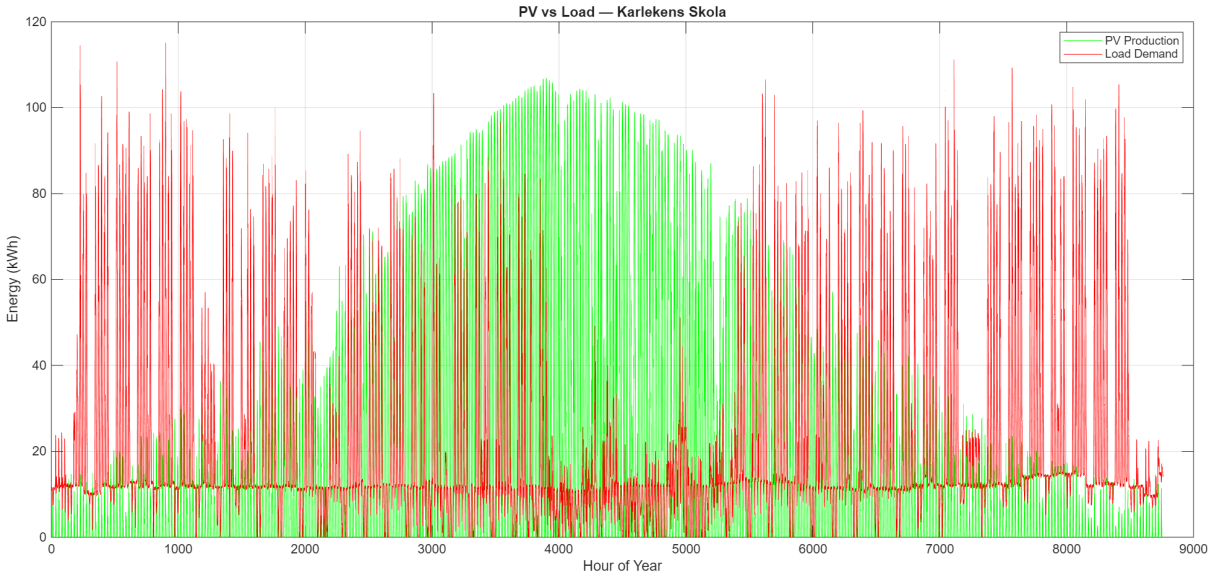


Figure 22. PV production versus load for the reference battery scenario at Kärlekens Skola. Source: Author's own simulation results based on measured load data (MATLAB).

6.1.3 Battery Operation and State of Charge

Battery behavior in the reference scenario is illustrated by the annual state-of-charge (SOC) profile shown in Figure 23. During periods of high solar availability, the battery undergoes regular daily charging and discharging cycles, indicating effective short-term load shifting. The battery is typically charged during midday PV peaks and discharged during late afternoon hours.

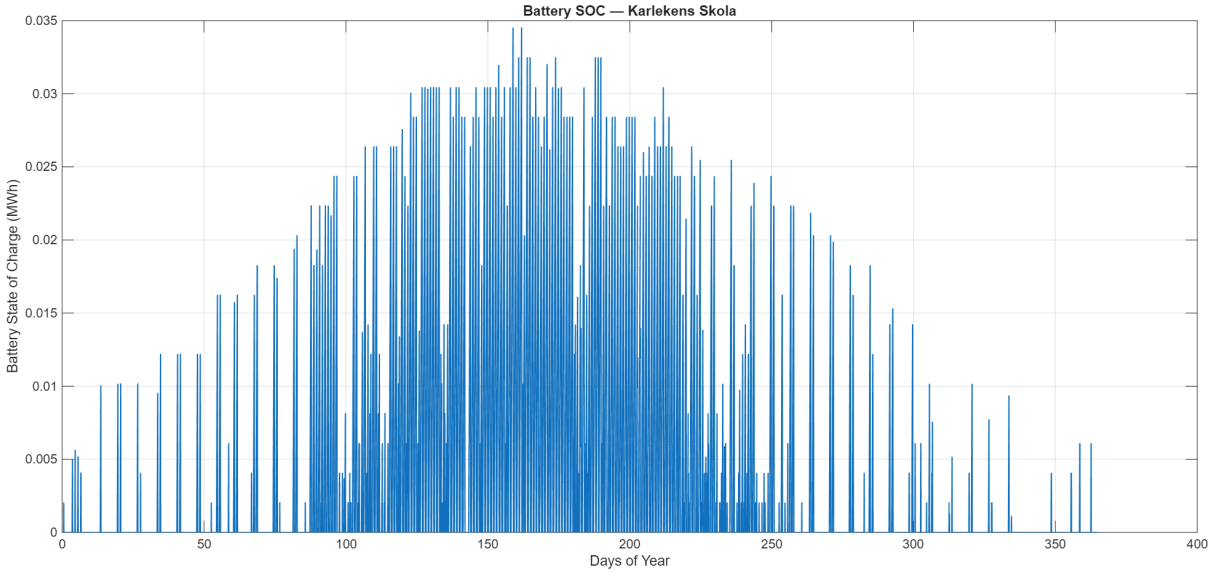


Figure 23. Annual battery state-of-charge (SOC) profile for the reference scenario. Source: Author's own simulation results based on measured load data (MATLAB).

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6.1.4 Comparative Results of Battery Capacity Scaling

To evaluate the impact of storage scaling, additional scenarios were analyzed by progressively increasing battery capacity while keeping PV capacity constant. The quantitative outcomes are summarized in Table 4.

Table 4. Summary of battery scenarios for Kärlekens Skola. Source: Author's own simulation results based on measured load data (MATLAB).

Scenario (PV factor_Batt factor)	PV capacity (kWp)	Battery (MWh)	Wasted energy (kWh/year)	Wasted (%)	Unmet load (kWh/year)	Unmet (%)
12_3 (Reference)	1,380.00	4.140	86,295.1	56.53	117,402.6	64.49
12_5	1,380.00	6.900	83,275.4	54.56	116,922.3	64.22
12_8	1,380.00	11.040	78,902.1	51.69	116,645.9	64.07
14_3	1,610.00	4.830	106,132.5	59.60	112,519.0	61.81
14_5	1,610.00	8.050	102,429.0	57.52	112,067.6	61.56
16_3	1,840.00	5.520	126,628.9	62.22	108,446.2	59.57
16_8	1,840.00	14.720	115,790.0	56.89	107,805.6	59.22

As battery capacity increases, unmet load decreases, particularly at lower storage levels. At higher battery capacities, further reductions in unmet load become less pronounced, while wasted energy increases. These results demonstrate a non-linear relationship between storage capacity, unmet load, and surplus energy under the defined scenario conditions.

6.2 Nymansgatan 23 — Urban Case Study

6.2.1 Baseline System Context and Load Characteristics

Nymansgatan 23 is a municipal building located in Halmstad, analyzed in this thesis based on measured load data and system information provided by the Halmstad Municipality (2023). The building's actual electricity consumption profile, system configuration, and solar production characteristics are documented in detail in the baseline case study data presented earlier in this thesis.

The site is equipped with two distinct solar PV systems:

- a rooftop-mounted PV system, and
- a ground-mounted tracking PV system,

Both of which were operational before the scenario-based analysis. These systems differ in orientation and production behavior and together define the site's baseline solar generation characteristics.

No battery storage was present in the original configuration. Due to the urban context and site constraints, wind energy systems were not considered, and all scenarios focus exclusively on PV–battery system integration based on the existing solar infrastructure.

The objective of this case study is to evaluate how battery storage sizing, when added to a site with mixed PV technologies, influences unmet load reduction and surplus (wasted) energy, using real measured load and production data (Halmstad Municipality, 2023).

6.2.2 Reference Scenario — PV–Load Interaction (with Battery Storage)

Figure 24 presents the measured annual electricity load profile of Nymansgatan 23, based on municipal operational data provided by municipal geospatial data from the City of Halmstad (2023). The load profile reflects the building's actual operational behavior, including its characteristic daily and weekly demand patterns.

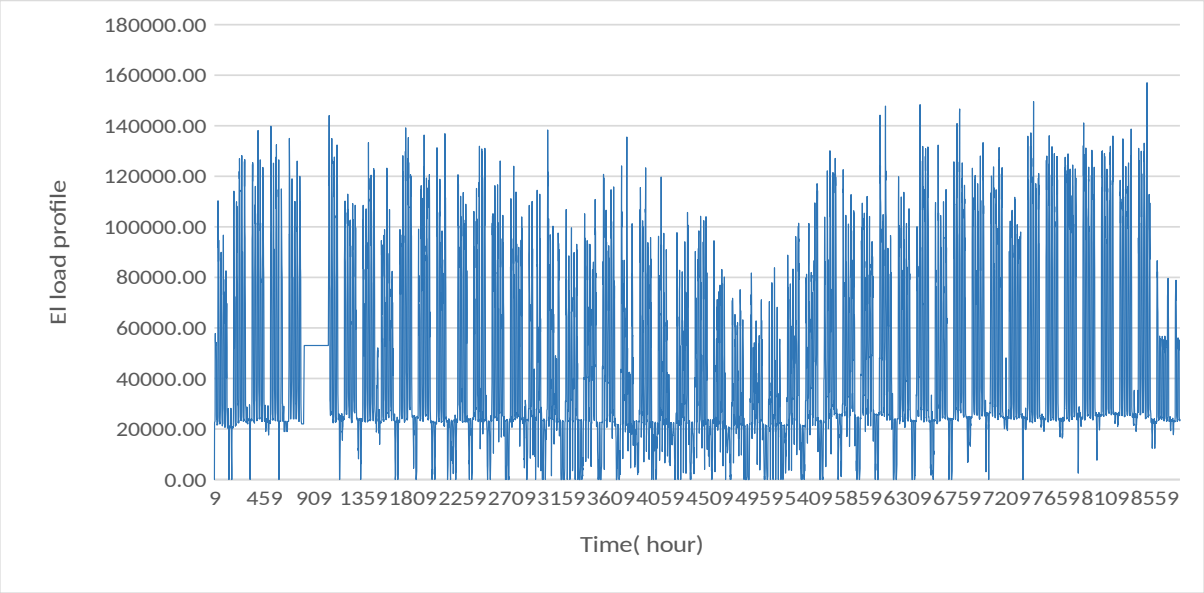


Figure 24. Annual measured electricity load profile of Nymansgatan 23 (baseline operation) based on municipal geospatial data from the City of Halmstad.

Compared to educational buildings such as Kärlekens Skola, Nymansgatan 23 exhibits a more continuous electricity demand, with a noticeable base load persisting outside peak operating hours. Seasonal variations in electricity demand are relatively limited, indicating that the primary challenge for renewable integration is temporal alignment between production and consumption, rather than large seasonal demand shifts.

6.2.3 Reference Scenario — PV–Load Interaction (with Battery Storage)

The reference scenario (12_3) represents the introduction of a battery system with the lowest storage capacity considered in this study, added to the existing rooftop and tracking PV systems without modifying their installed capacities.

Figure 25 compares the combined PV generation from both solar systems with the building load for the reference scenario. The presence of a tracking PV system leads to extended production hours compared to fixed rooftop PV alone; however, solar generation remains concentrated around daylight hours.

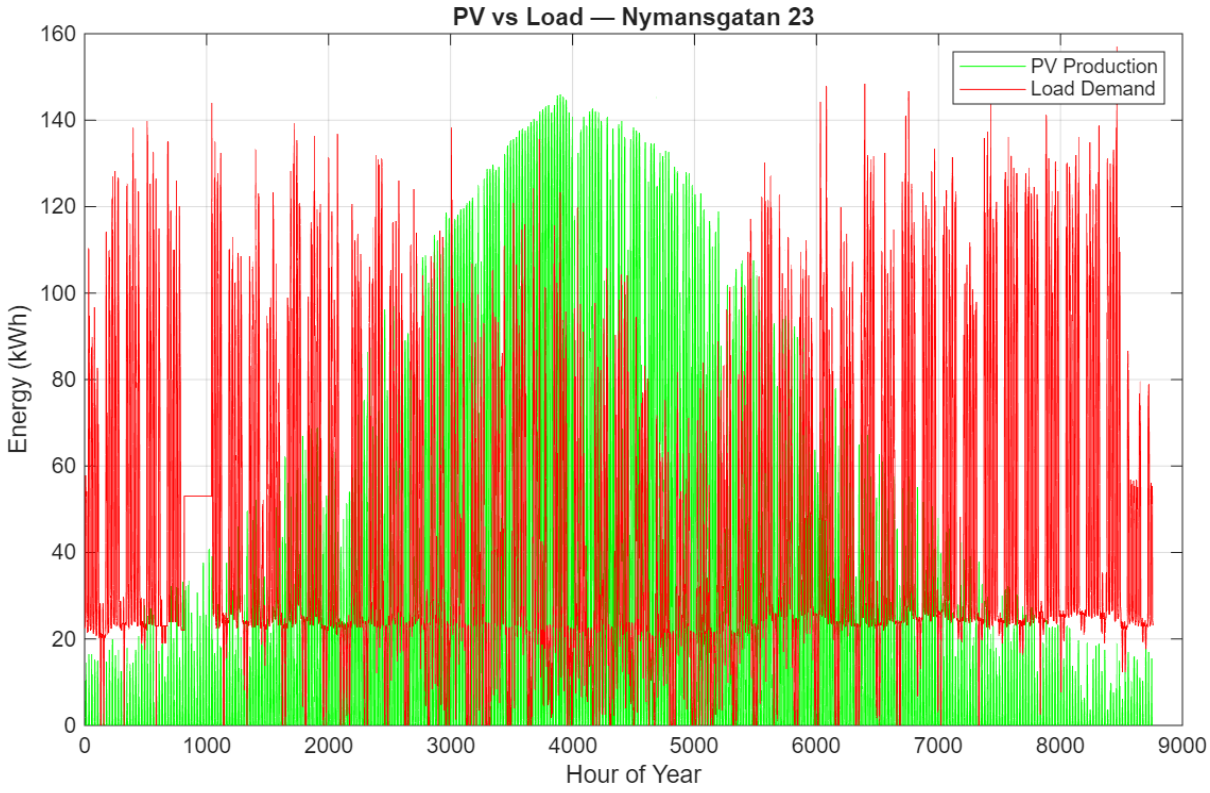


Figure 25. Combined PV generation (rooftop + tracking) versus load for the reference battery scenario (12_3) at Nymansgatan 23. Source: Author's own simulation results based on measured load data (MATLAB).

Surplus PV generation occurs during periods of high irradiance, while grid imports persist during early mornings, evenings, and winter periods, reflecting remaining temporal mismatches between electricity generation and demand under the reference configuration.

6.2.4 Battery Operation and State of Charge

Battery operation in the reference scenario is illustrated in Figure 26 through the annual state-of-charge (SOC) profile. The SOC pattern shows regular daily cycling, with charging during PV peak hours and discharging during late afternoon and evening periods. Compared to school buildings, the higher base load enables more frequent battery discharge events across the day.

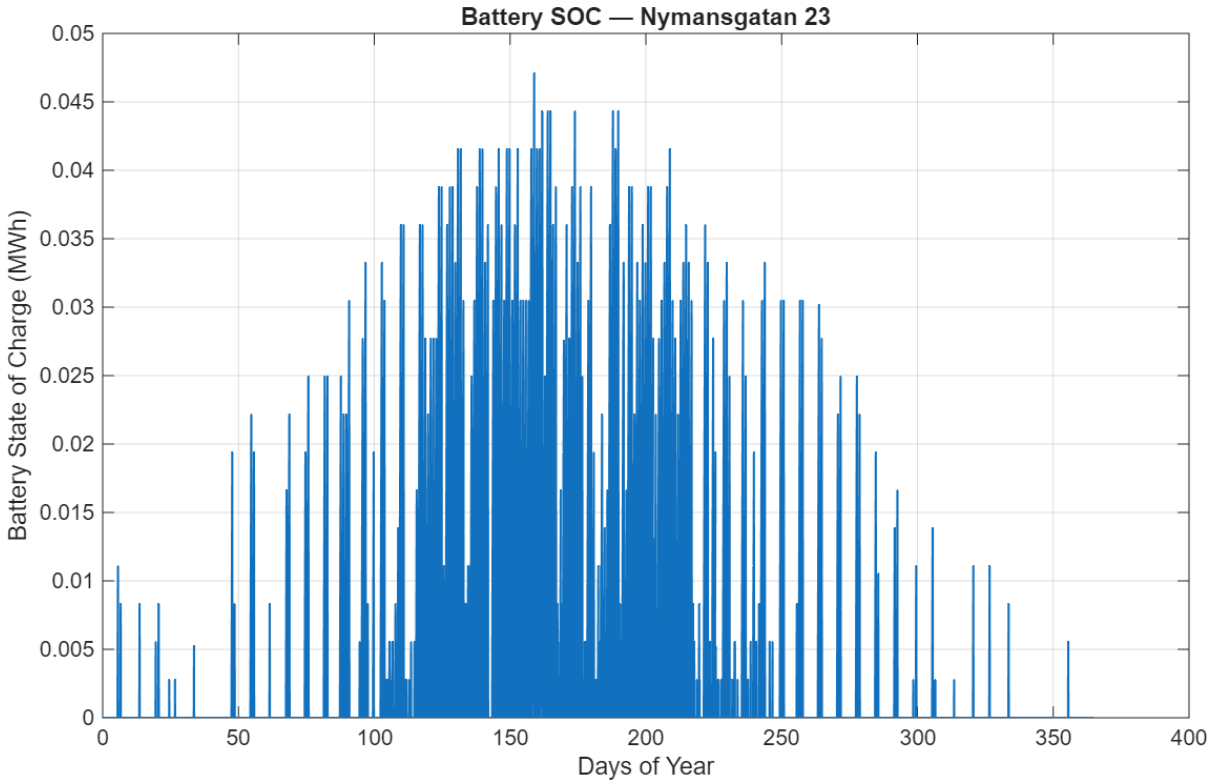


Figure 26. Annual battery state-of-charge (SOC) profile for the reference scenario (12_3). Source: Author's own simulation results based on measured load data (MATLAB).

The SOC profile further indicates reduced battery activity during extended low-irradiance periods, reflecting limited battery contribution during such intervals under the reference configuration.

6.2.5 Comparative Results of Battery Capacity Scaling

The quantitative results of all PV factor–battery factor scenarios for Nymansgatan 23 are summarized in Table 5.

Table 5. Summary of PV–battery scenarios for Nymansgatan 23 Source: Author's own simulation results based on measured load data (MATLAB).

Scenario (PV factor_Batt factor)	PV capacity (kWp)	Battery (MWh)	Wasted energy (kWh/year)	Wasted (%)	Unmet load (kWh/year)	Unmet (%)
12_3 (Reference)	1,884.10	5.652	88,392.8	42.41	269,303.6	69.30
14_3	2,198.11	6.594	112,746.0	46.37	259,289.7	66.72
16_3	2,512.13	7.536	138,305.8	49.77	250,629.7	64.49
12_5	1,884.10	9.420	85,086.5	40.83	267,947.9	68.95

Scenario (PV factor_Batt factor)	*PV capacity (kWp)*	*Battery (MWh)*	*Wasted energy (kWh/year)*	*Wasted (%)*	*Unmet load (kWh/year)*	*Unmet (%)*
12_8	*1,884.10*	*15.073*	*80,283.0*	*38.52*	*267,310.7*	*68.78*
14_5	*2,198.11*	*10.991*	*108,603.9*	*44.67*	*258,106.9*	*66.42*
16_8	*2,512.13*	*20.097*	*126,011.4*	*45.35*	*248,946.1*	*64.06*

At a fixed PV size (12_x scenarios), increasing battery capacity is associated with reductions in unmet load, most notably between scenarios 12_3 and 12_8. Over the same range, wasted energy increases, corresponding to higher levels of surplus PV generation.

At lower battery capacities (e.g., 14_3 and 16_3), increasing PV capacity is associated with increased surplus generation, while reductions in unmet load remain limited. Scenarios combining higher PV and battery capacities (e.g., 16_8) exhibit the lowest unmet load alongside the highest surplus energy values.

6.3 Sofiebergs äldreboende — Urban Case Study

6.3.1 Baseline System Context and Load Characteristics

Sofiebergs äldreboende is a municipal elderly care facility in Halmstad. The analysis presented in this case study is based exclusively on measured electricity consumption data and system information provided by municipal geospatial data from the City of Halmstad (2023), as documented in the baseline case study section of this thesis.

The building is characterized by continuous electricity demand, reflecting the operational requirements of elderly care services, including lighting, ventilation, medical equipment, and auxiliary systems. Unlike educational or office buildings, electricity consumption at Sofiebergs äldreboende remains significant throughout the day and night, resulting in a relatively high and persistent base load.

The site is equipped with an existing rooftop PV system (detail is reported in Appendix C), which was operational before the scenario-based analysis. No battery storage was present in the original configuration. Due to the urban context and site-specific constraints, wind energy systems were excluded, and all scenarios focus on PV–battery integration.

The objective of this case study is to assess how battery storage integration, when added to an existing PV system at a facility with continuous demand, influences unmet load reduction and surplus (wasted) energy, under realistic urban conditions.

Figure 27 presents the measured annual electricity load profile of Sofiebergs äldreboende, based on municipal operational data provided by municipal geospatial data from the City of Halmstad (2023). The load profile clearly demonstrates a stable and continuous consumption pattern, with limited variation between daytime and nighttime demand.

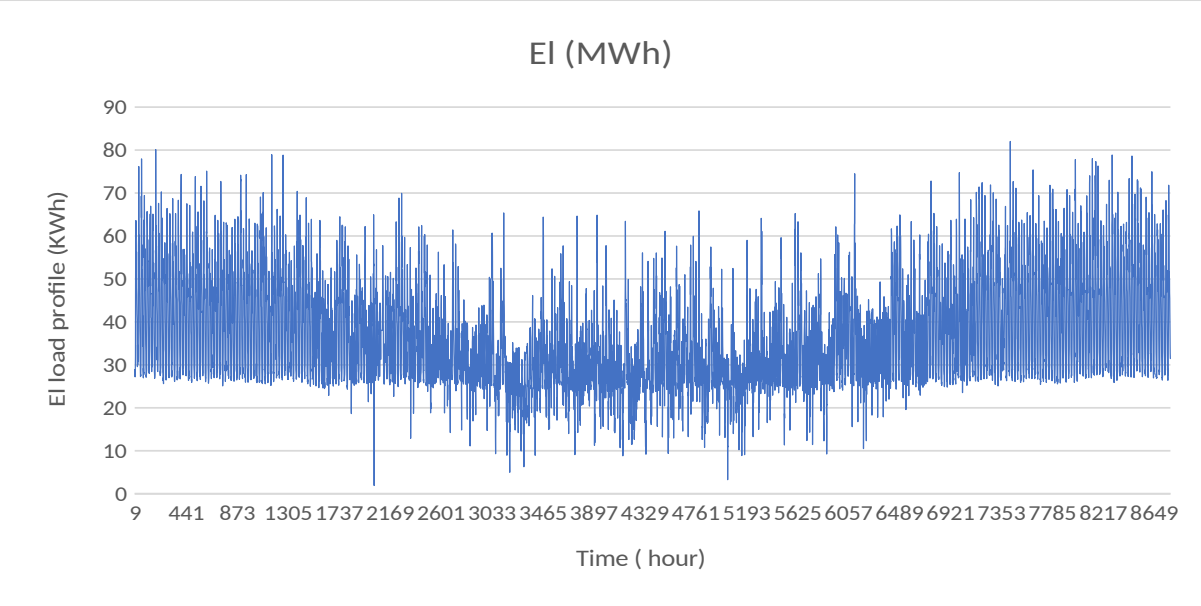


Figure 27. Annual measured electricity load profile of Sofiebergs äldreboende (baseline operation) (based on municipal geospatial data from the City of Halmstad).

Seasonal variations in electricity demand are present but less pronounced than in educational buildings, indicating that electricity use is primarily driven by continuous operational requirements rather than occupancy schedules. This load profile is associated with higher daytime and nighttime demand continuity, which affects on-site solar utilization and battery discharge patterns.

6.3.2 Reference Scenario — PV–Load Interaction (with Battery Storage)

The reference scenario (12_3) represents the introduction of battery storage with the lowest capacity considered in this study, added to the existing rooftop PV system without modifying the installed PV capacity.

Figure 28 compares PV generation with the building electricity load for the reference scenario, based on measured operational data provided by Halmstad Municipality (2023). Due to the continuous base load characteristic of elderly care facilities, a larger fraction of PV generation coincides with on-site electricity demand compared to more intermittent building types.

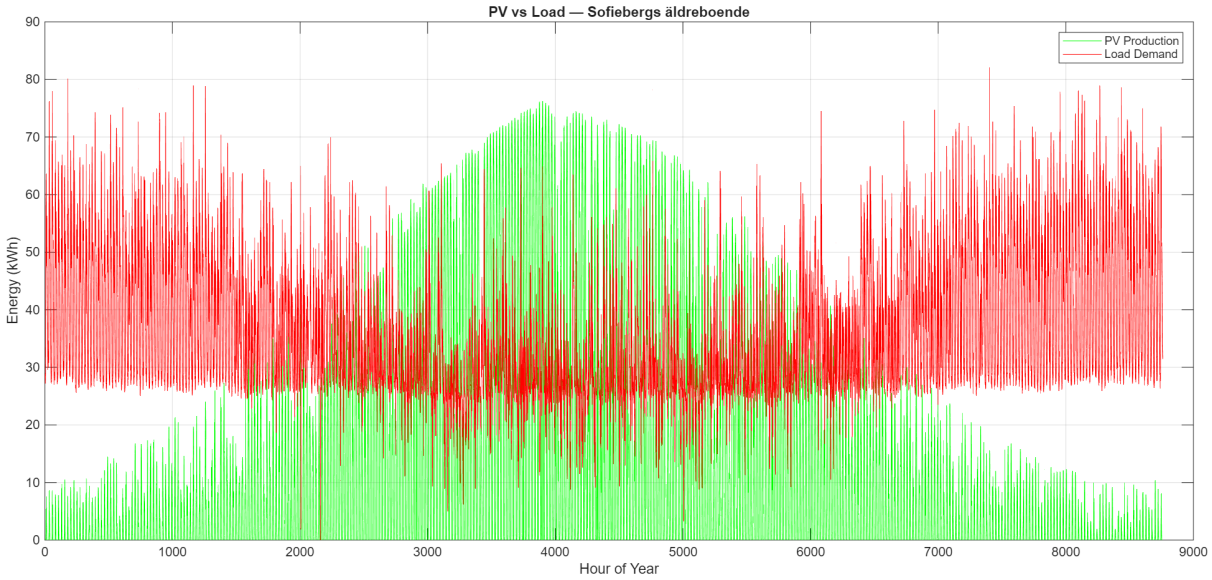


Figure 28. PV generation versus load for the reference battery scenario (12_3) at Sofiebergs äldreboende Source: Author's own simulation results based on measured load data (MATLAB).

PV production remains concentrated around daylight hours, resulting in periods of surplus generation, particularly during summer months, while grid imports persist during nighttime and low-irradiance periods. These patterns reflect remaining temporal mismatches between electricity generation and demand under the reference configuration.

6.3.3 Battery Operation and State-of-Charge Dynamics

Battery behavior in the reference scenario is illustrated by the annual state-of-charge (SOC) profile shown in Figure 29, derived from the scenario simulation results. The SOC pattern demonstrates regular daily cycling, with charging during periods of PV surplus and discharging during evening and nighttime hours.

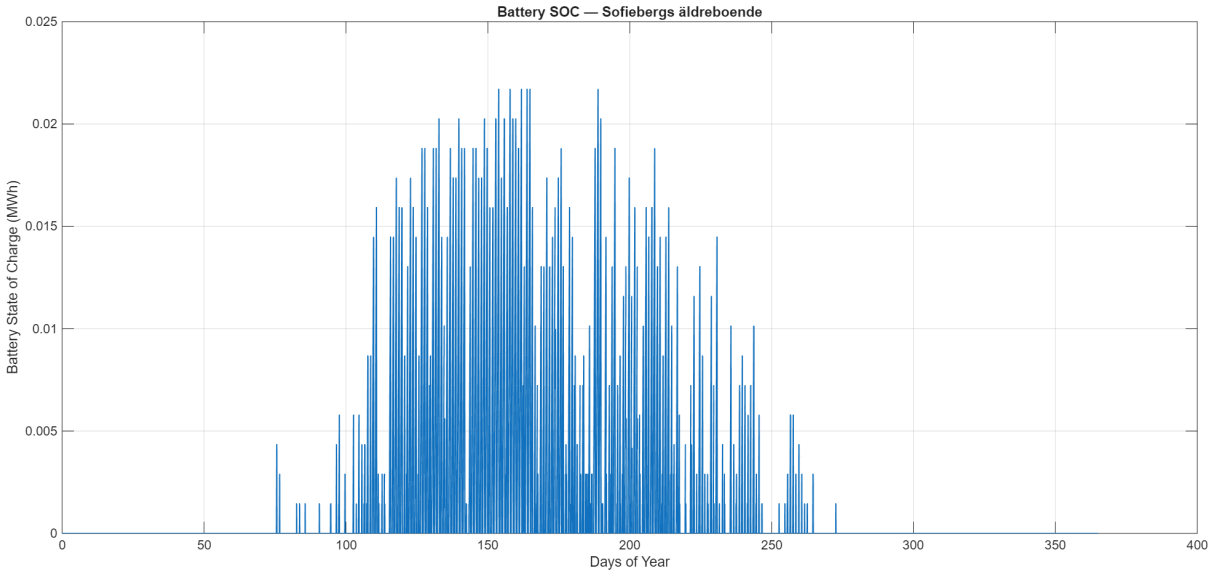


Figure 29. Annual battery state-of-charge (SOC) profile for the reference scenario (12_3). Source: Author's own simulation results based on measured load data (MATLAB).

Relative to educational and office buildings, the more continuous electricity demand enables more frequent battery discharge events across the day. The SOC profile further indicates reduced battery activity during consecutive low-irradiance periods, reflecting limited battery contribution under such conditions within the reference configuration.

6.3.4 Comparative Scenario Results

The quantitative performance of all PV–battery scenarios for Sofiebergs äldreboende is summarized in Table 6.

Table 6. Summary of PV–battery scenarios for Sofiebergs äldreboende. Source: Author's own simulation results based on measured load data (MATLAB).

Scenario (PV factor_Batt factor)	PV capacity (kWp)	Battery (MWh)	Wasted energy (kWh/year)	Wasted (%)	Unmet load (kWh/year)	Unmet (%)
12_3 (Reference)	984.02	2.952	30,053.9	27.61	236,331.5	75.06
12_5	984.02	4.920	28,749.5	26.41	235,488.3	74.79
12_8	984.02	7.872	26,849.7	24.67	234,578.1	74.50
14_3	1,148.03	3.444	42,039.4	33.11	230,206.5	73.12
14_5	1,148.03	5.740	40,290.4	31.73	229,015.8	72.74
16_3	1,312.03	3.936	54,837.5	37.79	224,980.3	71.46
16_8	1,312.03	10.496	49,498.9	34.11	221,739.1	70.43

Scenario (PV factor_Batt factor)	PV capacity (kWp)	Battery (MWh)	Wasted energy (kWh/year)	Wasted (%)	Unmet load (kWh/year)	Unmet (%)

For scenarios with fixed PV capacity (12_x), increasing battery size from the reference case (12_3) to higher storage levels leads to a substantial reduction in unmet load, decreasing from approximately 23 MWh/year to below 14 MWh/year in Scenario 12_8. This confirms that the continuous demand for elderly care facilities allows battery storage to be utilized more effectively than in buildings with intermittent operation.

At higher battery capacities, further reductions in unmet load become less pronounced, while wasted energy increases and exceeds 100 MWh/year in the largest storage configurations. Under these conditions, surplus PV generation remains present despite increased battery capacity.

Increasing PV capacity at low battery levels (e.g., 14_3 and 16_3) results in higher surplus generation without proportional reductions in unmet load, demonstrating that PV oversizing alone does not resolve temporal mismatch. The combined increase of PV and battery capacity (e.g., 16_8) achieves the lowest unmet load but also produces the highest amount of wasted energy, highlighting a clear saturation effect.

The results reveal a pronounced trade-off between reliability improvement and surplus generation, emphasizing the importance of balanced PV–battery sizing rather than aggressive system expansion.

6.4 Trönningeskolan — Urban Case Study

6.4.1 Baseline System Context and Load Characteristics

Trönningeskolan is a municipal school building in Halmstad. The analysis of this case study is based exclusively on measured electricity consumption data and building information provided by based on municipal geospatial data from the City of Halmstad, as documented in the baseline case study section of this thesis.

In contrast to the other case studies, no photovoltaic (PV) system was installed at Trönningeskolan in the baseline configuration. The building, therefore, represents a reference case without on-site renewable generation, making it particularly suitable for evaluating the impact of proposed PV and battery integration.

Due to the urban location and planning constraints, wind energy systems were not considered. All scenarios focus on the introduction of PV systems combined with battery storage.

The objective of this case study is to assess how different PV–battery configurations, introduced to a building without existing solar generation, affect unmet load reduction and surplus (wasted) energy, based on real measured demand.

Figure 27 presents the measured annual electricity load profile of Trönningeskolan, based on municipal operational data provided by based on municipal geospatial data from the City of Halmstad. The load profile exhibits a pronounced time-dependent pattern, with high electricity demand during teaching hours and substantially lower demand during evenings, nights, and weekends.

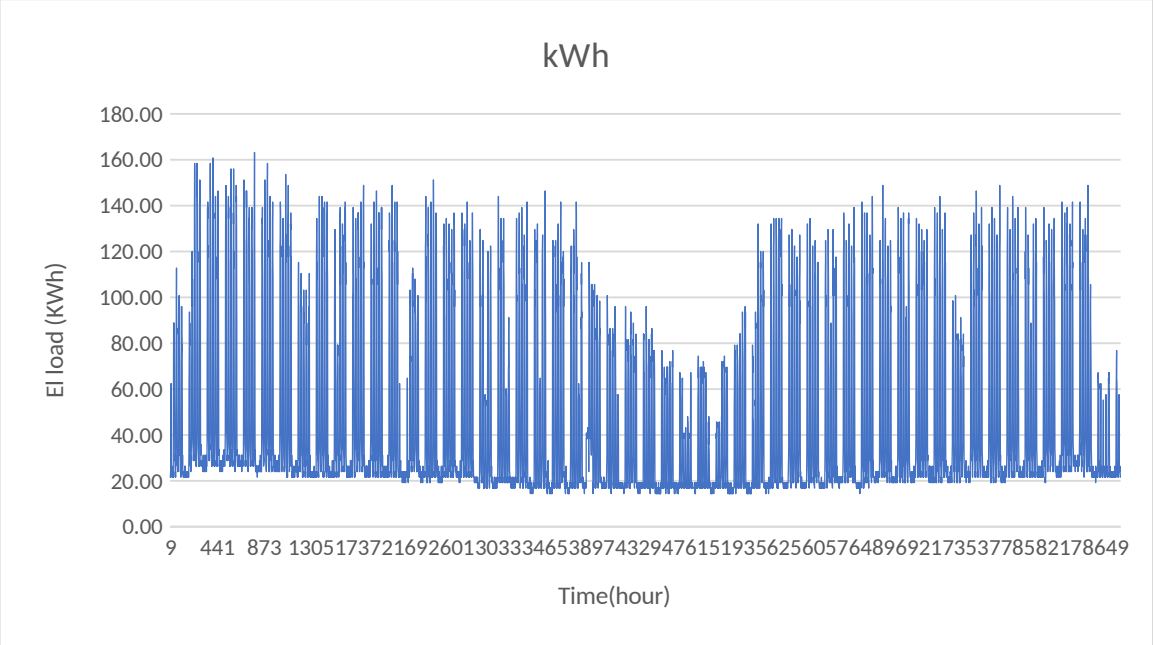


Figure 27. Annual measured electricity load profile of Trönningeskolan (baseline operation) (based on municipal geospatial data from the City of Halmstad).

Seasonal variations in electricity demand are present, reflecting differences in school operation and auxiliary loads over the year. This intermittent load profile is associated with limited temporal alignment between solar electricity production and demand, establishing a challenging baseline for subsequent PV–battery scenario evaluation.

6.4.2 Reference Scenario — PV–Load Interaction (with Battery Storage)

For Trönningeskolan, the reference scenario (12_3) represents the introduction of a first-level PV system combined with the minimum battery capacity defined in the scenario framework, applied to a building without prior solar generation.

Figure 28 compares PV generation with the building load for the reference scenario. PV production is concentrated around midday hours, while electricity demand peaks occur primarily during school operating hours. Although some degree of overlap exists, a significant PV surplus is generated during periods of low demand, particularly outside teaching hours.



Figure 28. PV generation versus load for the reference scenario (12_3) at Trönningeskolan. Source: Author's own simulation results based on measured load data (MATLAB).

Although partial temporal overlap between PV generation and electricity demand is observed, surplus PV generation occurs during periods of lower demand, particularly outside teaching hours. These patterns reflect remaining temporal mismatches between electricity production and consumption under the reference configuration.

6.4.3 Battery Operation and State-of-Charge Dynamics

Battery behavior in the reference scenario is illustrated in Figure 29 through the annual state-of-charge (SOC) profile. The SOC pattern indicates intermittent charging and discharging cycles, driven by the limited temporal overlap between PV generation and building demand.

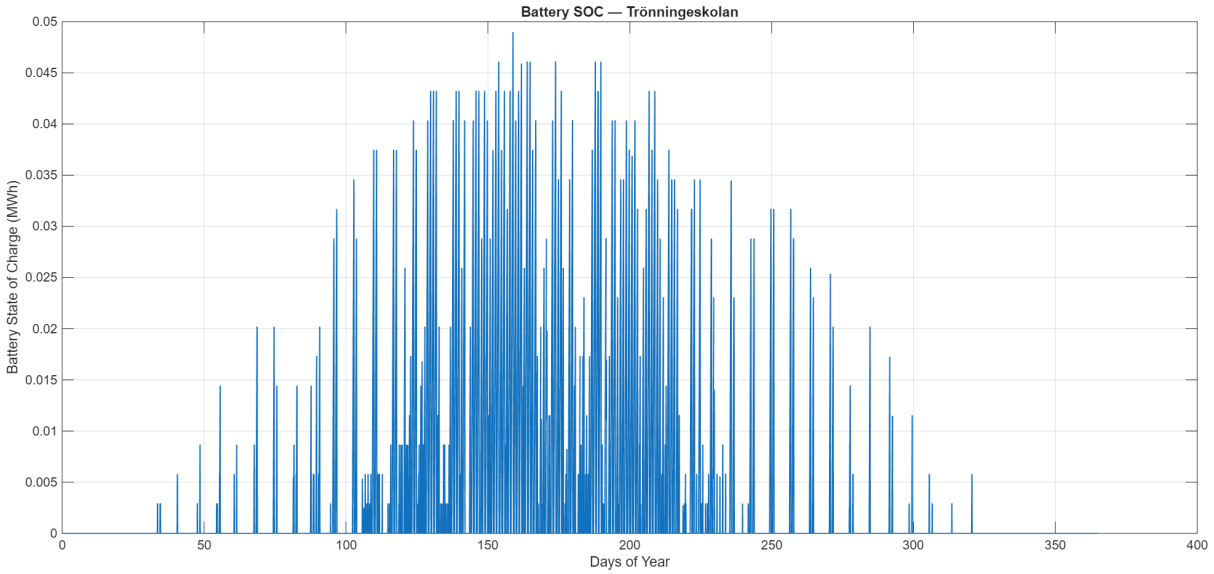


Figure 29. Annual battery state-of-charge (SOC) profile for the reference scenario (12_3) Source: Author's own simulation results based on measured load data (MATLAB).

Relative to Sofiebergs äldreboende, the battery at Trönningeskolan shows longer idle periods, particularly during nights and weekends, as observed in the SOC profile.

6.4.4 Comparative Scenario Results

The performance of all PV–battery scenarios for Trönningeskolan is summarized in Table 7. The results indicate that, at a fixed PV capacity (12_x scenarios), increasing battery size leads to only marginal reductions in unmet load, while wasted energy decreases slightly as storage capacity increases.

Table 7. Summary of PV–battery scenarios for Trönningeskolan Source: Author's own simulation results based on measured load data (MATLAB).

Scenario (PV factor_Batt factor)	PV capacity (kWp)	Battery (MWh)	Wasted energy (kWh/year)	Wasted (%)	Unmet load (kWh/year)	Unmet (%)
12_3 (Reference)	1,958.40	5.875	74,819.5	34.54	253,268.7	64.23
12_5	1,958.40	9.792	71,753.4	33.12	252,444.5	64.02
12_8	1,958.40	15.667	67,312.4	31.07	251,910.8	63.88
14_3	2,284.80	6.854	98,716.6	39.06	241,621.0	61.27
14_5	2,284.80	11.424	94,778.2	37.50	240,811.8	61.07
16_3	2,611.20	7.834	124,217.2	43.01	231,774.4	58.78
16_8	2,611.20	20.890	112,304.0	38.88	230,300.5	58.40

Scenario (PV factor_Batt factor)	*PV capacity (kWp)*	*Battery (MWh)*	*Wasted energy (kWh/year)*	*Wasted (%)*	*Unmet load (kWh/year)*	*Unmet (%)*

At higher PV capacities (14_x and 16_x scenarios), unmet load decreases more noticeably; however, this improvement is accompanied by a substantial increase in wasted energy, particularly at lower battery capacities. Scenarios combining higher PV and battery capacities (e.g. 16–8) achieve the lowest unmet load values while still exhibiting high levels of surplus generation.

6.5 City-Level Results: Urban Renewable Energy Scenarios

6.5.1 Typical Urban Load Profile (Common Assumption)

A single typical industrial load profile was applied consistently across all urban scenarios. This profile was constructed by scaling and extending a representative weekly industrial demand pattern reported in the literature, combined with monthly demand normalization, to enable full-year simulation under controlled conditions (Dehning et al., 2019).

The load profile used in this study is based on factory-level measurements from a medium-sized manufacturing facility in the automotive sector operating under a structured, shift-based production schedule. The original dataset is reported at a 15-minute temporal resolution and differentiates between production and non-production periods. As such, it captures typical characteristics of industrial electricity demand, including regular weekday operation, reduced activity during weekends, and lower electricity consumption during overnight hours (Dehning et al., 2019).

The original load profile reported by Dehning et al. represents a typical weekly industrial demand pattern at factory level. In the present study, this weekly profile was extended to an annual hourly load time series by systematic replication and scaling based on the Swedish calendar. The annualization process accounts for working days, weekends, and seasonal demand normalization, enabling full-year simulation while preserving the characteristic production and non-production patterns of the original profile.

This approach allows the representative weekly industrial behavior observed in the literature to be consistently applied in long-term, hourly energy system simulations, while maintaining transparency and comparability across all urban scenarios.

Previous research shows that industrial electricity demand characteristics depend strongly on factors such as the industrial sector, the organization of operations, and the regional context. As a result, electricity load profiles cannot be directly transferred between different industrial

sectors or geographic locations, and industrial demand patterns are not expected to be identical across regions within Sweden.

In the present study, a load profile from the automotive sector was deliberately applied consistently across all urban scenarios. This approach was chosen to limit model complexity and to establish a coherent basis for comparison between regions. While this simplification supports clearer interpretation of renewable energy system behavior, it is acknowledged that real-world industrial electricity demand may vary across locations and sectors.

By keeping the load profile identical across all cities, observed differences in system performance can be attributed primarily to regional climate conditions and renewable resource availability rather than variations in demand structure.

The constructed load profile exhibits:

• a persistent base demand represented throughout the year,

• limited seasonal variability,

• pronounced short-term fluctuations relevant for storage operation.

This profile is introduced once in Figure 30 (Dehning et al., 2019) and serves as a common reference for all subsequent city-level analyses.

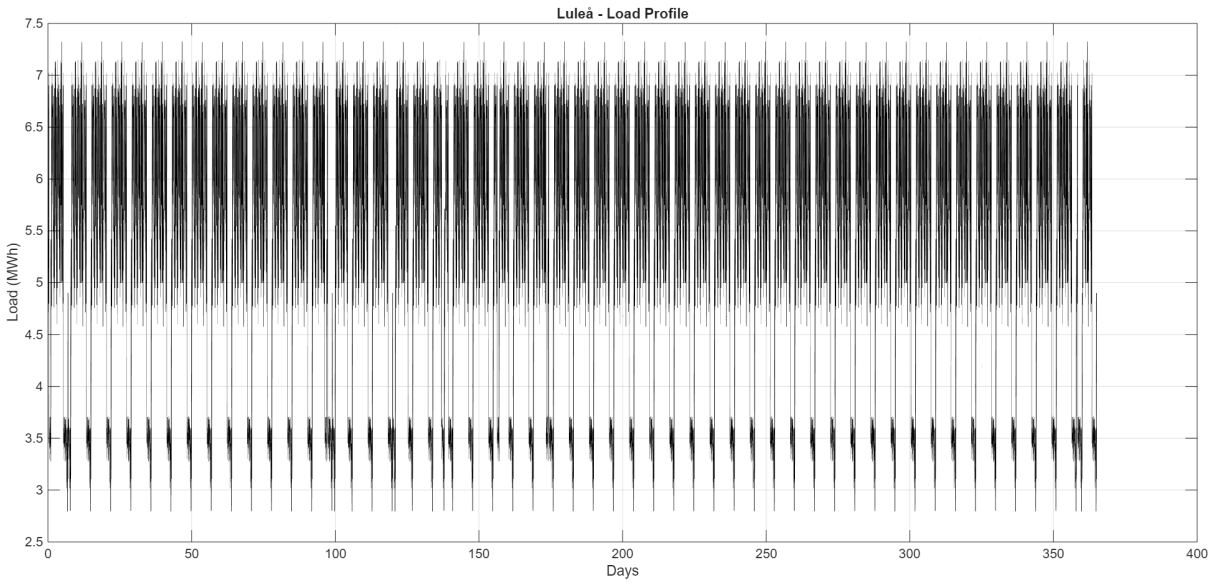


Figure 30. Typical urban/industrial load profile used for all cities. Source: Author's own construction based on a representative weekly industrial load profile reported in Dehning et al. (2023).

6.6 Reference City I — Luleå (SE1)

6.6.1 Climatic and Energy Context of Luleå

Luleå is located in the northernmost electricity zone of Sweden (SE1) (Swedish Energy Markets Inspectorate, 2024) and represents a high-latitude urban energy system characterized

by long and cold winters, limited solar availability during the heating season, and comparatively strong and stable wind resources. From a renewable system design perspective, this city provides a technically favorable environment for wind-dominated configurations, while simultaneously exposing the structural limitations of solar-based solutions under extreme seasonal conditions.

The normalized photovoltaic output for Luleå is illustrated in Figure 31 (PVGIS, 2023), showing a pronounced seasonal asymmetry in solar generation. PV production is highly concentrated in late spring and summer months, whereas winter output remains extremely low despite non-zero irradiation. This behavior confirms that PV scaling alone cannot resolve winter reliability constraints in high-latitude regions.

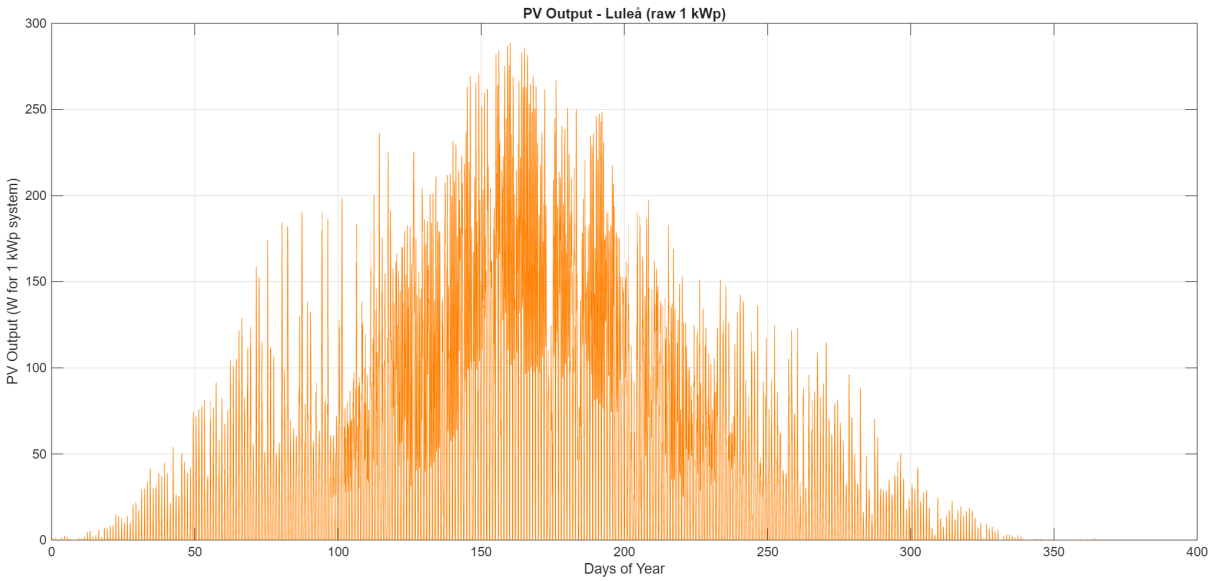


Figure 31. Hourly photovoltaic (PV) power output profile for Luleå, normalized to 1 kWp installed capacity over one year. Source: PVGIS, European Commission, Joint Research Centre (2023).

In contrast, wind energy exhibits a markedly different temporal profile. As shown in Figure 32 (PVGIS, 2023), wind power generation at hub height remains significant throughout the year, including during winter months when electricity demand is highest, and PV output is minimal. This complementary behavior establishes wind as the primary reliability backbone of renewable systems in Luleå.

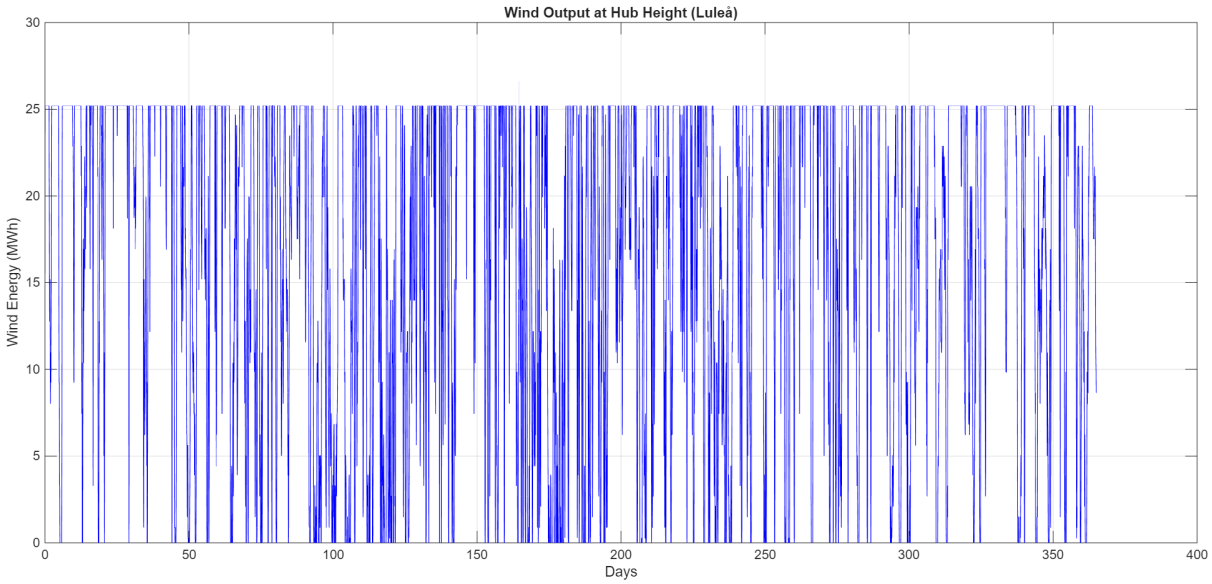


Figure 32. Hourly wind energy output at turbine hub height for Luleå over one year. Source: PVGIS, European Commission, Joint Research Centre (2023).

Together, these figures provide the physical context necessary to interpret the scenario results presented in the following sections.

6.6.2 Six-Turbine Configuration — Representative Scenario Analysis

The six-turbine configuration is analyzed first, as it represents the structurally most robust system design evaluated for Luleå. The six-turbine configuration is analyzed first, as it represents the structurally most robust system design evaluated for Luleå. Within this configuration group, scenarios are defined using a combined notation that reflects both photovoltaic and battery scaling. Specifically, the notation *PV factor, Battery factor* is adopted, where the first value denotes the PV scaling factor and the second value denotes the battery scaling factor. For example, Scenario 35–6 represents a configuration with a PV factor of 35 and a battery factor of 6. This notation is used consistently throughout the city-level scenario analysis.

Among the four six-turbine scenarios, Scenario 35_6 is selected as the representative reference, as it reflects the dominant system behavior without being excessively oversized or under-dimensioned.

Although this scenario is not the numerical midpoint of the tested range, it represents the first configuration where the six-turbine system exhibits its dominant operational behavior. Subsequent scenarios primarily amplify the same dynamics through further PV and battery scaling rather than introducing qualitatively different system responses.

In subsequent six-turbine scenarios, further increases in PV and battery capacity primarily amplify these same dynamics rather than introducing qualitatively different system responses. As such, Scenario 35–6 provides a suitable reference for interpreting the broader behavior of wind-supported configurations in Luleå.

The interaction between renewable generation and demand for this configuration is illustrated in Figure 33, where total annual renewable production significantly exceeds the load demand, while periods of unmet load persist due to temporal mismatch. This result demonstrates that annual energy sufficiency does not imply temporal adequacy across all time scales.

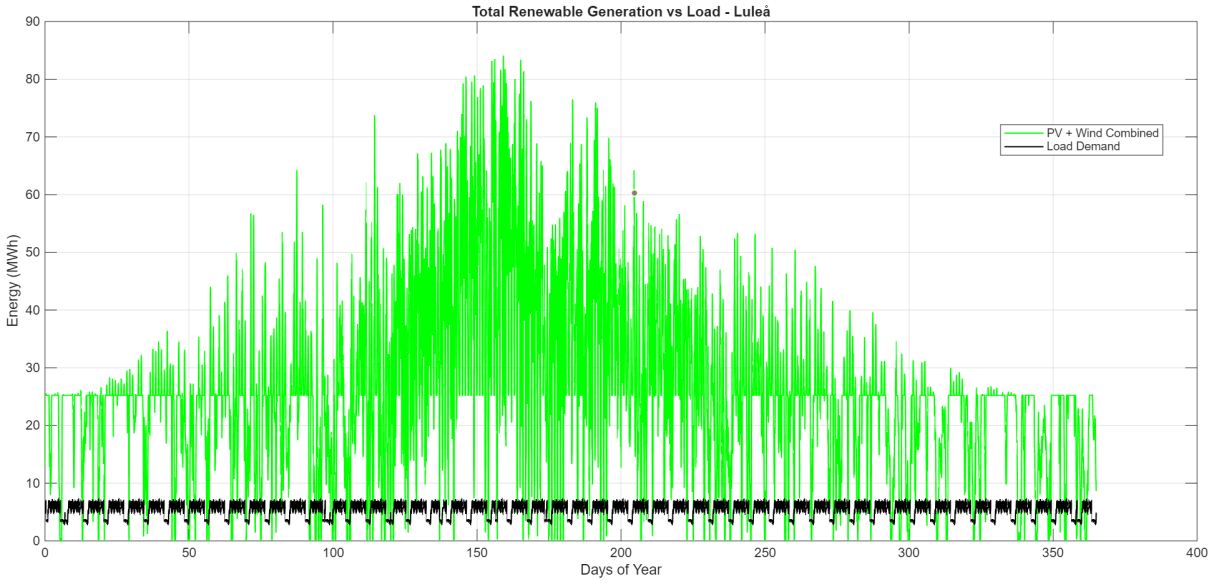


Figure 33. Comparison of combined renewable electricity generation (PV + wind) and hourly load demand in Luleå over a full year, highlighting periods of surplus generation and supply–demand mismatch. Source: Author's own simulation results based on measured load data (MATLAB).

Battery operation over the year is shown in Figure 34, revealing frequent saturation during summer months and prolonged low state-of-charge levels during winter. This pattern indicates that additional storage capacity provides limited reliability benefits once renewable availability during critical winter periods becomes the dominant constraint.

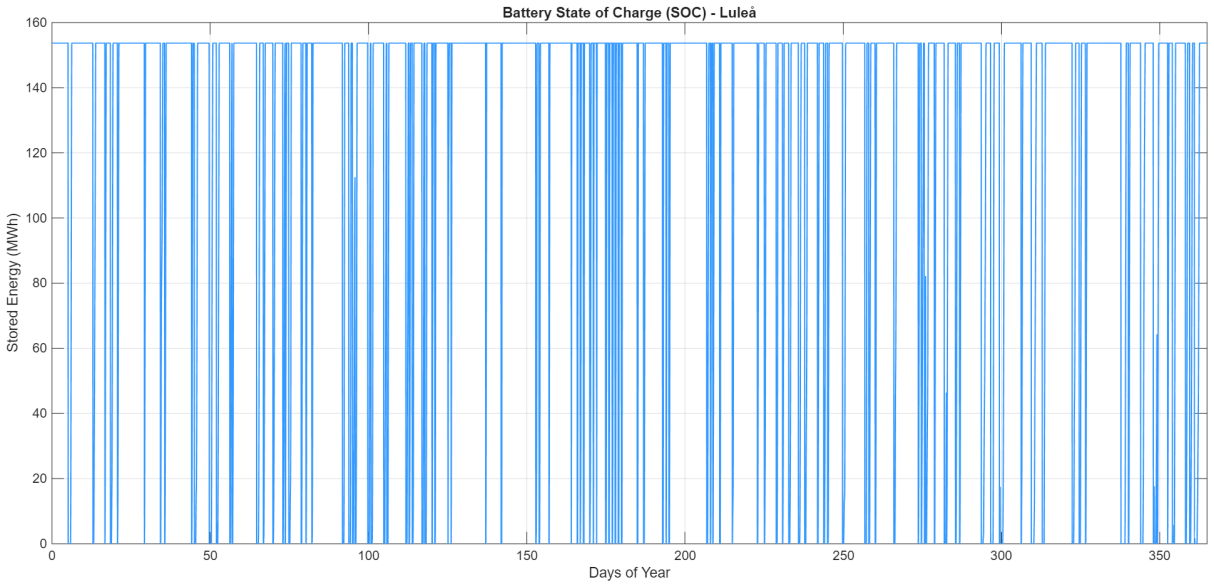


Figure 34. Hourly battery state of charge (SOC) for the Luleå case, illustrating battery charging and discharging behavior in response to renewable generation variability and load demand. Source: Author's own simulation results based on measured load data (MATLAB).

The numerical results for all six-turbine scenarios are summarized in Table 7. Across Scenarios 35_6 to 50_7.5, unmet load remains consistently low, approximately in the range of 5–6%, while renewable coverage reaches several hundred percent. Increasing PV and battery capacity leads primarily to higher wasted energy rather than meaningful reductions in shortage.

Table 7. Summary of six-turbine renewable energy scenarios for Luleå. Source: Author's own simulation results based on measured load data (MATLAB).

Scenario	PV Factor	Battery Factor	PV Generation (MWh/year)	Wind Generation (MWh/year)	Battery Capacity (MWh)	Unmet Load (MWh/year)	Unmet Load (%)	Wasted Energy (MWh/year)	Renewable Coverage (%)
S6-1	35	6.0	71,608	156,780	153.7	2,430.8	5.16	166,855	484.7
S6-2	40	6.5	81,838	156,780	190.3	2,405.9	5.11	173,071	506.4
S6-3	45	7.0	92,068	156,780	230.6	2,392.9	5.08	179,141	528.1
S6-4	50	7.5	102,298	156,780	274.5	2,385.0	5.06	185,010	549.8

These results indicate that the six-turbine configuration in Luleå operates close to the lowest unmet load levels observed among the evaluated scenarios under the applied modeling assumptions.

6.6.3 Three-Turbine Configuration — Representative Scenario Analysis

The three-turbine scenarios represent a deliberate reduction in wind capacity, compensated by substantial increases in PV and battery sizing. This configuration tests whether storage and solar over-dimensioning can replace missing wind generation in a high-latitude context.

As in the six-turbine configuration, scenarios are denoted using the PV_factor–Battery_factor notation. Accordingly, Scenario 85–9.5 represents a configuration with a PV factor of 85 and a battery factor of 9.5. For this group, Scenario 85_9.5 is selected as the representative reference, as it captures the characteristic behavior of PV-dominated systems without reflecting extreme saturation effects.

The annual energy balance for this configuration is presented in Figure 35, demonstrating that although total renewable generation remains significantly higher than demand on an annual basis, unmet load increases markedly compared to the six-turbine configuration. This reflects the reduced contribution of wind generation during winter periods.

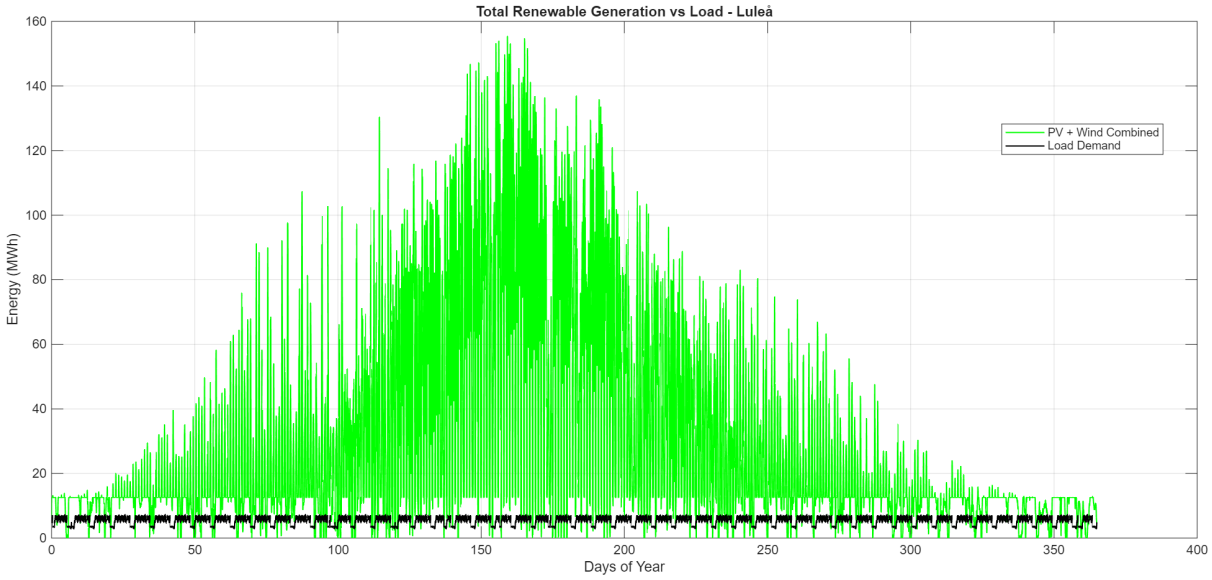


Figure 35. Total combined renewable electricity generation (PV + wind) versus hourly load demand for Luleå over the full annual simulation period, illustrating seasonal generation variability and persistent mismatch between production and demand. Source: Author's own simulation results based on measured load data (MATLAB).

The corresponding battery state-of-charge profile is shown in Figure 36, confirming that even substantially enlarged storage systems fail to prevent prolonged winter shortages when both PV output and wind generation are insufficient. These periods coincide with high demand and low renewable availability, showing that even substantially increased storage capacity does not eliminate seasonal shortages when primary generation is insufficient.

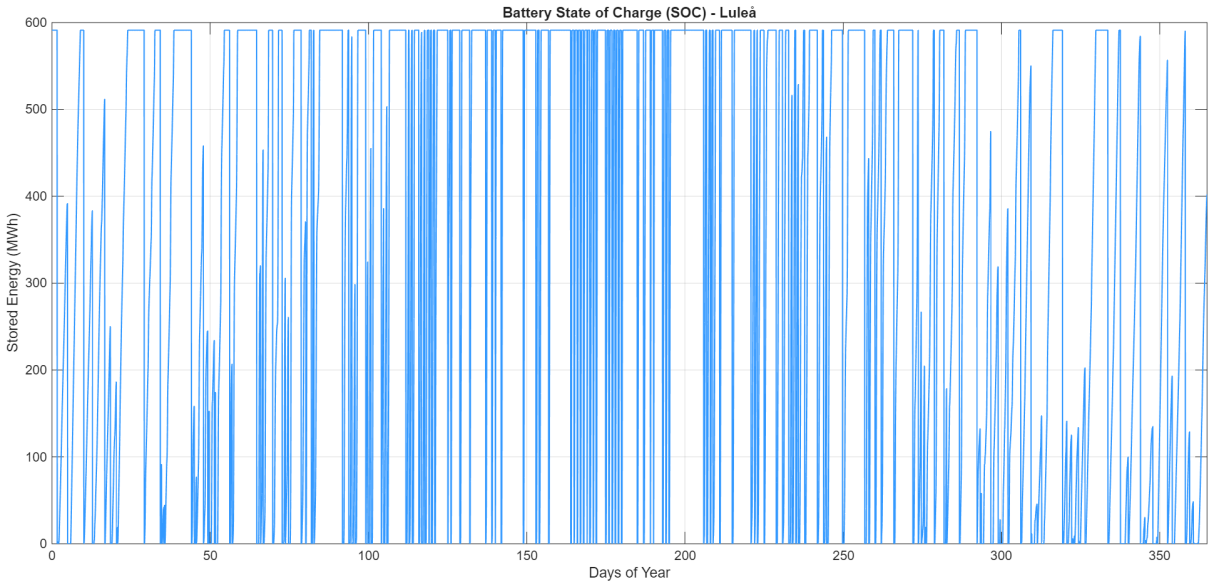


Figure 36. Battery state of charge (SOC) for the Luleå case study over the annual simulation period, showing charging–discharging dynamics driven by renewable generation surplus and load demand balance. Source: Author's own simulation results based on measured load data (MATLAB).

Table 8 summarizes the numerical results for all three-turbine scenarios. Unmet load increases to values exceeding 10%, while wasted energy continues to rise with increasing PV and battery capacity.

Table 8. Summary of three-turbine renewable energy scenarios for Luleå. Source: Author's own simulation results based on measured load data (MATLAB).

Scenario	PV Factor	Battery Factor	PV Generation (MWh/year)	Wind Generation (MWh/year)	Battery Capacity (MWh)	Unmet Load (MWh/year)	Unmet Load (%)	Wasted Energy (MWh/year)	Renewable Coverage (%)
S3-1	80	9.0	163,676	78,390	527.0	3,173.7	6.74	134,374	513.7
S3-2	85	9.5	173,906	78,390	591.1	3,168.7	6.72	137,975	535.4
S3-3	90	10.0	184,136	78,390	658.8	3,155.2	6.70	141,529	557.2
S3-4	95	10.5	194,365	78,390	730.2	3,152.7	6.69	144,985	578.9

These trends indicate diminishing returns, where additional system investment primarily increases surplus generation without substantially reducing unmet load.

6.7 Halmstad — City-scale Scenario Analysis

6.7.1 Climatic and renewable context of Halmstad

Halmstad is located in southern Sweden, within electricity price area SE4, a region characterized by comparatively high solar irradiation within the Swedish context, while wind conditions remain moderate and less dominant than in northern regions (Swedish Energy Markets Inspectorate, 2024). This climatic setting results in system configurations where photovoltaic generation contributes a substantial share of electricity production, while wind power primarily supports system balance by complementing periods of lower solar availability.

The seasonal behavior of solar irradiation is illustrated in Figure 37 (PVGIS, 2023), which presents the hourly PV output profile normalized to 1 kWp. The figure shows a pronounced summer peak and a near-zero production level during winter months, confirming that PV generation alone cannot address seasonal supply deficits.

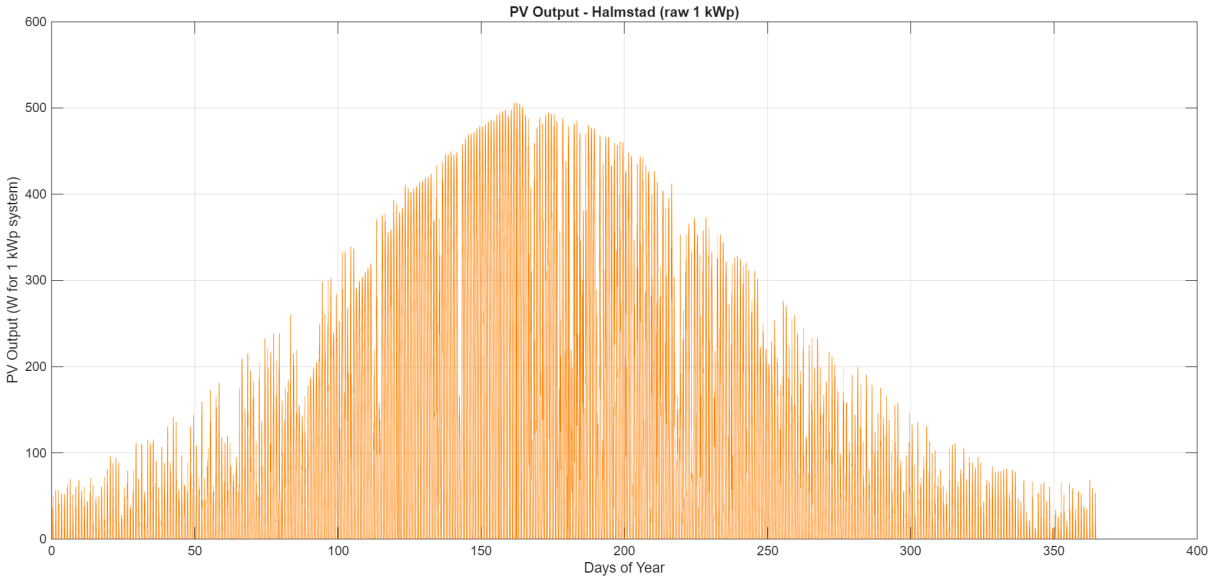


Figure 37. Hourly photovoltaic (PV) electricity generation per 1 kWp installed capacity in Halmstad over one year, illustrating seasonal and daily variability in solar resource availability. Source: PVGIS, European Commission, Joint Research Centre (2023).

Wind conditions at hub height are shown in Figure 38 (PVGIS, 2023), representing the corresponding wind power output profile. In contrast to PV, wind availability remains more evenly distributed throughout the year, although at a lower average level compared to northern cities such as Luleå. This indicates that wind contributes primarily to reliability smoothing rather than to large-scale energy surplus.

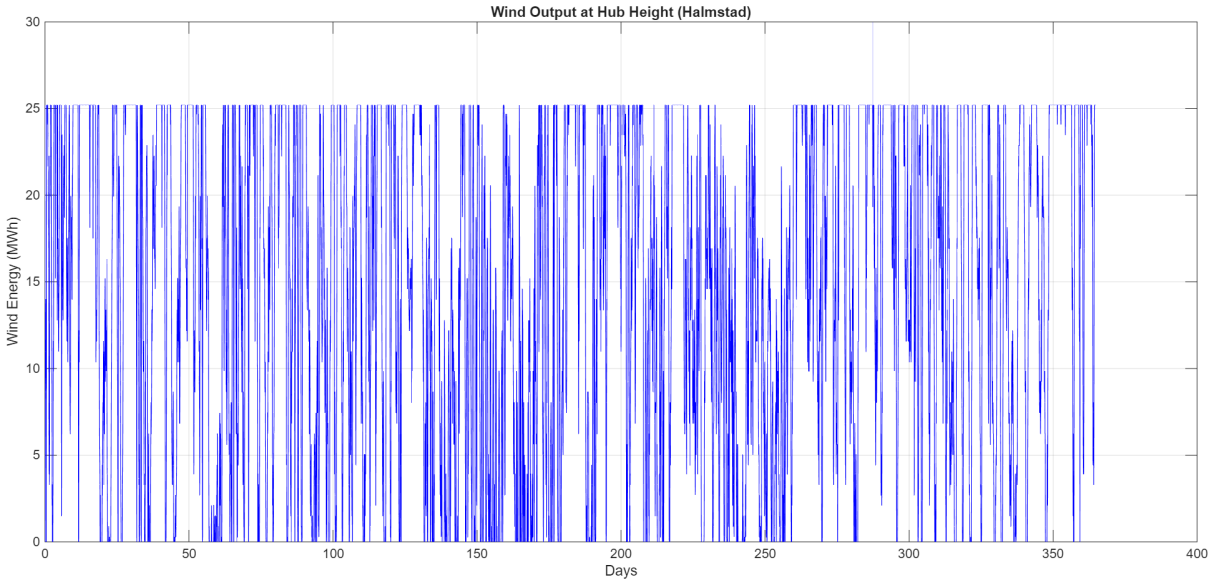


Figure 38. Hourly wind power output at turbine hub height in Halmstad over one year, showing temporal variability in wind energy generation potential. Source: PVGIS, European Commission, Joint Research Centre (2023).

The combined interpretation of Figures 37 and 38 demonstrates a fundamental structural characteristic of Halmstad: while solar resources are strong but highly seasonal, wind resources are comparatively stable but insufficient to fully compensate for winter PV scarcity. This resource interaction forms the physical basis for the system behavior observed in the subsequent scenario analysis and explains why PV oversizing alone does not eliminate unmet load during winter periods.

6.7.2 Load characteristics and system demand

All Halmstad scenarios are evaluated against the same industrial load profile, characterized by a peak demand of approximately 7 MW and an average load of around 5 MW. The hourly load profile used in the simulations is shown in Figure 30, previously introduced as the common reference load for all cities.

The relatively flat demand profile, combined with limited seasonal variation, implies that system performance in Halmstad is primarily constrained by renewable resource availability and storage limitations, rather than by load volatility.

The combined contribution of photovoltaic and wind generation relative to the load demand is illustrated in Figure 39, which presents the total renewable output over the year. The figure highlights substantial periods of energy surplus alongside persistent temporal mismatches, confirming that high annual renewable production does not directly translate into full load coverage.

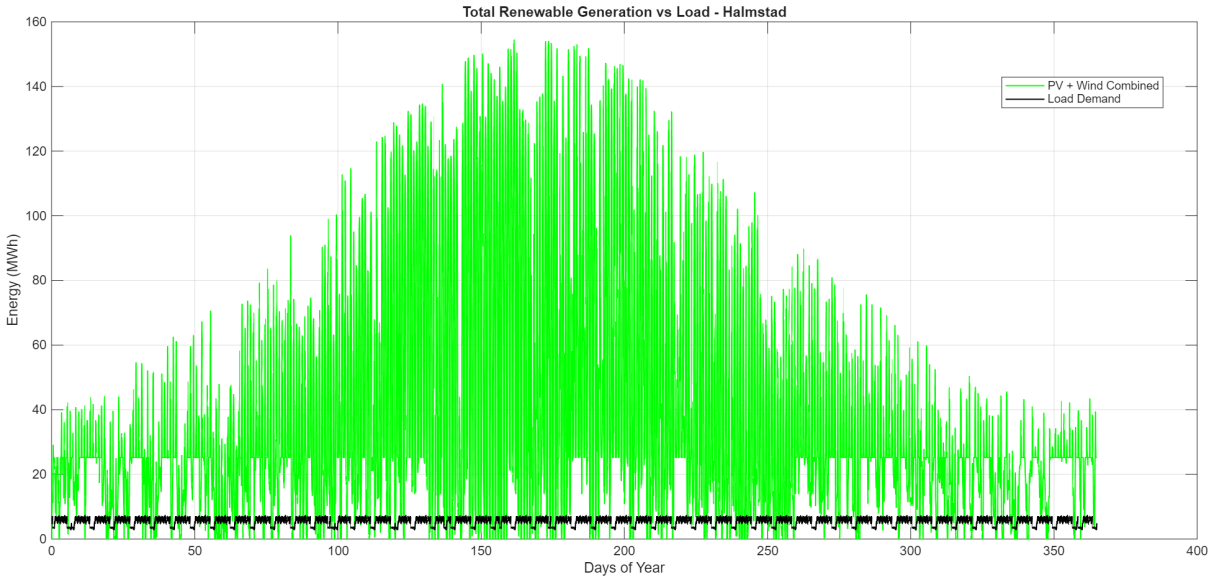


Figure 39. Hourly combined renewable electricity generation (PV + wind) compared with hourly electricity load demand for Halmstad over one full year. Source: Author's own simulation results based on measured load data (MATLAB).

Battery state-of-charge dynamics further highlight this limitation (Figure 40). The battery reaches full charge frequently during high-production periods, indicating saturation, while still failing to cover prolonged winter deficits.

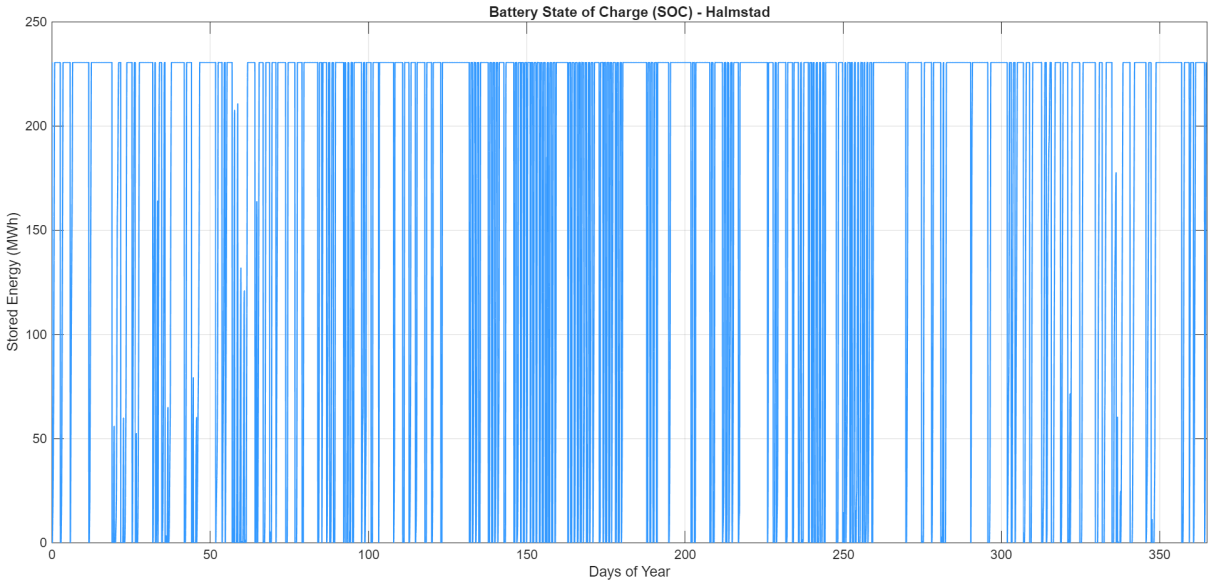


Figure 40. Battery state of charge (SOC) variation over one full year for the Halmstad case, illustrating charging and discharging behavior under the selected PV–wind–battery scenario. Source: Author's own simulation results based on measured load data (MATLAB).

Quantitatively, unmet load remains between 8.3% and 8.6% across all six-turbine scenarios (Table 9), demonstrating that increasing PV capacity and battery size yields only marginal reliability improvements once a structural threshold is reached.

Table 9. Summary of six-turbine scenarios for Halmstad. Source: Author's own simulation results based on measured load data (MATLAB).

Scenario	PV_factor	Battery_factor	Battery (MWh)	Unmet Load (MWh)	Shortage (%)	Wasted Energy (MWh)	Renewable Coverage (%)
S6-1	35	6.0	153.72	4030.7	8.55	214,799	599.8
S6-2	40	6.5	190.32	3996.4	8.48	230,040	644.8
S6-3	45	7.0	230.58	3951.9	8.39	244,738	689.7
S6-4	50	7.5	274.50	3931.8	8.34	259,162	734.6

6.7.3 Three-turbine scenarios (reduced wind support)

As in the three-turbine configuration, scenarios are defined using the PV factor–Battery factor notation. For this group, Scenario 80-9 is selected as the representative reference case, as it captures the characteristic behavior of PV-dominated systems under reduced wind support without reflecting extreme saturation effects.

The total renewable output relative to load for the reference three-turbine configuration is illustrated in Figure 41 (Total renewable generation vs load – Halmstad. Despite extremely high renewable coverage, values approaching 1000% in the most oversized cases, significant unmet load persists.

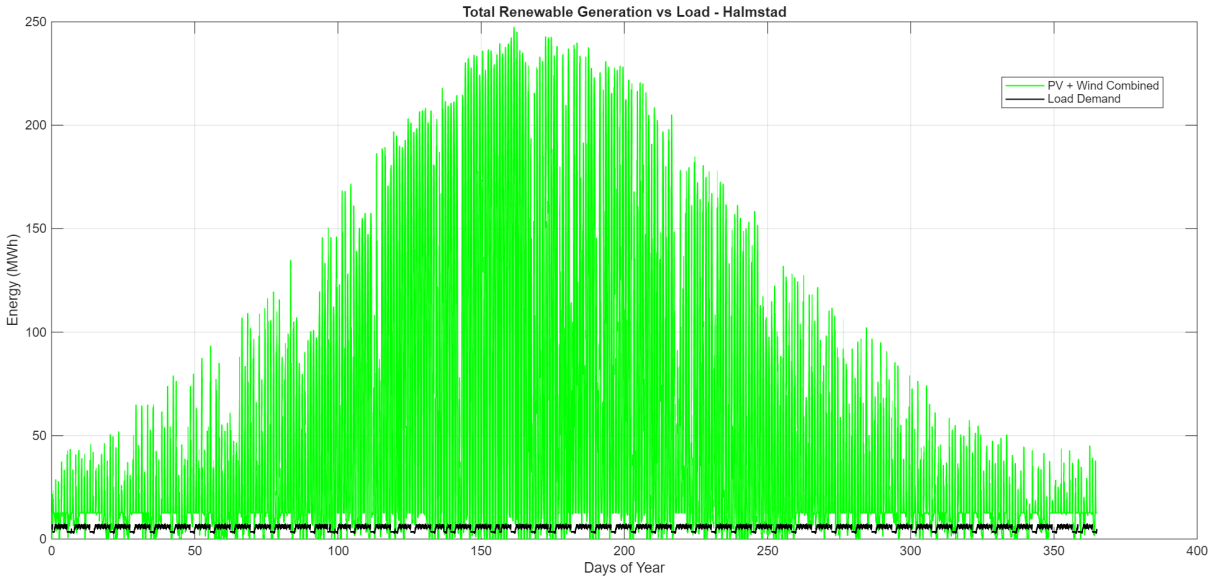


Figure 41. Hourly comparison of total renewable electricity generation (PV + wind) and load demand in Halmstad over one year, illustrating seasonal variation and periods of surplus and deficit. Source: Author's own simulation results based on measured load data (MATLAB).

Battery behavior in these scenarios (Figure 42) shows long periods of underutilization during winter, followed by extended saturation during summer, highlighting the mismatch between storage capacity and seasonal energy availability.

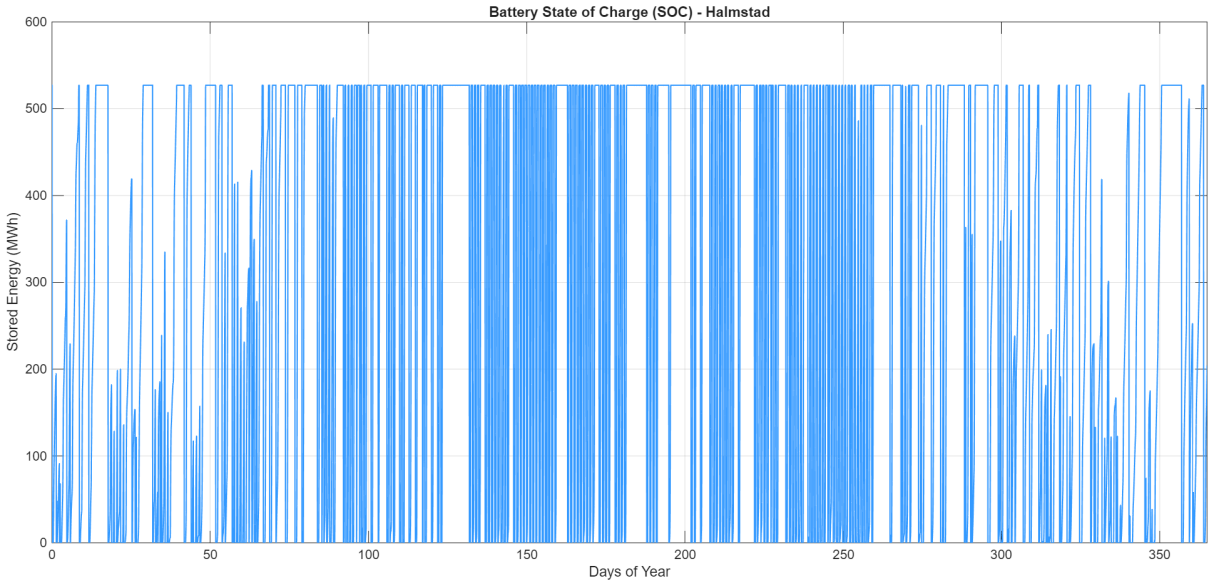


Figure 42. Hourly battery state of charge (SOC) in Halmstad, showing charging–discharging behavior in response to renewable generation variability and load demand over the annual simulation period. Source: Author's own simulation results based on measured load data (MATLAB).

As shown in Table 10, unmet load remains consistently high, between 10.7% and 10.9%, across all three-turbine scenarios. This demonstrates that battery scaling cannot compensate for insufficient wind capacity, regardless of PV oversizing.

Table 10. Summary of three-turbine scenarios for Halmstad. Source: Author's own simulation results based on measured load data (MATLAB).

Scenario	PV_factor	Battery_factor	Battery (MWh)	Unmet Load (MWh)	Shortage (%)	Wasted Energy (MWh)	Renewable Coverage (%)
S3-1	80	9.0	527.04	5110.6	10.85	262,741	861.5
S3-2	85	9.5	591.09	5104.9	10.83	272,717	906.4
S3-3	90	10.0	658.80	5079.8	10.78	282,258	951.3
S3-4	95	10.5	730.17	5072.4	10.76	291,438	996.2

6.8 Pre-analysis of climatic conditions — Malmö (SE4)

Malmö is located in southern Sweden and belongs to the electricity price area SE4, representing one of the most solar-favorable urban contexts considered in this study (Swedish Energy Markets Inspectorate, 2024). Compared to northern cities, Malmö benefits from higher annual solar irradiation, while wind conditions are moderate and less persistent.

The solar resource profile for Malmö is shown in Figure 43 (PVGIS, 2023), which illustrates the hourly photovoltaic output normalized to 1 kWp.

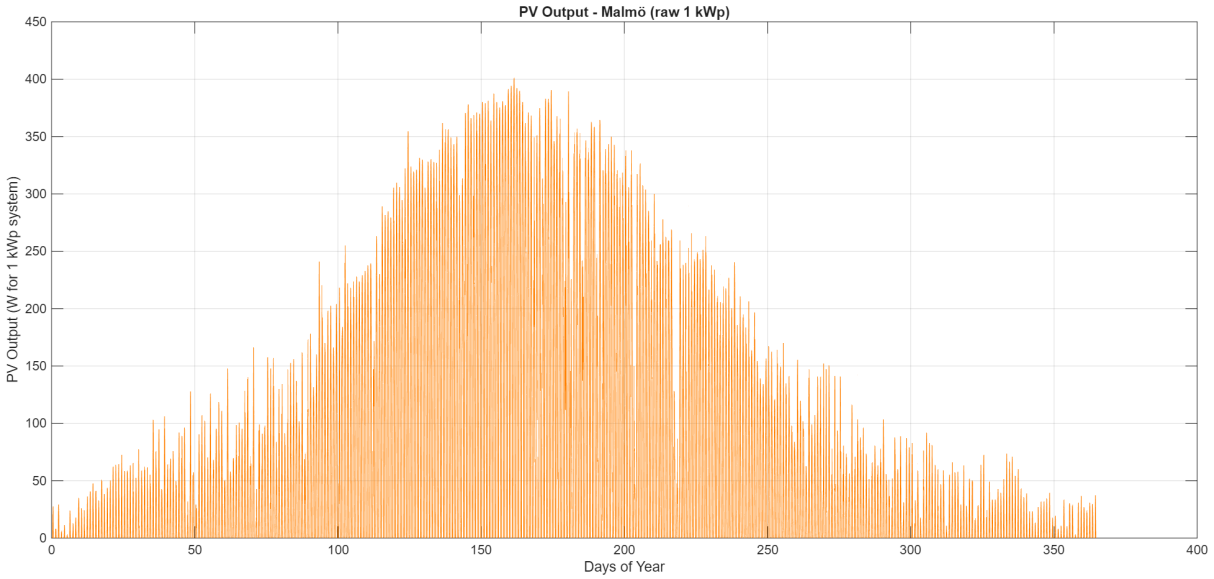


Figure 43. Total renewable electricity generation (combined PV and wind) compared with hourly load demand in Malmö over one year. Source: PVGIS, European Commission, Joint Research Centre (2023).

The figure illustrates a strong seasonal contrast, with high production during summer months and sharply reduced output during winter. This pronounced seasonality suggests that photovoltaic generation alone cannot ensure a reliable supply throughout the year.

Wind resource availability at hub height is presented in Figure 44 (PVGIS, 2023). Compared to northern coastal cities, wind production in Malmö remains relatively moderate and exhibits limited capacity to compensate for winter PV scarcity. Although wind output is more evenly distributed over the year than solar generation, its magnitude is insufficient to act as a dominant reliability driver.

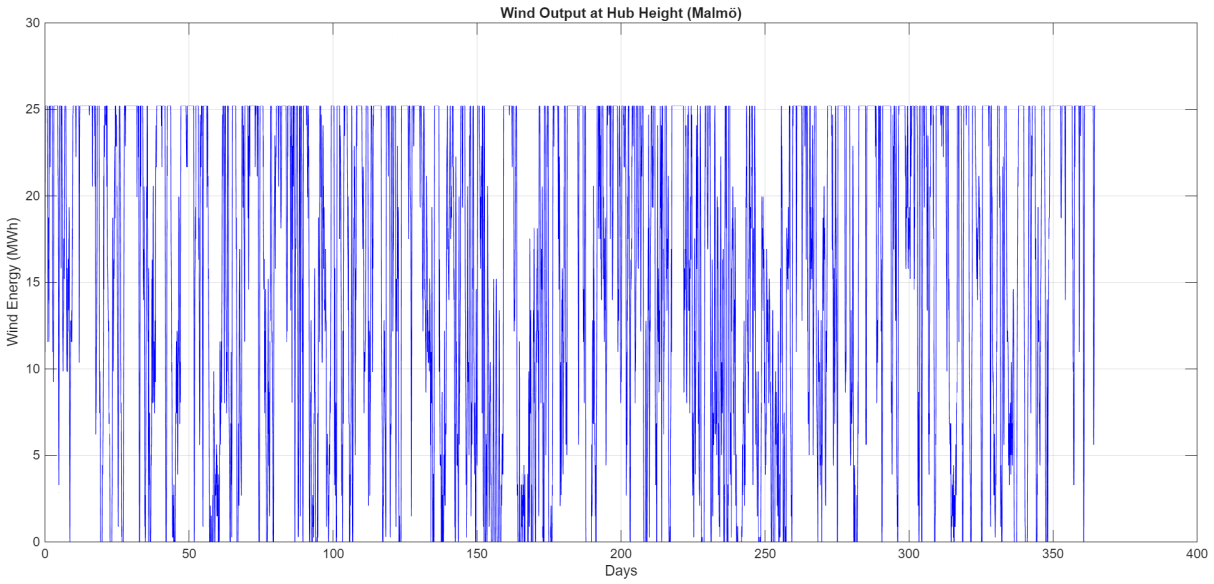


Figure 44. Hourly wind power generation at turbine hub height in Stockholm over a full year, based on regional wind speed data. Source: PVGIS, European Commission, Joint Research Centre (2023).

The combined interpretation of Figures 43 and 44 highlights a structural characteristic of Malmö: while the city is well suited for PV-driven systems in terms of annual energy production, system reliability remains constrained by seasonal mismatches and limited wind support. As a result, scenarios relying primarily on PV and battery scaling are expected to exhibit increasing energy surplus without proportional reductions in unmet load.

This climatic context explains the system behavior observed in the Malmö city-scale scenarios and provides a clear basis for comparison with both Halmstad and northern reference cities in the subsequent analysis.

As shown in Table 11, differences between six-turbine and three-turbine configurations are observed in terms of unmet load and renewable coverage. Six-turbine systems achieve lower unmet load (6.2–6.4%) with moderate PV and battery scaling, while exhibiting high levels of wasted energy. In contrast, three-turbine configurations require substantially larger PV and battery capacities yet maintain higher unmet load levels (9.25–9.35%) alongside increased renewable coverage.

Table 11. Summary of renewable system configurations and performance ranges for wind–PV–battery scenarios, including unmet load, wasted energy, and renewable coverage for Malmö. Source: Author's own simulation results based on measured load data (MATLAB).

Configuration	PV factor range	Battery factor range	Unmet load (%)	Wasted energy (%)	Renewable coverage (%)
6 Wind Turbines	35 → 50	6 → 7.5	6.2 → 6.4	73 → 75	545 → 646
3 Wind Turbines	80 → 95	9 → 10.5	9.25 → 9.35	58 → 60	695 → 797

6.9 Pre-analysis of climatic conditions _ Stockholm (SE3)

Stockholm is located in central Sweden and belongs to the electricity price area SE3, representing a transitional climatic context between the solar-favored southern regions and the wind-dominated northern regions (Swedish Energy Markets Inspectorate, 2024). Compared to Malmö and Halmstad, Stockholm receives lower annual solar irradiation, while wind conditions are stronger but less persistent than in northern coastal cities.

The photovoltaic resource profile for Stockholm is illustrated in Figure 45 (PVGIS, 2023), showing the hourly PV output normalized to 1 kWp.

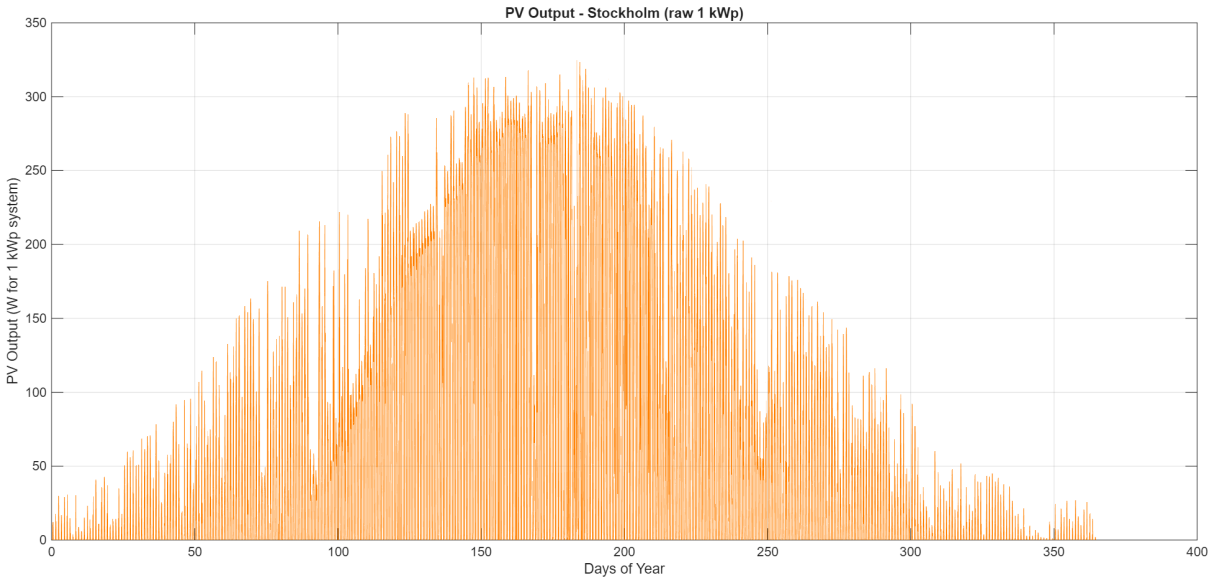


Figure 45. Hourly photovoltaic (PV) electricity generation profile in Stockholm for a normalized system capacity of 1 kWp over a full year. Source: PVGIS, European Commission, Joint Research Centre (2023).

The figure reveals a pronounced seasonal pattern, with moderate summer production and extended periods of low output during winter months. Compared to southern cities, the overall PV contribution is reduced, limiting the effectiveness of PV scaling as a standalone reliability strategy.

Wind resource availability at hub height is presented in Figure 46 (PVGIS, 2023).

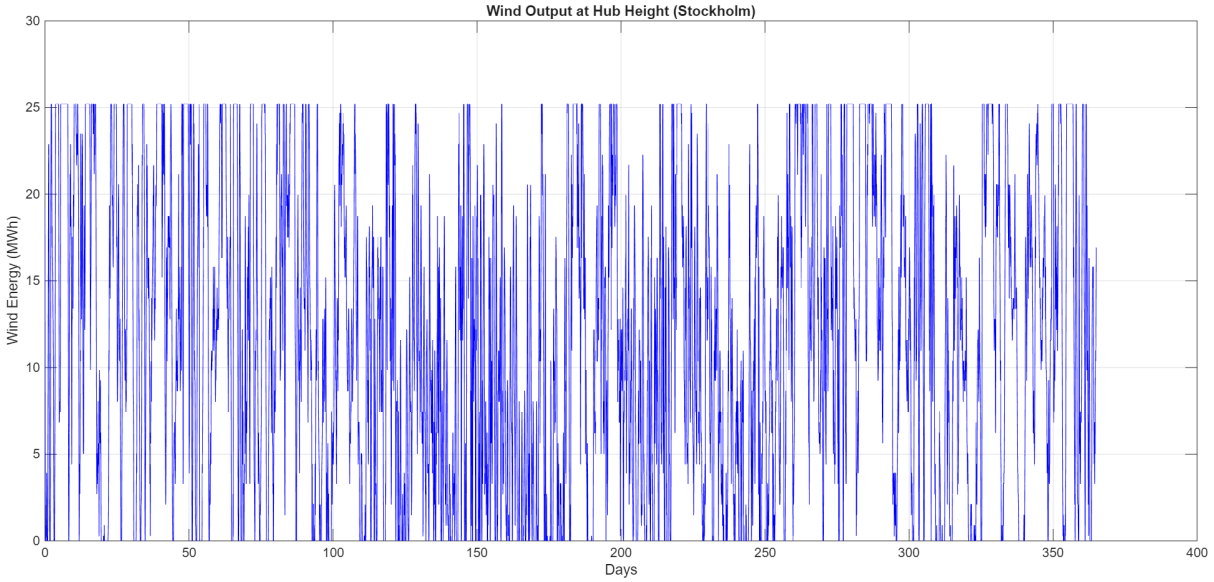


Figure 46. Hourly wind power generation at turbine hub height in Stockholm over a full year, based on regional wind speed data. Source: PVGIS, European Commission, Joint Research Centre (2023).

Wind generation is more substantial and more evenly distributed over the year than solar production yet remains weaker than in northern reference cities. This intermediate wind regime allows partial mitigation of winter PV scarcity but is insufficient to stabilize seasonal imbalances.

The combined interpretation of Figures 45 and 46 indicates that Stockholm occupies an intermediate system regime, neither strongly PV-driven nor fully wind-dominated. As a result, system performance remains sensitive to wind capacity, while PV and battery oversizing primarily increase annual energy surplus rather than delivering proportional reliability gains.

The results summarized in Table 12 show that system performance in Stockholm is strongly governed by wind capacity rather than by total renewable energy production.

Table 12. Summary of city-scale scenario results for Stockholm. Source: Author's own simulation results based on measured load data (MATLAB).

Configuration	PV factor range	Battery factor range	Unmet load (% of demand)	Wasted energy (% of production)	Renewable coverage (%)
6 Wind Turbines	35 → 50	6 → 7.5	10.8 → 11.1	64 → 66	427 → 513
3 Wind Turbines	80 → 95	9 → 10.5	~15.2	43 → 47	572 → 658

Six-turbine configurations achieve a substantially lower unmet load (approximately 10.8–11.1% of demand) compared to three-turbine systems, despite operating with lower renewable

coverage. This indicates that higher annual energy production alone is not sufficient to ensure reliability in Stockholm's climatic context.

In contrast, three-turbine scenarios rely on aggressive PV and battery oversizing, reaching renewable coverage values above 570%. However, this oversizing does not translate into acceptable reliability, as unmet load remains high (around 15.2%). The wasted energy fractions further highlight this structural inefficiency: while six-turbine systems exhibit higher fractional waste due to improved winter usability, three-turbine systems suffer from battery depletion during winter periods, confirming that insufficient wind capacity cannot be compensated for through PV and storage scaling alone.

6.10 Pre-analysis of climatic conditions — Umeå (SE2)

Umeå is located in northern Sweden and belongs to the electricity price area SE2, representing a distinctly wind-favorable climatic context compared to southern and central cities (Swedish Energy Markets Inspectorate, 2024). Due to its high latitude, solar irradiation levels are significantly lower than in Malmö and Stockholm, while wind conditions are stronger and more persistent throughout the year.

The photovoltaic resource profile for Umeå is illustrated in Figure 47 (PVGIS, 2023), showing the hourly PV output normalized to 1 kWp.

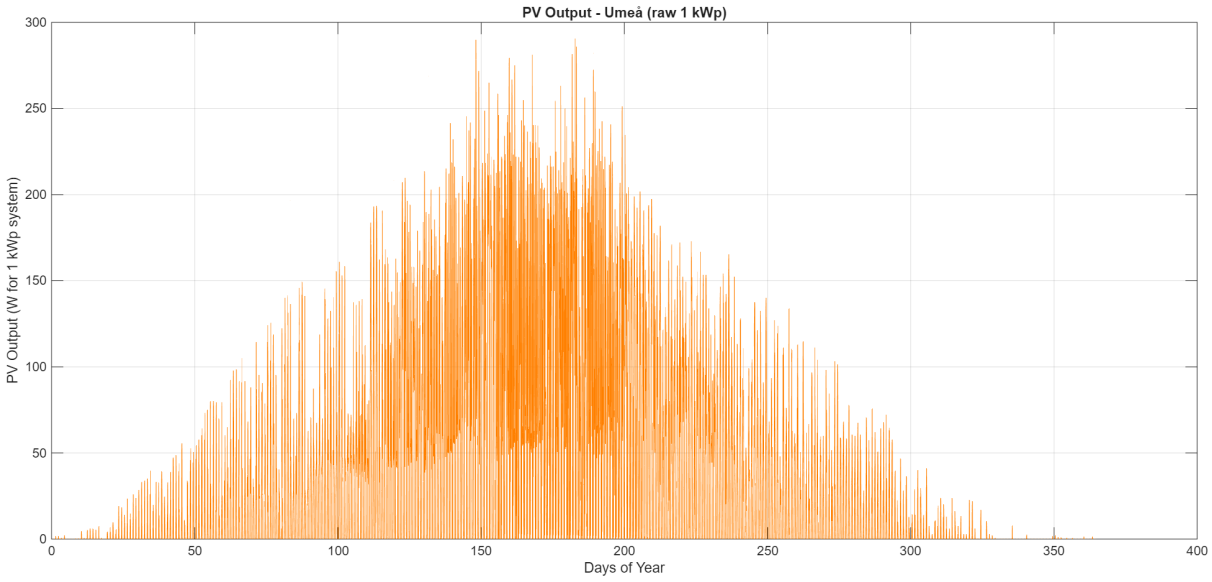


Figure 47. Hourly photovoltaic (PV) power output in Umeå. Source: PVGIS, European Commission, Joint Research Centre (2023).

The figure reveals very limited solar production during the winter months and a shorter effective production season overall. This confirms that photovoltaic generation plays a secondary role in ensuring system adequacy in northern regions and primarily contributes to summer surplus.

Wind resource availability at hub height is presented in Figure 48 (PVGIS, 2023). In contrast to PV, wind generation remains substantial and relatively stable across seasons, including winter periods when solar output is minimal. This characteristic allows wind power to act as the dominant reliability driver in Umeå's renewable system configurations.

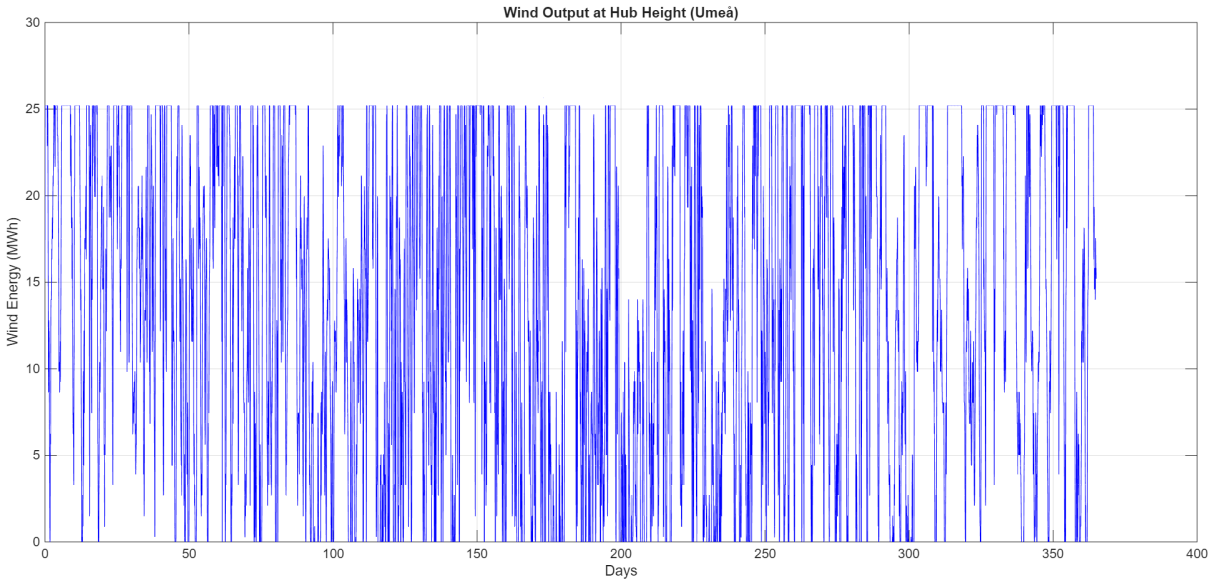


Figure 48. Hourly wind power output at turbine hub height in Umeå derived from site-specific wind speed data over an annual period. Source: PVGIS, European Commission, Joint Research Centre (2023).

The combined interpretation of Figures 47 and 48 indicates that Umeå operates within a fundamentally wind-dominated regime. Seasonal energy balance is therefore less constrained by PV scarcity and more dependent on wind capacity and system sizing. This climatic structure explains why Umeå achieves lower unmet load levels than southern cities under comparable wind-supported configurations and provides a clear contrast to PV-driven system behavior observed in Malmö and Stockholm.

Table 13 shows that both three- and six-turbine configurations in Umeå operate close to a structural reliability limit.

Table 13. Summary of Umeå city-scale scenarios. Source: Author's own simulation results based on measured load data (MATLAB).

Configuration	PV factor range	Battery factor range	Unmet load (% of demand)	Wasted energy (% of production)	Renewable coverage (%)
3 Wind Turbines	80 → 95	9 → 10.5	~8.1 → 8.0	~61 → 62	~410 → 430
6 Wind Turbines	35 → 50	6 → 7.5	~8.0 → 7.9	~62 → 63	~427 → 445

Increasing PV and battery capacities leads to higher renewable coverage and increased energy curtailment, while reductions in unmet load remain limited. Six-turbine configurations achieve slightly lower unmet load values compared to three-turbine cases. Additional details on the city-scale scenarios are provided in Appendix F.

6.11 Cross-city synthesis and structural interpretation of scenario results

This phase synthesizes the results obtained across all analyzed cities and scenario groups to identify the structural drivers behind system performance rather than reiterate city-specific numerical outcomes. Photovoltaic generation, wind power, and battery storage are interpreted as a coupled system whose behavior is governed primarily by temporal alignment and seasonal resource availability.

A central finding of the cross-city analysis is that annual energy sufficiency is a poor indicator of operational reliability. Across all cities, scenarios with very high renewable coverage continue to exhibit non-negligible unmet load, highlighting the dominant role of temporal mismatch rather than absolute generation capacity.

A clear structural contrast emerges between northern cities (Luleå, Umeå) and southern cities (Stockholm, Malmö, Halmstad). Northern locations benefit from stronger and more persistent winter wind resources, which substantially reduce seasonal shortages and stabilize battery operation. Southern cities, despite higher annual solar irradiation, remain vulnerable to winter deficits due to the concentration of photovoltaic output in summer months. These differences explain why similar scenario structures yield systematically different reliability outcomes across regions.

The comparison between six-turbine (6WT) and three-turbine (3WT) configurations further reinforces this interpretation. In 6WT systems, wind generation provides a reliability backbone that limits winter shortages and reduces the degree of photovoltaic and battery oversizing required. However, even in these systems, improvements in reliability diminish as PV and battery capacities increase, indicating that seasonal mismatch, not installed capacity, becomes the governing constraint.

In contrast, 3WT systems display structurally constrained behavior across all cities. Large increases in photovoltaic capacity and battery size lead to substantial energy oversupply and cost escalation, while reductions in unmet load remain marginal. This demonstrates that the limitation of these systems is not insufficient energy production, but insufficient winter generation combined with infeasible seasonal storage requirements.

Battery storage plays a consistent but bounded role in both configurations. Batteries effectively mitigate short-term fluctuations and daily mismatches yet fail to resolve seasonal deficits. As

PV capacity increases, battery utilization becomes increasingly inefficient, with frequent summer saturation and winter depletion. Consequently, additional storage capacity primarily increases cost and wasted energy rather than improving reliability.

These results collectively challenge the use of renewable coverage as a proxy for system adequacy. High coverage values predominantly reflect annual oversupply and mask persistent temporal gaps between supply and demand. Without sufficient winter generation, further photovoltaic or battery scaling yields diminish or result in negligible reliability gains.

The cross-city synthesis reveals that wind availability sets the lower bound of achievable reliability, photovoltaic capacity governs surplus magnitude, and battery storage modulates—but does not fundamentally alter—system behavior. The observed inefficiencies and apparent oversizing in several scenarios are therefore not modelling artefacts, but structural consequences of attempting to address seasonal reliability constraints using technologies optimized for short-term balancing.

This interpretation provides the foundation for the final synthesis, where methodological validity, scientific contribution, and implications for urban renewable energy system design are explicitly articulated.

6.12 Final Synthesis and System-Level Interpretation

This phase consolidates the findings from both the building-scale case studies and the city-scale scenario analyses, with the objective of extracting system-level insights that transcend individual locations or configurations. Rather than revisiting numerical results, this synthesis focuses on identifying consistent behavioral patterns, structural limitations, and economic dominance across all evaluated scenarios.

Across all scales, the combined PV–wind–battery systems exhibit a recurring trade-off between annual energy sufficiency and temporal reliability. While renewable coverage values frequently exceed several hundred percent, unmet load remains persistent, particularly during winter periods. This demonstrates that energy abundance alone is not a sufficient indicator of system adequacy and that temporal alignment between generation, storage, and demand remains the primary constraint.

A key structural insight emerging from both case studies and city-scale systems is the limited effectiveness of PV oversizing beyond a certain penetration level. Increasing PV capacity substantially raises annual production and renewable coverage but yields only marginal reductions in unmet load. This behavior is consistently observed in both three-turbine and six-turbine configurations and across all climatic zones. The underlying cause is not insufficient installed capacity but seasonal mismatches, particularly the inability of PV-dominated systems to supply winter demand without adequate wind support.

Wind power plays a decisive role in improving structural reliability, especially at the city scale. Scenarios with six wind turbines consistently outperform their three-turbine counterparts in terms of unmet load reduction, despite comparable or even lower PV capacities. This confirms that wind acts as a reliability backbone rather than an energy contributor alone, stabilizing winter supply and reducing the dependence on extreme PV and battery oversizing.

The role of battery storage emerges as both critical and constrained. Batteries significantly improve short-term load shifting and enhance self-consumption at the building scale, particularly in case studies with existing PV systems. However, at the city scale, battery effectiveness rapidly saturates. Even very large battery capacities fail to eliminate winter shortages once seasonal deficits dominate system behavior. This reveals a fundamental limitation of energy-based storage when applied to seasonal reliability challenges. In several scenarios, additional battery capacity primarily increases capital cost without delivering proportional reliability gains.

Economic evaluation reinforces these technical observations. Battery-related capital expenditure and annualized cost dominate the total system cost in PV-heavy and wind-poor configurations. In contrast, wind-supported systems achieve a lower cost per unit of reliability improvement, despite higher upfront investment in generation infrastructure. This cost–performance asymmetry explains why scenarios with extreme PV and battery sizing appear energy-rich but economically inefficient. Importantly, these economic patterns are consistent across both case studies and city-scale analyses, indicating scale-independent system behavior.

The comparison between building-scale case studies and city-scale systems reveals a notable consistency in dominant constraints. While absolute values differ, the relative effectiveness of PV, wind, and battery components follows the same hierarchy. Batteries are effective for short-term balancing but ineffective for seasonal gaps; PV oversizing leads to waste rather than reliability; and wind availability fundamentally determines winter performance. This cross-scale consistency strengthens the general validity of the proposed scenario framework.

The analysis demonstrates that fully renewable electricity systems based solely on PV and battery scaling face inherent structural limits in northern climates. Reliable system design requires a balanced integration of wind generation, moderate PV sizing, and storage deployed within its effective operating domain. Attempts to compensate for missing wind capacity through extreme PV and battery oversizing result in diminishing technical returns and escalating economic penalties.

This thesis, therefore, contributes a system-level understanding of why certain renewable configurations consistently fail to deliver reliability, despite high renewable coverage metrics. By explicitly linking technical behavior with economic dominance, the results provide actionable insights for energy system designers, planners, and policymakers operating under realistic climatic and infrastructural constraints.

6.13 Building-Scale Results: Case Study Buildings in Halmstad

The building-scale analysis demonstrates clear differences in renewable energy performance depending on system configuration, installed capacity, and building demand characteristics. Across all case studies, solar photovoltaic (PV) systems contribute meaningfully to on-site electricity supply during high-irradiance periods, while their contribution remains limited during winter months due to seasonal solar constraints.

For buildings equipped with rooftop PV systems (e.g., Kärlekens Skola and Nymansgatan 23), a strong temporal alignment is observed between daytime electricity demand and PV generation, resulting in relatively high levels of direct self-consumption. In these cases, the majority of PV production is utilized on-site during spring and summer months, with surplus electricity exported to the grid primarily during peak solar hours.

In contrast, Sofiebergs äldreboende exhibits a more complex production profile due to the combination of rooftop-mounted PV and facade-integrated solar shading systems. The installation of additional facade-mounted systems in 2024 leads to a noticeable increase in annual solar electricity generation compared to 2023. This expansion results in higher absolute levels of self-consumed solar energy and increased electricity export to the grid, particularly during summer months. Despite lower energy yield per installed area for facade systems, their contribution remains significant due to their extended production period and dual functional role.

Across all building-scale cases, solar electricity supplies a substantial share of daytime demand during high-production months but does not eliminate grid dependency on an annual basis. Seasonal mismatch between generation and demand remains evident, confirming that PV-only systems are insufficient to achieve full electricity self-sufficiency at the building level under current configurations.

6.14 City-Scale Results: Renewable Energy System Scenarios

At the city scale, results are derived from scenario-based simulations combining solar PV, wind power, and battery energy storage systems (BESS) to meet a representative industrial electricity demand profile. By applying an identical load profile across all cities, observed performance differences are attributed primarily to regional climate conditions and system configuration choices.

6.14.1 Renewable Generation Profiles

Solar PV generation exhibits pronounced seasonal variability across all cities, with peak output occurring during late spring and summer. Southern cities (Malmö and Halmstad) demonstrate higher annual PV yields per installed capacity compared to northern cities (Umeå and Luleå), reflecting differences in solar irradiance. In northern regions, PV generation remains minimal during winter months but does not drop to zero, contributing marginally to annual supply.

Wind energy production follows a temporal behavior that differs substantially from solar-based generation, with greater availability during the winter period and pronounced short-term

fluctuations. In northern locations, especially in Luleå and Umeå, wind conditions are generally stronger and more stable, which leads to higher yearly electricity production compared to southern cities when the same turbine capacity is installed.

The complementary seasonal behavior of solar and wind generation is evident across all locations, with wind partially compensating for low solar availability during winter periods.

6.14.2 Energy Balance and Unmet Load

Across all simulated scenarios, unmet load decreases as total renewable capacity increases; however, the rate of improvement diminishes beyond certain capacity thresholds. Scenarios with high renewable penetration achieve substantial reductions in unmet load but do not fully eliminate it, even when large battery capacities are introduced.

In wind-dominant configurations, lower unmet load levels are achieved with comparatively smaller PV capacities, particularly in northern cities. Conversely, PV-dominant scenarios require significantly larger storage capacities to achieve similar reliability levels due to the diurnal nature of solar generation.

Battery storage plays a critical role in reducing short-term mismatches between generation and demand but shows limited effectiveness in addressing long-duration seasonal deficits. As battery capacity increases, unmet load decreases gradually, while wasted energy often increases due to excess generation during high-production periods.

6.14.3 Wasted Energy and Renewable Coverage

Wasted energy emerges as a significant trade-off in high-renewable scenarios. Configurations designed to minimize unmet load frequently result in high levels of surplus electricity that cannot be stored or consumed, particularly in wind-heavy systems. This effect is most pronounced in scenarios with large wind capacities combined with limited storage.

Renewable coverage increases monotonically with installed capacity but exhibits diminishing returns at higher system sizes. Scenarios with moderate renewable capacities often achieve a balanced compromise between coverage, wasted energy, and storage utilization, while extreme configurations prioritize reliability at the expense of efficiency.

6.14.4 Comparative Insights Across Spatial Scales

A clear distinction emerges between building-scale and city-scale results. At the building level, renewable systems primarily enhance self-consumption and reduce grid imports during specific periods but remain constrained by limited installation area and seasonal variability. At the city scale, larger system sizes and access to wind resources enable higher renewable penetration, though full supply autonomy remains challenging without substantial overcapaci

The results highlight that renewable energy system performance is strongly dependent on regional resource availability, system composition, and the chosen balance between reliability and efficiency. These findings provide a robust quantitative basis for the subsequent discussion of design trade-offs and policy implications.

6.14.5 Economic Implications of Scenario Design (Qualitative Assessment)

The economic implications of the investigated renewable energy system configurations are assessed qualitatively, based on the cost structure and total annualized cost (TAC) formulation defined in the methodology. The objective is not to perform numerical cost optimization, but to identify dominant cost components and structural economic trade-offs across different system designs.

Across the building-scale case studies, where wind energy is excluded due to urban constraints, battery energy storage emerges as the dominant contributor to system CAPEX and TAC. At the city scale, battery dominance becomes particularly pronounced in PV-heavy and wind-poor configurations.

Photovoltaic oversizing exhibits similar economic behavior. Although higher PV capacities increase annual energy production, they contribute primarily to surplus generation that cannot be fully utilized. As a result, PV-related CAPEX increases without corresponding improvements in reliability, leading to reduced cost-effectiveness under seasonal constraints.

In contrast, wind-supported configurations demonstrate a more favorable cost–performance relationship. Despite higher upfront CAPEX per installed unit, wind turbines directly address winter-time supply deficits and reduce reliance on excessive PV and battery expansion. Consequently, wind-supported systems achieve larger reductions in unmet load per unit of additional TAC compared to PV–battery-heavy configurations.

The qualitative economic assessment confirms that system designs characterized by extreme PV and battery scaling are economically inefficient, whereas balanced configurations with adequate wind capacity yield more robust reliability outcomes at lower marginal cost. These findings are consistent with the economic logic embedded in the scenario framework and support the technical conclusions drawn from the simulation results.

7 Discussion

This chapter interprets and contextualizes the results presented in the previous sections, with an emphasis on understanding system behavior, key trade-offs, and practical implications rather than repeating numerical outcomes. The discussion brings together findings from both building-scale case studies and city-scale simulations to examine how renewable energy systems perform under different spatial scales, climatic conditions, and system design strategies.

7.1 Building-Scale System Insights

Taken together, the building-scale case studies reveal that the effectiveness of PV–battery integration is primarily governed by load continuity rather than installed capacity.

The Kärlekens Skola case study demonstrates that PV–battery integration in urban educational buildings can significantly improve short-term load matching but remains fundamentally limited by temporal and seasonal mismatches. Battery storage enhances daily operational flexibility and increases on-site utilization of solar energy, yet it does not eliminate grid dependency nor resolve winter-time supply gaps.

The observed trade-off between unmet load reduction and increasing surplus energy highlights the structural limits of storage-based solutions under realistic urban constraints. These findings provide an important reference for interpreting the city-level scenario analyses presented in subsequent chapters, where similar dynamics emerge despite substantially larger system capacities. Compared to buildings with more continuous demand, these limitations are more pronounced in educational facilities with strong temporal intermittency.

The Nymansgatan 23 case study demonstrates that buildings with a higher and more continuous base load benefit more from moderate battery integration than highly intermittent users. Battery storage effectively improves short-term self-consumption and reduces unmet load in the reference and medium scenarios.

Nevertheless, the observed growth in wasted energy across higher-capacity scenarios confirms that battery oversizing primarily shifts the system from unmet demand toward surplus generation, rather than achieving balanced operation. These findings reinforce the structural limitations of PV–battery systems in dense urban contexts and provide a critical reference for the subsequent city-level scenario analysis. In contrast to educational buildings, Nymansgatan 23 demonstrates that a higher and more continuous base load improves the effectiveness of moderate battery integration.

The Sofiebergs äldreboende case study demonstrates that buildings with continuous electricity demand are particularly well-suited for PV–battery integration at moderate storage levels. Battery storage effectively enhances on-site utilization of solar generation and reduces unmet load more consistently than in buildings with intermittent demand patterns.

However, the observed increase in surplus energy at high storage capacities confirms that even favorable load profiles cannot fully eliminate temporal mismatches. These findings reinforce the importance of careful storage sizing and provide a valuable benchmark for interpreting city-level scenarios in which continuous demand profiles are present. Among the analyzed case studies, Sofiebergs äldreboende represents the most favorable load profile for PV–battery integration due to its continuous demand.

The Trönningeskolan case study highlights the structural challenges of PV–battery integration in buildings with highly intermittent demand profiles. While the introduction of PV and battery storage reduces unmet load in the reference and medium scenarios, improvements remain limited compared to buildings with more continuous demand.

As PV and battery capacities increase, wasted energy grows rapidly, reflecting extended periods during which solar generation exceeds on-site demand and storage capacity. This indicates that, for school buildings without existing PV, aggressive system sizing primarily shifts the system toward surplus generation rather than achieving balanced operation.

These findings emphasize that load characteristics play a decisive role in determining the effectiveness of PV–battery systems and provide a critical contrast to the elderly care and mixed-use case studies analyzed earlier. By contrast, Trönningeskolan illustrates the structural limitations of PV–battery systems in buildings with highly intermittent demand.

The building-scale analysis confirms that load characteristics, rather than system size alone, determine whether PV–battery integration leads to balanced operation or merely shifts the system from unmet demand toward surplus generation.

7.2 Comparison Between Building-Scale and City-Scale System Performance

A clear distinction emerges between building-scale and city-scale renewable energy systems in terms of achievable performance and overall system flexibility. At the building level, system design is strongly constrained by physical factors such as available roof area, existing infrastructure, and relatively fixed demand profiles. As a result, solar PV installations mainly function as demand-offsetting technologies, reducing grid imports during periods of high solar irradiance but rarely achieving high levels of annual self-sufficiency.

City-scale systems, by contrast, benefit from greater spatial flexibility and access to a broader mix of renewable resources, particularly wind energy. This allows for higher levels of renewable penetration and more effective mitigation of seasonal mismatches. Nevertheless, the results indicate that even at the city scale, complete elimination of unmet load is difficult to achieve without substantial system overcapacity. This highlights structural limitations that

remain inherent in renewable-dominated systems operating without firm dispatchable generation.

Taken together, these findings suggest that conclusions drawn from building-scale analyses cannot be directly transferred to urban or regional contexts. While building-level systems primarily support local efficiency and resilience, city-scale systems provide a more suitable framework for examining system-wide trade-offs between reliability, efficiency, and resource utilization.

7.3 Impact of Climatic Conditions and Resource Availability

Regional climatic conditions have a strong influence on system performance across all analyzed scenarios. Northern cities located in electricity zones SE1 and SE2 consistently show higher wind energy yields, particularly during winter months when solar PV generation is minimal. In contrast, southern cities in zones SE3 and SE4 benefit from higher solar irradiance during spring and summer, while experiencing lower overall renewable availability during winter periods.

The seasonal complementarity between wind and solar resources is evident across all locations, which supports the relevance of hybrid system configurations. However, the extent of this complementarity differs by region. In northern cities, wind energy plays a dominant role in reducing unmet load, while solar PV contributes mainly to improving annual energy coverage. In southern cities, PV systems provide a larger share of annual electricity supply, yet wind energy remains critical for meeting demand during winter months.

These findings highlight the importance of region-specific system design approaches rather than uniform deployment of renewable technologies across different geographic contexts.

7.4 Trade-Offs Between Reliability, Oversizing, and Energy Waste

A key outcome of the scenario analysis is the clear trade-off between system reliability, renewable capacity oversizing, and wasted energy. Scenarios that aim to minimize unmet load generally require large renewable capacities, which in turn result in considerable surplus generation during periods of high resource availability.

Wind-dominated system configurations are particularly associated with high levels of wasted energy, especially in northern regions where wind resources are strong and persistent. While these configurations achieve lower shortage levels, they do so at the cost of reduced efficiency in energy utilization. In contrast, PV-dominated systems tend to generate lower levels of surplus energy but face greater challenges in maintaining reliability during periods of limited solar availability.

These results illustrate that minimizing unmet load and minimizing wasted energy are inherently conflicting objectives. Approaching near-zero shortage levels requires accepting significant surplus generation, whereas reducing wasted energy implies a higher tolerance for unmet demand.

7.5 Role and Limitations of Battery Energy Storage

Battery energy storage systems play an important role in addressing short-term mismatches between renewable electricity generation and demand. Across the analyzed scenarios, the inclusion of battery storage leads to lower levels of unmet load and improved renewable self-consumption, particularly in PV-dominated system configurations.

At the same time, the results show that battery storage is not an effective solution for long-duration or seasonal mismatches. Increasing battery capacity beyond moderate levels results in diminishing improvements in system reliability, while significantly increasing system costs and leaving a substantial share of storage capacity underutilized.

This outcome is consistent with the inherent physical limitations of electrochemical storage technologies. It suggests that batteries should primarily be regarded as tools for short-term operational flexibility rather than as comprehensive solutions for ensuring long-term system reliability. Seasonal balancing therefore, remains largely dependent on resource complementarity and broader system-level design choices.

7.6 Economic Implications of Scenario-Based Design

Although economic optimization was not the primary objective of this thesis, economic indicators still offer important insights into the feasibility of different system configurations. Scenarios characterized by extreme renewable oversizing and large battery capacities tend to result in substantially higher total annualized costs, without delivering proportional improvements in overall system performance.

In contrast, more moderate configurations that balance renewable energy coverage with acceptable levels of unmet load consistently show more favorable cost–performance relationships. This finding indicates that striving for absolute reliability through extensive renewable oversizing is neither economically efficient nor operationally realistic under current technological conditions.

The adopted scenario-based, non-optimized approach further improves transparency by making cost–performance trade-offs directly observable, rather than embedding them within algorithmic optimization processes. In this way, the approach supports more informed decision-making for planners and stakeholders who must balance technical ambition with economic realism.

7.7 Implications for System Design and Energy Planning

The findings of this thesis carry several implications for renewable energy system design and energy planning. First, hybrid configurations that combine wind, solar, and energy storage consistently outperform single-technology systems across all analyzed regions. Second, effective system design must be adapted to local climatic conditions to achieve meaningful performance improvements. Third, policy and planning frameworks need to acknowledge that high levels of renewable penetration inherently involve trade-offs between reliability, efficiency, and cost.

This phase compares the city-scale scenarios to identify common structural patterns rather than re-discussing individual city results. Luleå and Halmstad are implicitly used as contrasting references, representing wind-dominated northern conditions and solar-favored southern conditions, respectively.

Across all cities, wind capacity emerges as the primary determinant of system reliability. Cities with stronger wind resources consistently achieve lower unmet load levels, while systems relying predominantly on PV—even with large battery capacities—exhibit persistent winter shortages.

Increasing PV and battery capacity leads to rapidly diminishing returns. Although renewable coverage and annual surplus increase significantly, reductions in unmet load remain marginal. This confirms that PV and battery oversizing alone cannot resolve seasonal mismatches.

Rather than treating maximal renewable coverage or zero unmet load as isolated design objectives, system development should aim for robust, interpretable, and context-appropriate solutions. In this respect, scenario-based analysis offers a practical way to explore these trade-offs and supports more nuanced decision-making than approaches that rely solely on mathematical optimization.

The results obtained in this study are largely consistent with findings reported in previous research on hybrid renewable energy systems and industrial and service sector electricity demand. In particular, the pronounced seasonal mismatch between photovoltaic generation and electricity demand, especially at the city scale, has been widely documented in the literature (Lund and Mathiesen, 2009; Connolly et al., 2010).

In a similar way, the complementary behavior of wind and solar resources observed across regions in Sweden aligns with conclusions reported by Roy et al. (2022), who highlight the importance of integrating multiple renewable technologies to mitigate the effects of resource intermittency.

At the building scale, the limited contribution of solar PV during winter months, together with the resulting reliance on alternative electricity sources, reflects patterns reported by Dehning et al. (2019) for industrial and institutional load profiles in northern European climates. In addition, the observed effects of system oversizing—expressed through high levels of surplus renewable generation—are consistent with findings from hybrid system assessments by Diaf

et al. (2008), where non-optimized configurations were shown to expose practical limitations that are often obscured in optimization-driven studies.

In contrast to approaches that prioritize mathematical optimality, the present study adopts a transparent, scenario-based framework. This allows for a clearer interpretation of real-world system behavior under varying climatic and operational conditions.

8 Conclusions and Recommendations

This thesis set out to examine how renewable energy systems can be designed and evaluated to meet industrial and service sector electricity demand across different spatial scales and climatic regions in Sweden. Using a scenario-based, time-resolved simulation framework, the study evaluated the performance of solar photovoltaic systems, wind power, and battery energy storage at both building and city scales. The emphasis was placed on system behavior, key trade-offs, and practical feasibility, rather than on algorithmic optimization.

8.1 Main Conclusions

This thesis demonstrates that renewable energy systems can substantially reduce dependence on the electricity grid and support a more sustainable electricity supply for industrial and urban contexts. However, the results consistently show that system performance is not governed by installed capacity alone. Instead, it is strongly shaped by spatial scale, regional resource availability, and the structural balance between generation technologies and storage.

At the building scale, on-site solar PV systems prove effective as demand-offsetting measures. They reduce grid imports during high-irradiance periods and improve local energy efficiency, particularly during spring and summer months. Nevertheless, physical constraints such as limited installation area and pronounced seasonal variability restrict their overall impact. Even when combined with battery storage, building-scale systems remain fundamentally constrained and are unable to achieve high levels of annual self-sufficiency. Their contribution is therefore incremental rather than transformative.

At the city scale, higher renewable penetration becomes technically feasible due to larger system sizes and access to wind resources. Across all analyzed cities, wind power emerges as the most influential factor in reducing unmet load, particularly during winter periods when photovoltaic generation is limited. Solar PV plays an important complementary role, especially in southern regions, but its contribution remains highly seasonal. As a result, the complete elimination of unmet load requires substantial renewable oversizing, which in turn leads to large volumes of surplus electricity and declining overall system efficiency.

A consistent finding across both spatial scales is the limited role of battery energy storage in addressing long-duration mismatches. Batteries significantly improve short-term flexibility and enhance self-consumption, especially in PV-dominated configurations. However, beyond moderate storage capacities, additional battery sizing delivers diminishing reliability benefits while increasing system costs and leaving a growing share of storage capacity underutilized. Seasonal deficits, therefore, remain largely unaffected by storage expansion alone.

From an economic perspective, this technical limitation translates directly into cost dominance effects. As battery capacity increases beyond moderate levels, the additional reliability gained per unit of installed storage becomes progressively smaller, while capital expenditure continues to rise steeply. In several scenarios, large battery systems remain underutilized for significant

parts of the year, particularly during summer saturation and winter depletion periods. This results in system configurations that are energy-rich but economically inefficient. In contrast, wind-supported systems achieve comparable or superior reliability with substantially lower reliance on battery oversizing, indicating a more favorable cost–performance balance. These findings highlight that pursuing near-zero unmet load through aggressive PV and battery scaling leads to diminishing economic returns and reinforces the importance of evaluating reliability improvements in relation to their associated costs rather than as isolated technical targets.

The scenario analysis further reveals that increasing renewable capacity does not lead to proportional improvements in reliability. While higher PV and wind capacities raise annual renewable coverage, reductions in unmet load quickly saturate. This behavior is observed across all climatic zones, although its severity differs regionally. In northern, wind-rich contexts, relatively low unmet load levels can be achieved with balanced system configurations. In contrast, southern cities require more aggressive oversizing to approach similar reliability levels, at the cost of substantially higher energy waste.

Taken together, these results confirm that renewable energy system design is governed by structural constraints rather than by energy abundance. Reliable system performance depends on the availability of winter generation, particularly wind, while photovoltaic oversizing primarily increases surplus production. Battery storage acts as a valuable operational tool but cannot compensate for a missing seasonal supply. Consequently, balanced hybrid systems consistently outperform extreme, single-technology solutions.

The findings show that fully renewable electricity systems based solely on PV and battery scaling face inherent limitations in northern European climates. Achieving robust and economically reasonable performance requires the coordinated integration of wind power, moderate PV sizing, and storage deployed within its effective operating range. Attempts to replace missing wind capacity through excessive PV and battery expansion result in diminishing technical returns and escalating economic penalties.

8.2 Answers to the Research Questions

The study provides clear responses to the research questions formulated at the outset:

- Renewable energy systems can make a substantial contribution to meeting industrial and service sector electricity demand; however, their effectiveness depends on the integration of multiple technologies and the application of region-specific system design.
- System performance differs significantly between building-scale and city-scale applications, with city-scale systems offering greater flexibility and higher achievable levels of renewable penetration.

- Increasing renewable capacity through oversizing reduces unmet load but also introduces considerable energy waste and economic inefficiencies.
- Scenario-based analysis allows for transparent examination of trade-offs between reliability, efficiency, and cost, and offers clear advantages over approaches that rely solely on optimization.

8.3 Practical Implications and Recommendations

From a practical system design perspective, the results of this thesis clearly indicate that pursuing extreme configurations aimed at eliminating unmet load entirely is neither technically necessary nor economically justified. Across both building-scale and city-scale analyses, attempts to drive unmet load toward zero consistently require disproportionate increases in photovoltaic capacity and battery storage. While such configurations appear technically ambitious, they introduce significant surplus generation and rapidly escalating costs, without delivering commensurate improvements in operational reliability.

This highlights a critical distinction for planners and decision-makers: economically robust renewable energy systems are not synonymous with technically extreme ones. The results demonstrate that balanced system configurations—those that accept a limited level of unmet load while avoiding excessive oversizing—consistently outperform highly oversized systems when evaluated from a cost–performance perspective. In practice, a small residual unmet load may be more economically and operationally manageable than the substantial capital investment required to eliminate it entirely.

For policymakers and energy planners, these findings emphasize the importance of resisting one-dimensional performance targets such as zero unmet load or maximum renewable coverage. Instead, system design should be guided by an explicit recognition of trade-offs between reliability, cost, and energy efficiency. Planning frameworks that acknowledge acceptable levels of surplus generation and residual grid interaction are better aligned with real-world constraints than approaches that prioritize technical self-sufficiency as an isolated objective.

At the building scale, solar PV installations remain a highly effective measure for reducing electricity costs and emissions, even when full self-sufficiency is neither feasible nor economically rational. Battery storage can further enhance short-term flexibility, but its role should remain proportionate to the temporal characteristics of demand. At the city scale, the integration of wind power emerges as a decisive factor for improving system robustness, while excessive reliance on PV and battery scaling alone leads to diminishing returns.

The practical implication of this work is that renewable energy systems should be designed for resilience and economic robustness rather than theoretical completeness. Accepting controlled imperfections—such as limited unmet load or surplus generation—enables more realistic, affordable, and scalable energy system solutions. This perspective is particularly relevant for

long-term energy planning in regions with strong seasonal variability, where structural constraints cannot be overcome through capacity expansion alone.

8.3.1 Building-scale practical recommendations

For Kärlekens Skola, the results suggest that the existing photovoltaic system contributes effectively to reducing grid electricity consumption during daytime operation and provides a reliable starting point for on-site renewable electricity generation. At the same time, the simulations indicate that increasing PV capacity without changes to the building's electricity use pattern mainly results in higher levels of surplus generation, while the impact on reducing unmet demand remains relatively small.

Under these conditions, further expansion of PV capacity would only be justified if it is combined with measures that improve the timing between electricity demand and solar production, for example through increased daytime load utilization or basic load management strategies. Battery storage can offer short-term flexibility and help smooth peak demand, but on its own it does not represent an effective solution for significantly reducing annual unmet electricity demand in this building.

For Kärlekens Skola, the preferred approach is to concentrate on making better use of the existing PV system and improving the temporal alignment between electricity demand and solar generation, rather than treating additional capacity expansion as a standalone solution.

For Nymansgatan 23, it is recommended to maintain the current strategy based on photovoltaic generation combined with grid interaction. The existing system shows satisfactory performance, and relying on the grid to handle surplus electricity remains a simpler and more effective option than implementing on-site energy storage under current operating conditions.

Further expansion of PV capacity is not advised, as the analyzed scenarios indicate that larger PV systems mainly increase surplus electricity generation without leading to a substantial reduction in annual grid dependence. Battery installation should therefore only be considered if there is a clearly defined operational objective, such as peak-load management, changes in the load profile, or a significant increase in daytime electricity demand. Under the present conditions, battery storage adds limited value to overall system performance.

Battery installation should be considered only when a clearly defined operational purpose is identified, such as peak-load management, modifications to the load structure, or an increase in daytime electricity demand. Under the current operating conditions, battery storage offers limited additional benefits to overall system performance.

For Sofiebergs äldreboende, maintaining the existing photovoltaic capacity is recommended. The current PV system supplies approximately 14% of the building's annual electricity demand, while more than 70% of the generated solar electricity is consumed directly on site. This indicates a relatively good match between PV generation and the building's fairly continuous electricity load profile.

Further expansion of PV capacity is not advised, as the analyzed scenarios show that larger PV systems mainly increase the share of surplus electricity—exceeding 30% of annual PV generation—without substantially reducing overall reliance on the electricity grid. Even with increased PV capacity, more than 70% of the annual electricity demand remains unmet by on-site generation.

In this case, battery storage can serve a functional role. Increasing battery capacity helps reduce surplus electricity and improves the on-site use of solar generation, although its effect on lowering annual grid dependence remains limited. Battery deployment is therefore best justified as a means of supporting daily energy management and operational flexibility, rather than as a solution for achieving full energy self-sufficiency.

For Sofiebergs äldreboende, the recommended approach is to retain the existing photovoltaic system and to apply battery storage selectively in order to improve self-consumption and short-term operational flexibility. Extensive expansion of generation capacity is not advised unless it is linked to clearly defined operational objectives.

For Trönningeskolan, the installation of a photovoltaic system is recommended as an effective measure for reducing electricity purchases from the grid. The reference scenario indicates that a considerable share of the generated solar electricity—approximately 63%—can be consumed on site, which contributes to lower grid dependence during daytime operation.

Increasing PV capacity leads to a gradual reduction in annual unmet demand, but it also results in higher levels of surplus electricity. For this reason, any PV expansion should remain moderate and carefully sized, with the primary goal of reducing grid consumption rather than achieving full self-sufficiency.

Battery storage plays a supporting role in this case. While battery deployment can enhance daily energy management and provide additional operational flexibility, its influence on reducing annual grid dependence is limited. Battery systems are therefore best considered as a complementary measure rather than a primary solution.

The recommended strategy for Trönningeskolan is to prioritize reductions in grid electricity use through photovoltaic generation, supported by limited and targeted battery deployment, while avoiding unnecessary system complexity or excessive oversizing.

8.4 Limitations and Future Research

This thesis is subject to several limitations. At the city scale, electricity demand is represented using a standardized industrial load profile rather than measured city-wide consumption data. In addition, the economic analysis focuses on comparative cost–performance behavior and does not include full life-cycle cost assessment or market-based optimization.

Future research could build on this work by incorporating measured industrial electricity demand at the city scale and by examining sector-specific load profiles. Further extensions may

include the integration of additional flexibility options such as demand-side management, long-duration energy storage, or sector coupling. A more detailed representation of grid constraints, policy mechanisms, and market interactions would also enhance the applicability of scenario-based renewable energy system analyses in real planning contexts.

9 References

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10 Appendix

10.1 Appendix A – Building-Scale Energy Analysis: Kärlekens skola

(Measured load profiles, PV production, and simulation results)

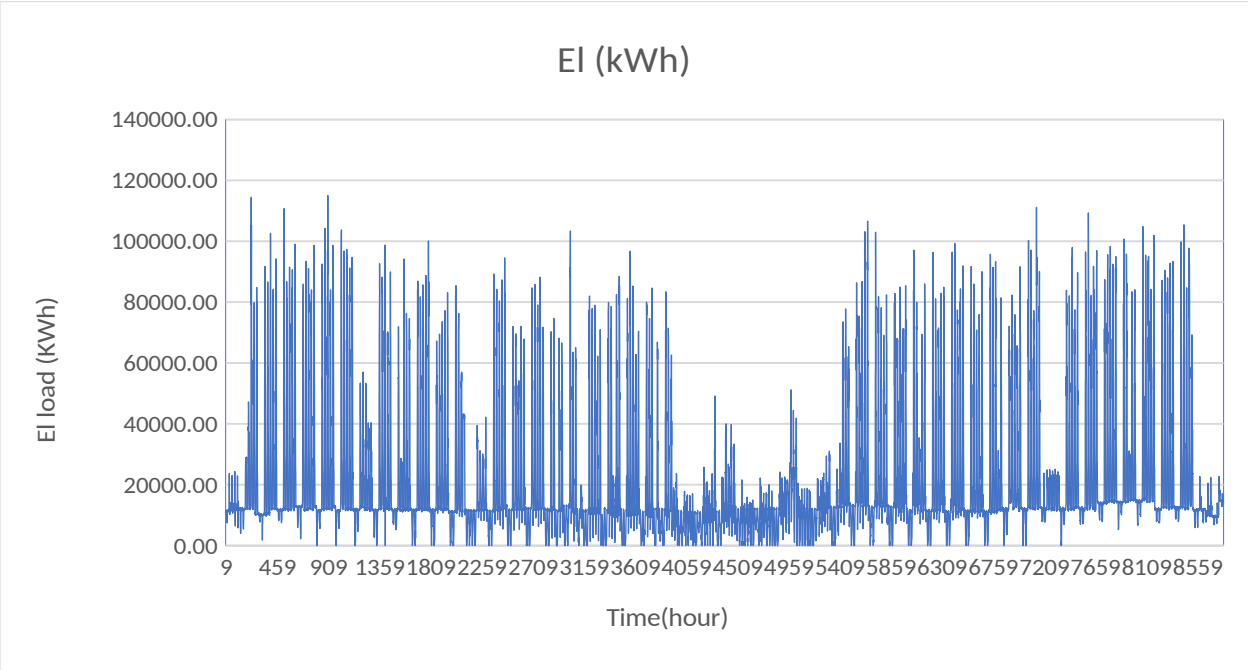


Figure A 1. Annual electricity load profile of Kärlekens Skola showing hourly electricity demand over one year (kWh), based on municipal geospatial data from the City of Halmstad.

Note: The figure represents measured hourly electricity consumption data for Kärlekens Skola over a full year (8,760 hours).

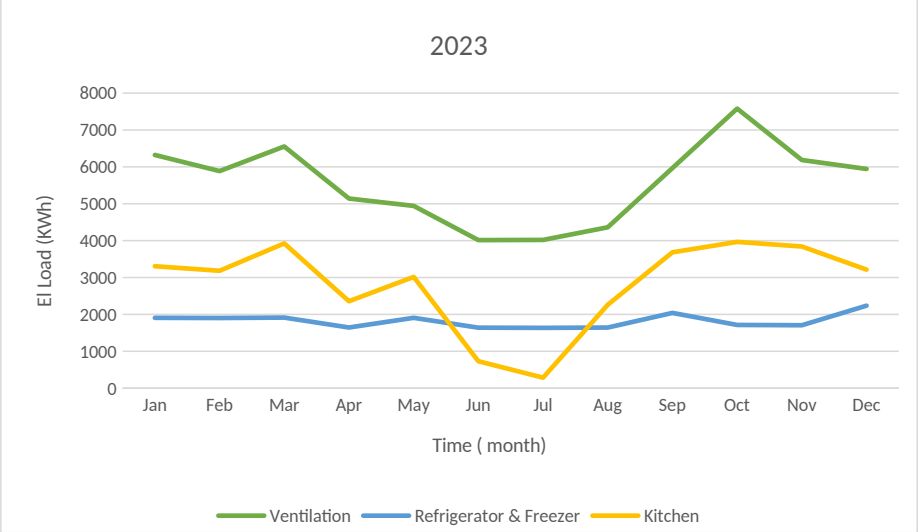


Figure A 2. Comparison of Electricity Consumption by Sector (Ventilation, Kitchen, Refrigeration) for 2023, based on municipal geospatial data from the City of Halmstad.

Kärlekens skola

Halmstads helt nya skola som skall ersätta bland annat gamla Stenstorpsskolan.
En ny och modern skola som självklart har solceller på taket från början.
Skolans tages i bruk hösten 2020.
Här testar vi för första gången en informationsskärm från eviShine. Denna kan även ses i webbläsare på adressen
https://evishine.dk/87022?display=true&id=82337 eller i en annan version på http://evishine.dk/sites/kärlekensskola/

Objektets adress:	Stenstorpsvägen 2 , Halmstad
Solceller installerade:	2020
Entreprenörer:	ELAB
Antal moduler / Total yta:	90 st / 154,2 m²
Typ av solcell:	Monokristallin kisel
Toppeffekt beräknad / verklig:	30 kWp / 28,1 kWp
Årsproduktion beräknad / verklig:	29 324 kWh / 24 261 kWh
Anläggningens årsförbrukning el:	183 856 kWh inkl. solel
Investeringskostnad:	430 930 kr
Nyckeltal kr/kWp, kr/m², kr/kWh:	14 293 kr, 2 795 kr, xx,x kr
Fabrikat solcellspaneler:	LG Electronics LG 335N1C-A5
Fabrikat växelriktare mm.:	Fronius

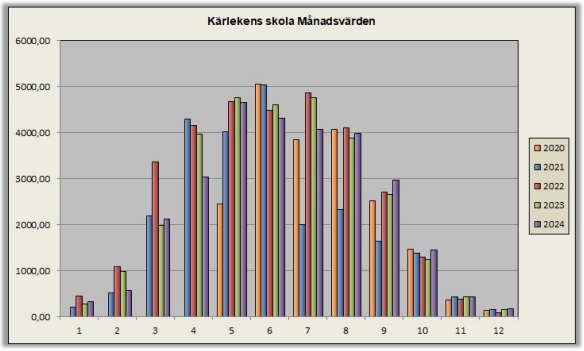










Figure A 3.Photovoltaic system characteristics and installation overview of Kärlekens Skola, including system specifications, annual electricity production, monthly generation trends, and on-site installation images (City of Halmstad, Technical and Property Management Department).

Table A 1. Comparative performance and economic indicators for Scenario 14_5 (PV factor = 14, Battery factor = 5) and Scenario 16_8 (PV factor = 16, Battery factor = 8) at Kärlekens Skola (Source: Author's own simulation results based on measured load data (MATLAB).

Parameter	Scenario 14_5 (PV factor14, Batt factor 5)	Scenario 16_8 (PV factor 16, Batt factor 8)
Peak load (kWh)	115.00	115.00
PV capacity (kWp)	1,610.00	1,840.00
Battery size (MWh)	8.050	14.720
Annual PV generation (kWh)	178,083.99	203,524.55
Self-consumed energy (kWh)	66,175.82	70,000.66
Self-consumption ratio (%)	37.16	34.39
Battery shortage (kWh)	112,067.6	107,805.6
Battery shortage (% of demand)	61.56	59.22
Battery wasted energy (kWh)	102,429.0	115,790.0
Battery wasted energy (% of PV)	57.52	56.89
PV CAPEX (MEUR)	4.186	4.784
Cost per kWp (SEK/kWp)	29,900	29,900

10.2 Appendix B – Building-Scale Energy Analysis: Nymansgatan 23

(Measured load profiles, PV production, and simulation results)

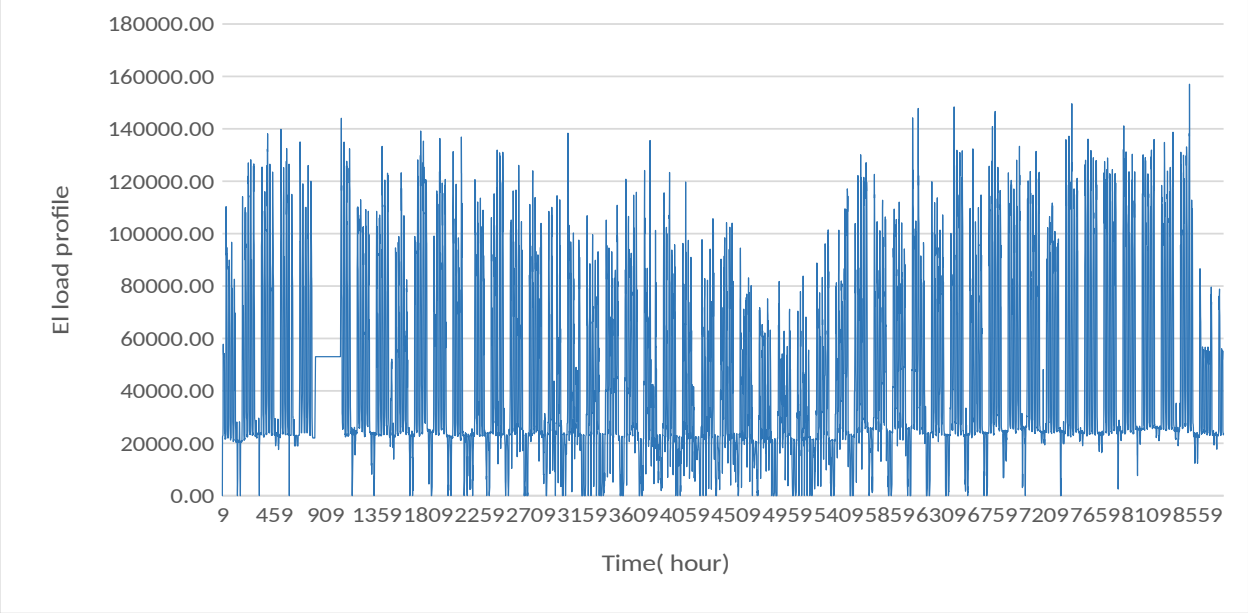


Figure B 1. Annual electricity load profile of the residential building at Nymansgatan 23, showing hourly electricity demand over one year (based on municipal geospatial data from the City of Halmstad).



Figure B 2. The comparison of different kitchen appliances—main oven, kitchen, and cooking pot, and secondary oven—shows different trends of electricity use in 2023. (based on municipal geospatial data from the City of Halmstad).

Nymansgatan 23, Solföljare

2015 monterade vi 191 m² solcellspaneler på en av byggnadens takytor och nu har vi kompletterat detta med tre sk. Solföljare av fabrikat DEGER modell DEGERtracker D80. Dessa skall ge upp mot 45% mer energi på årsbasis jämfört med stationärt monterade. Skall bli intressant att följa dessa.

Objektets adress:	Nymnasgatan 23 , Halmstad
Solceller installerade:	2018
Entreprenörer:	Solkompaniet
Antal moduler / Total yta:	3*30 st / 3*48,9 = 146,8 m²
Typ av solcell:	Monokristallin kisel
Toppeffekt beräknad / verklig:	3*10,05 kWp / 24,8 kWp
Årsproduktion beräknad / verklig:	42 558 kWh / 41 000 kWh
Anläggningens årsförbrukning el:	446 046 kWh inkl. solel
Investeringskostnad:	968 000 kr
Nyckeltal kr/kWp, kr/m², kr/kWh:	29 108 kr, 3 476 kr, 1,05 kr
Fabrikat solcellspaneler:	BenQ SunForte PM096B00
Fabrikat växelriktare mm.:	SMA

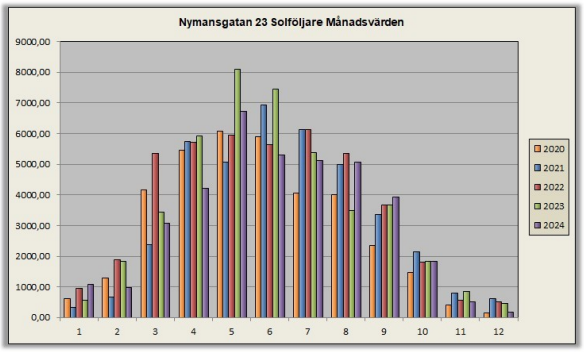










Figure B 3. Photovoltaic system with solar trackers at Nymansgatan 23, Halmstad, including technical specifications, installed capacity, and monthly electricity production data for the ground-mounted tracking system. Source: City of Halmstad, Technical and Property Management Department.

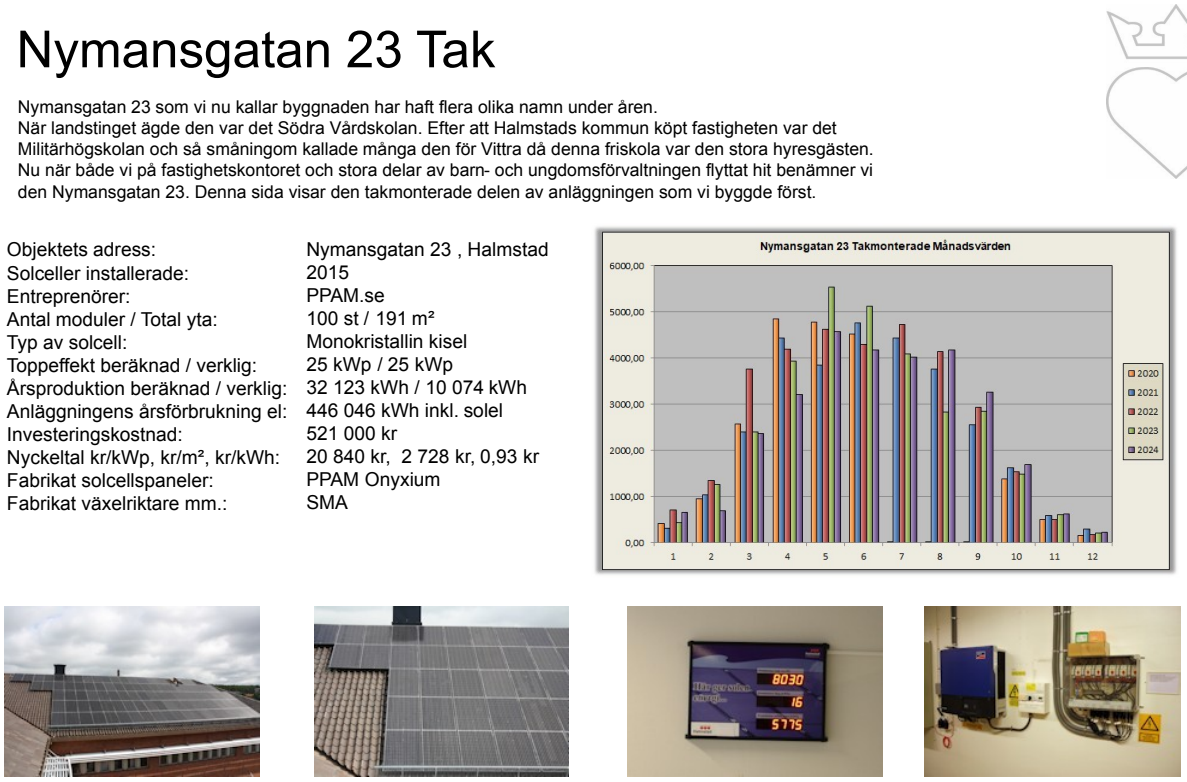


Figure B 4. Roof-mounted photovoltaic installation at Nymansgatan 23, Halmstad, showing system layout, technical characteristics, and monthly electricity production for the fixed rooftop PV system. (City of Halmstad, Technical and Property Management Department).

Table B 1. Comparative performance and cost indicators for Scenario 6 (PV factor = 14, Battery factor = 5) and Scenario 7 (PV factor = 16, Battery factor = 8) at Nymansgatan 23 (Source: Author's own simulation results based on measured load data (MATLAB).

Parameter	Scenario 6 (PV factor 14, Batt factor 5)	Scenario 7 (PV factor 16, Batt factor 8)
PV capacity (kWp)	2,198.11	2,512.13
Battery size (MWh)	10.991	20.097
Annual PV generation (kWh)	243,135.74	277,869.42
Self-consumed energy (kWh)	123,998.54	131,803.86
Self-consumption ratio (%)	51.00	47.43

Parameter	Scenario 6 (PV factor 14, Batt factor 5)	Scenario 7 (PV factor 16, Batt factor 8)
Shortage (kWh)	258,106.9	248,946.1
Shortage (%)	66.42	64.06
Wasted energy (kWh)	108,603.9	126,011.4
Wasted energy (%)	44.67	45.35
PV CAPEX (MEUR)	5.715	6.532
Cost per kWp (SEK/kWp)	29,900	29,900

10.3 Appendix C – Building-Scale Energy Analysis: Sofiebergs äldreboende

(Measured load profiles, PV production, and simulation results)

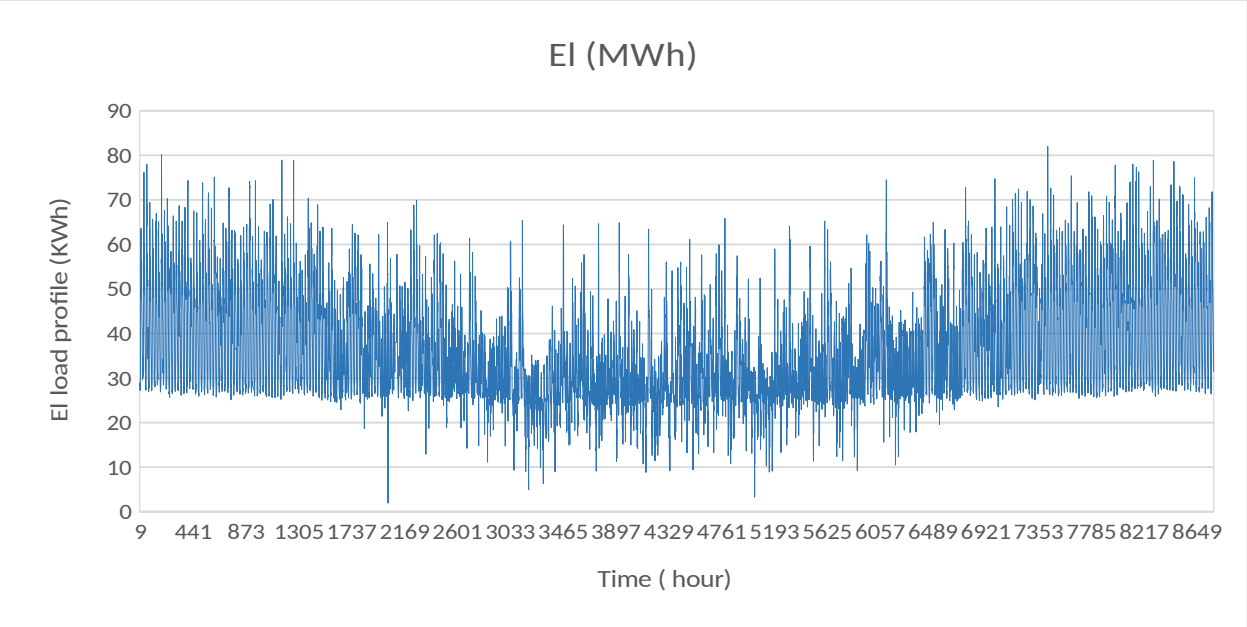


Figure C 1. Annual electricity load profile of Sofiebergs äldreboende, illustrating hourly electricity demand over one year (based on municipal geospatial data from the City of Halmstad).

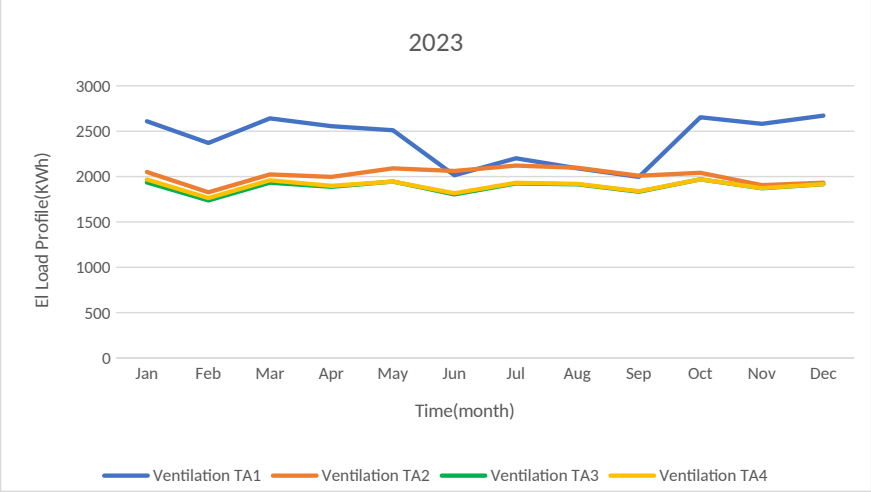


Figure C 2. Monthly electricity consumption of ventilation units (TA1–TA4) at Sofiebergs äldreboende in 2023, showing the distribution of energy use among different ventilation systems (based on municipal geospatial data from the City of Halmstad).

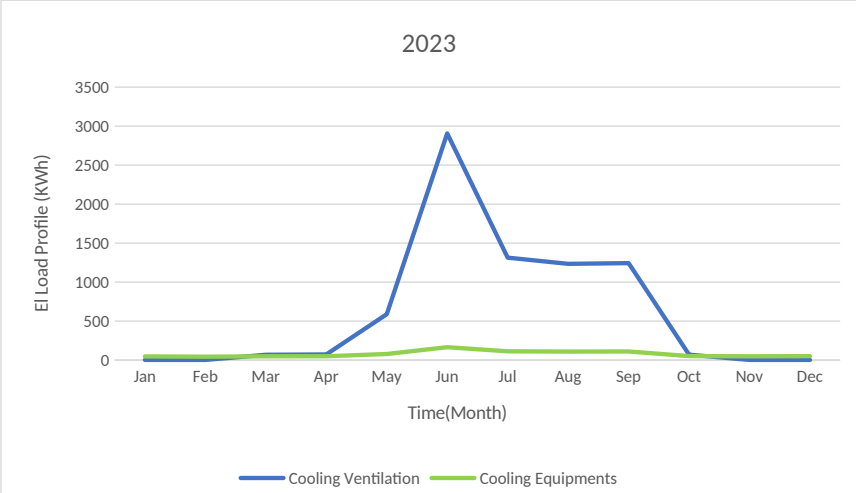


Figure C 3. Monthly electricity consumption of cooling systems at Sofiebergs äldreboende in 2023, including cooling ventilation and cooling equipment (based on municipal geospatial data from the City of Halmstad).

Sofiebergs Äldreboende Fasad

Sofiebergs Äldreboende stod klart 2012. Under sommaren 2018 med den värme vi hade då togs beslut att installera komfortkyla i våra äldreboende och som en första åtgärd beslöts att sätta upp yttre solavskärmning som saknades. Detta är nu gjort i form av solcellspaneler med cirka 30% genomsläpplighet.

Objektets adress:	Sofiebergsvägen 30 , Halmstad
Solceller installerade:	2020
Entreprenörer:	Nyedal
Antal moduler / Total yta:	146 st / 235,9 m²
Typ av solcell:	Monokristallin kisel
Investeringskostnad:	1 030 000 kr
Nyckeltal kr/kWp, kr/m², kr/kWh:	36 140 kr, 4 366 kr, 1,26 kr
Fabrikat solcellspaneler:	Biosol Lumina 40 cells Mono
Fabrikat växelriktare mm.:	SolarEdge

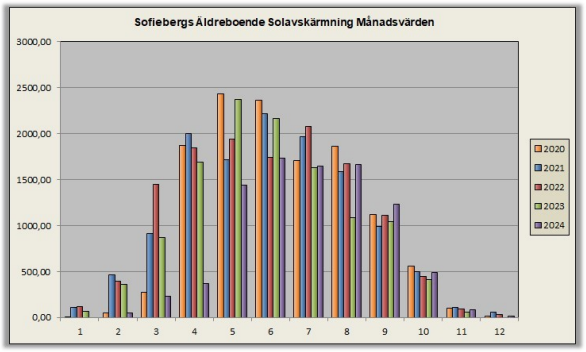










Figure C 4. Technical specifications and visual documentation of the photovoltaic installation at Sofiebergs äldreboende, including installed capacity, system configuration, and façade-integrated solar shading solution (City of Halmstad, Technical and Property Management Department).

Table C 1. Comparative performance and economic indicators for Scenario 14_5 (PV factor = 14, Battery factor = 5) and Scenario 16_8 (PV factor = 16, Battery factor = 8) at Sofiebergs äldreboend (Source: Author's own simulation results based on measured load data (MATLAB).

Parameter	Scenario 14_5 (PV factor 14, Batt factor 5)	Scenario 16_8 (PV factor 16, Batt factor 8)
PV capacity (kWp)	1,148.03	1,312.03
Battery size (MWh)	5.740	10.496
Annual PV generation (kWh)	126,984.72	145,125.40
Self-consumed energy (kWh)	82,252.62	–
Self-consumption ratio (%)	64.77	–
Battery shortage (kWh)	229,015.8	221,739.1

Parameter	Scenario 14_5 (PV factor 14, Batt factor 5)	Scenario 16_8 (PV factor 16, Batt factor 8)
Battery shortage (%)	72.74	70.43
Battery wasted energy (kWh)	40,290.4	49,498.9
Battery wasted energy (% of PV)	31.73	34.11
PV CAPEX (MEUR)	2.985	3.411
Cost per kWp (SEK/kWp)	29,900	29,900

10.4 Appendix D – Building-Scale Energy Analysis: Trönningeskolan

(Measured load profiles and simulated system performance)

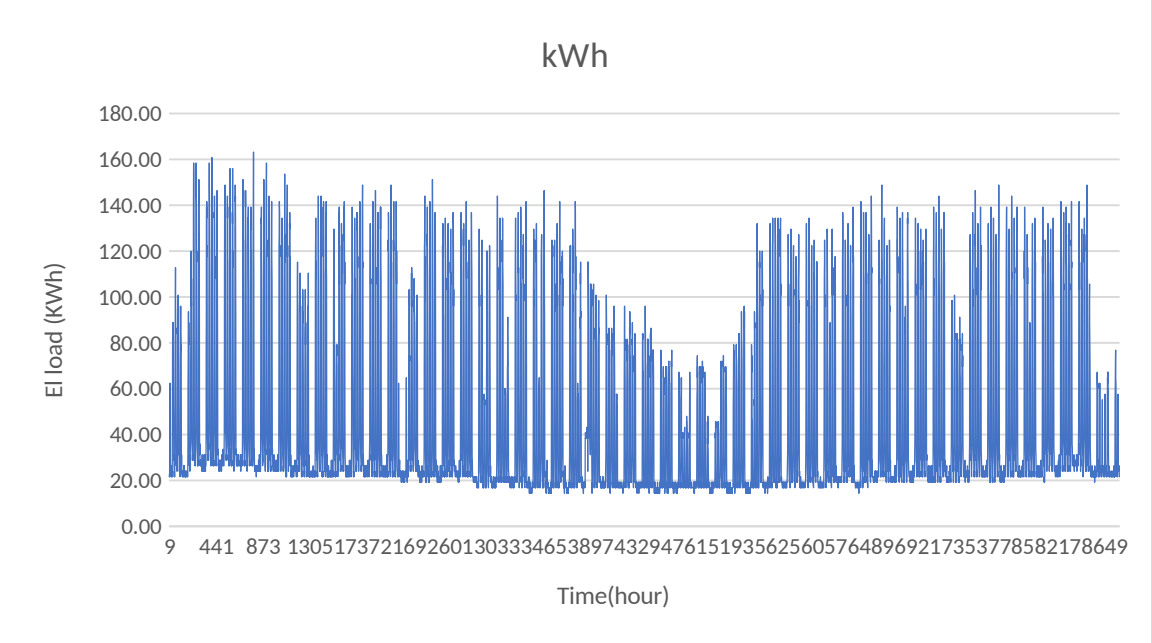


Figure D 1. Annual electricity load profile of Trönningeskolan, illustrating hourly electricity demand over one year (based on municipal geospatial data from the City of Halmstad).

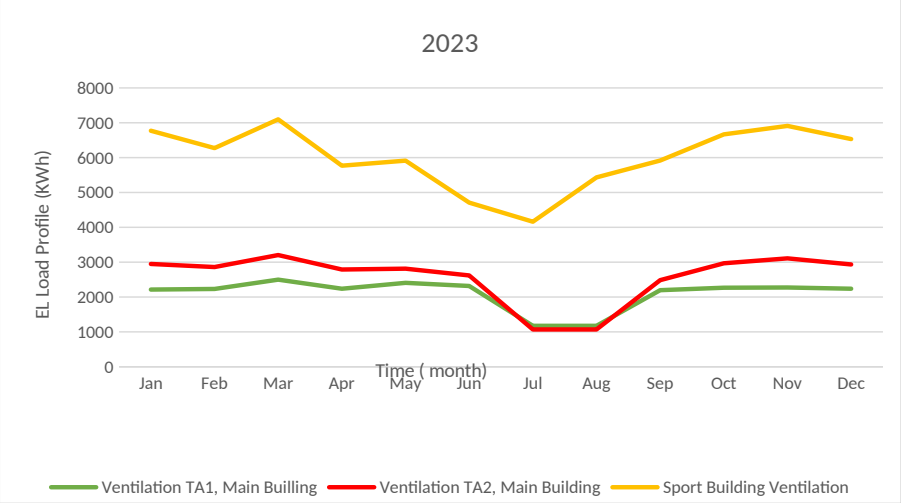


Figure D 2. Monthly electricity consumption of ventilation systems at Trönningeskolan in 2023, including main building ventilation units (TA1 and TA2) and sports building ventilation (based on municipal geospatial data from the City of Halmstad).

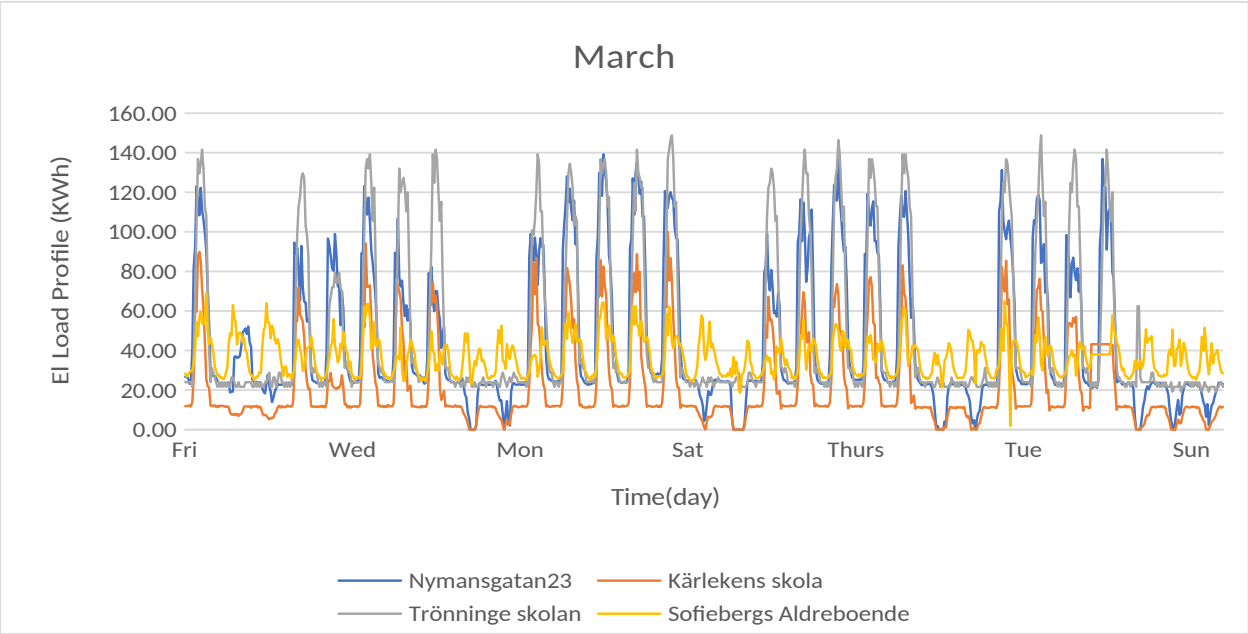
Table D 1. Comparative performance and economic indicators for Scenario 14_5 (PV factor = 14, Battery factor = 5) and Scenario 16_8 (PV factor = 16, Battery factor = 8) at Trönningeskolan (Source: Author's own simulation results based on measured load data (MATLAB).

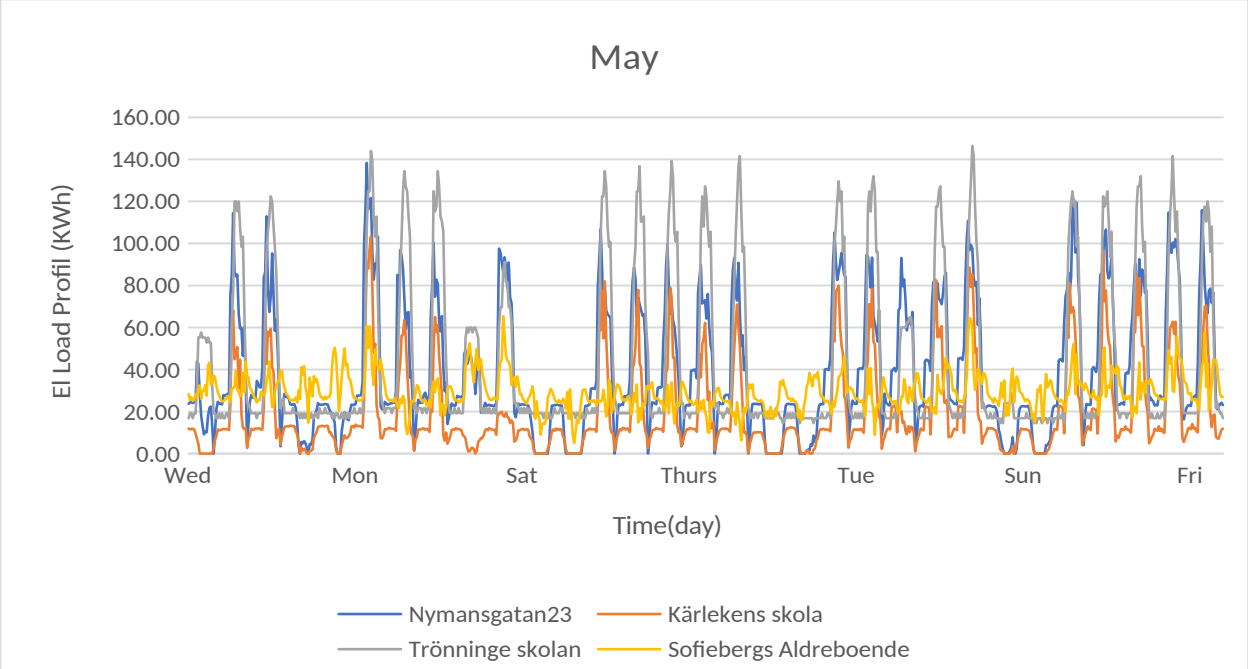
Parameter	**Scenario 14_5 (PV factor 14, Batt factor 5)**	**Scenario 16_8 (PV factor 16, Batt factor 8)**
PV capacity (kWp)	2,284.80	2,611.20
Battery size (MWh)	11.424	20.890
Annual PV generation (kWh)	252,724.40	288,827.89
Self-consumed energy (kWh)	147,851.76	156,951.15
Self-consumption ratio (%)	58.50	54.34
Battery shortage (kWh)	240,811.8	230,300.5
Battery shortage (% of demand)	61.07	58.40
Battery wasted energy (kWh)	94,778.2	112,304.0

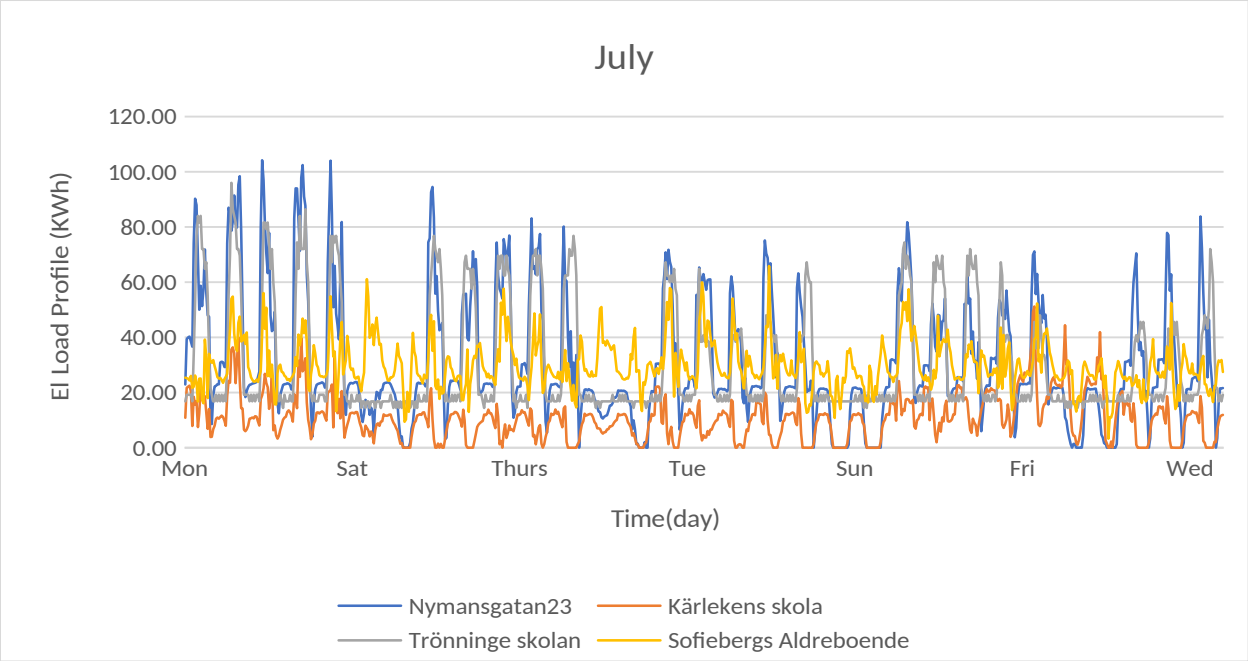
Parameter	Scenario 14_5 (PV factor 14, Batt factor 5)	Scenario 16_8 (PV factor 16, Batt factor 8)
Battery wasted energy (% of PV)	37.50	38.88
PV CAPEX (MEUR)	5.940	6.789
Cost per kWp (SEK/kWp)	29,900	29,900

10.5 Appendix E – Comparative Load Profile Analysis for Building-Scale Case Studies

(Monthly electricity demand patterns during representative high-, medium-, and low-load periods)




May
El Load Profil (KWh)
160.00
140.00
120.00
100.00
80.00
60.00
40.00
20.00
0.00
Wed
Mon
Sat
Thurs
Tue
Sun
Fri
Time(day)
Nymansgatan23
Kärlekens skola
Trönninge skolan
Sofiebergs Aldreboende


July
El Load Profile (KWh)
120.00
100.00
80.00
60.00
40.00
20.00
0.00
Mon
Sat
Thurs
Tue
Sun
Fri
Wed
Time(day)
Nymansgatan23
Kärlekens skola
Trönninge skolan
Sofiebergs Aldreboende

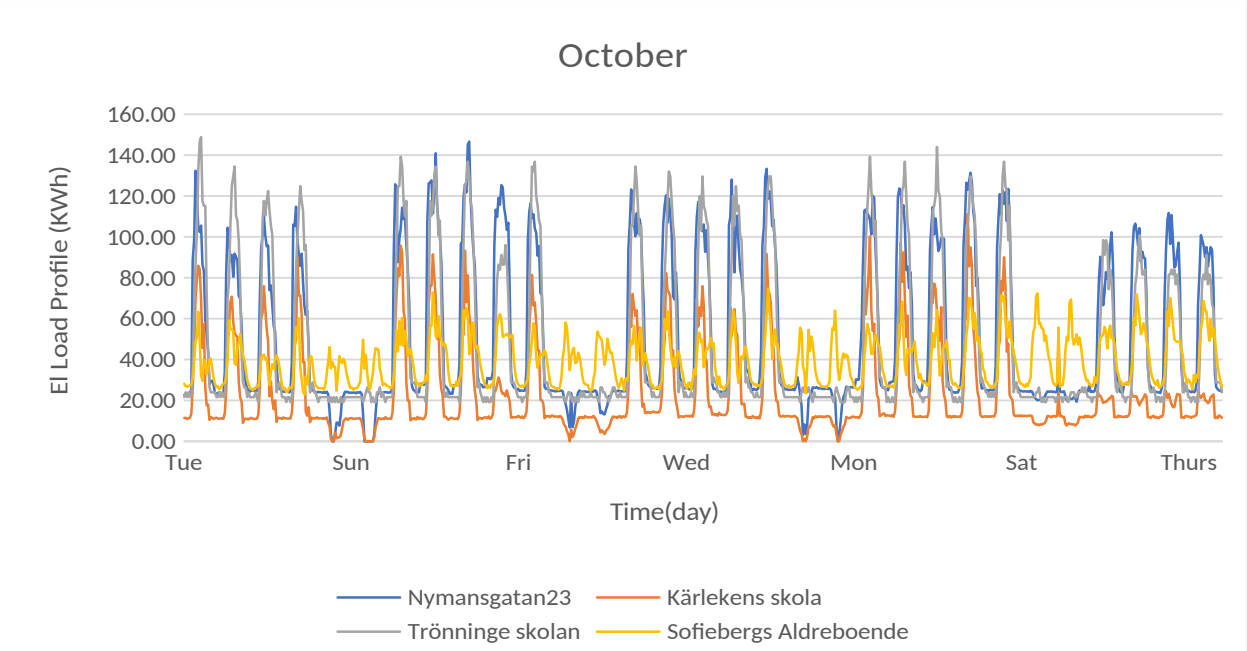


Figure E 1.Hourly electricity load profiles for the four building-scale case studies during representative months with contrasting demand characteristics. The figure shows load patterns for March (high-demand working month), May (typical working month), July (low-demand holiday period), and October (high-demand post- summer working month). The comparison highlights daily and weekly load variation across the four buildings and illustrates the influence of operational activity, occupancy, and seasonal behavior on electricity demand (based on municipal geospatial data from the City of Halmstad).

10.6 Appendix F – City-Scale Renewable Energy System Results Across Swedish Cities

(Scenario-based simulation results for Luleå, Umeå, Stockholm, Malmö, and Halmstad)

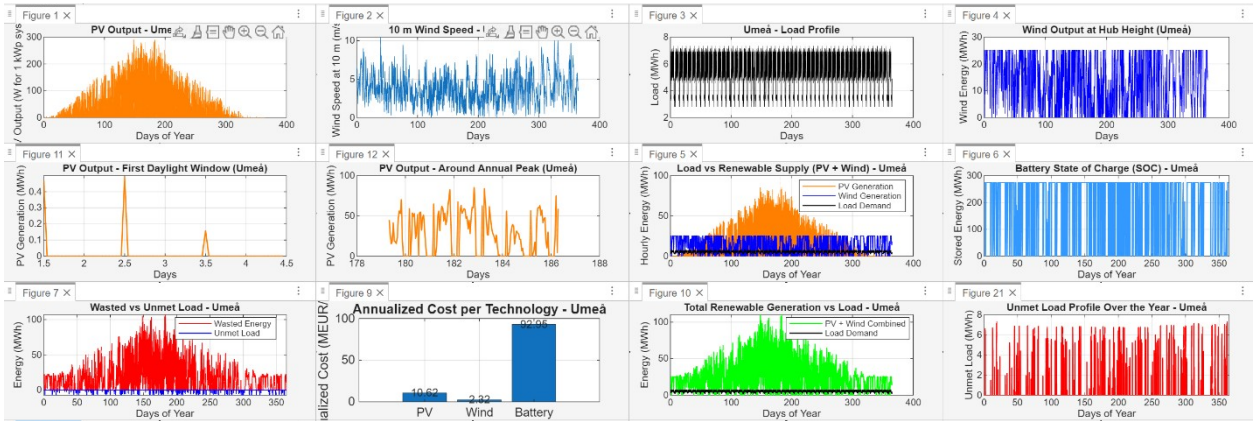


*Figure F 1. Comprehensive simulation results for the **Umeå** case study under Scenario 50–7.5, defined by a PV factor of 50 and a battery factor of 7.5, combined with **six wind turbines**. The figure illustrates the temporal behavior of solar PV output, wind speed, and wind power generation, industrial electricity demand, battery state of charge, renewable energy supply versus load, unmet load, and wasted energy over a full annual period. In addition, the annualized cost distribution across PV, wind, and battery technologies is presented. Together, these results capture the interactions between generation, storage, demand, and cost at the city scale under region-specific climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*

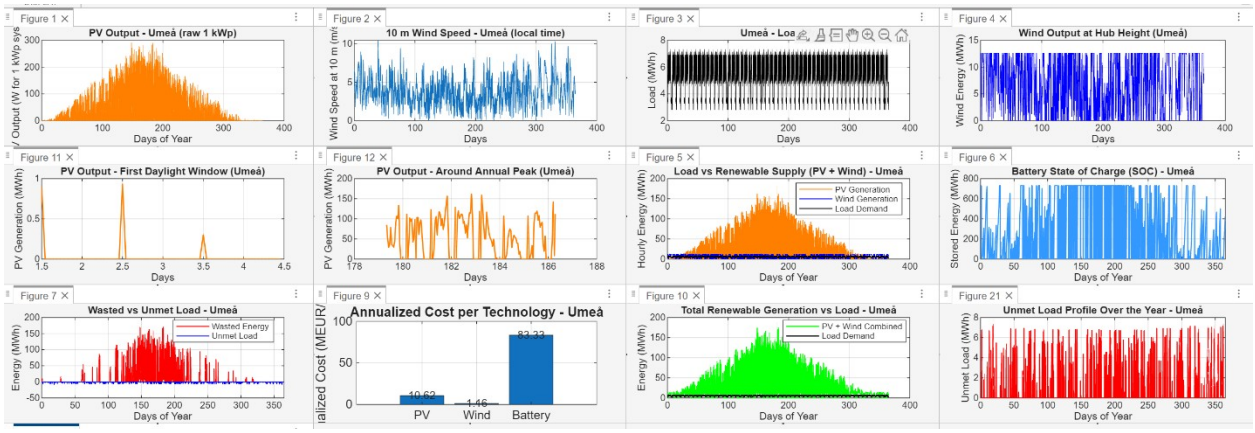


*Figure F 2. Comprehensive simulation results for the **Umeå** case study under Scenario 95–10.5, defined by a PV factor of 95 and a battery factor of 10.5, combined with **three wind turbines**. The figure presents the temporal behavior of solar PV output, wind speed and wind power generation, industrial electricity demand, battery state of charge, renewable electricity supply versus load, unmet load, and wasted energy over a full annual simulation period. In addition, the annualized cost distribution across PV, wind, and battery technologies is shown. Together, these results illustrate the interactions between generation, storage, demand, and cost at the city scale under region-specific climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*

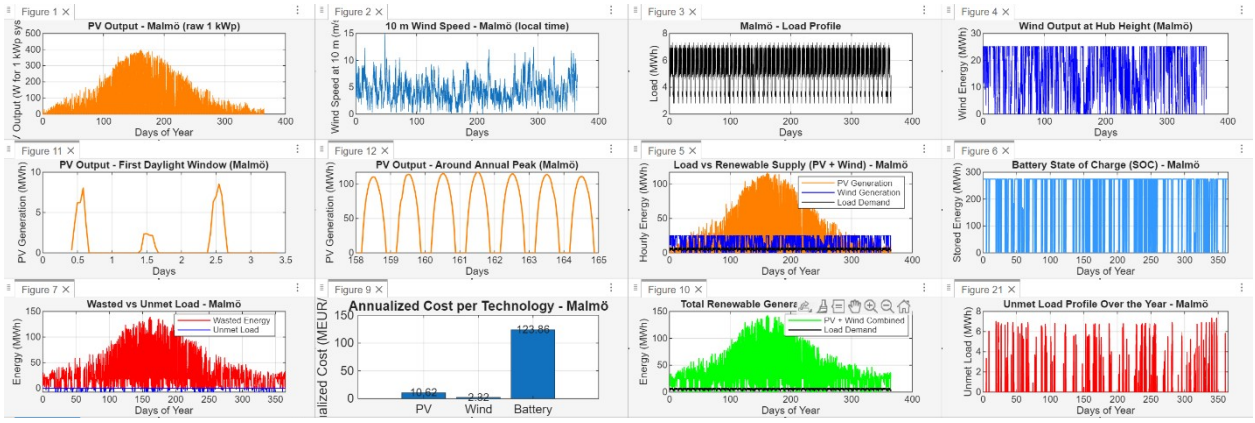


*Figure F 3. Comprehensive simulation results for the **Malmö** case study under Scenario 50–7.5, defined by a PV factor of 50 and a battery factor of 7.5, combined with **six wind turbines**. The figure illustrates the temporal behavior of solar PV output, wind speed, and wind power generation, industrial electricity demand, battery state of charge, renewable electricity supply versus load, unmet load, and wasted energy over a full annual simulation period. In addition, the annualized cost distribution across PV, wind, and battery technologies is presented. Together, these results highlight the interactions between generation, storage, demand, and cost at the city scale under southern Swedish climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*

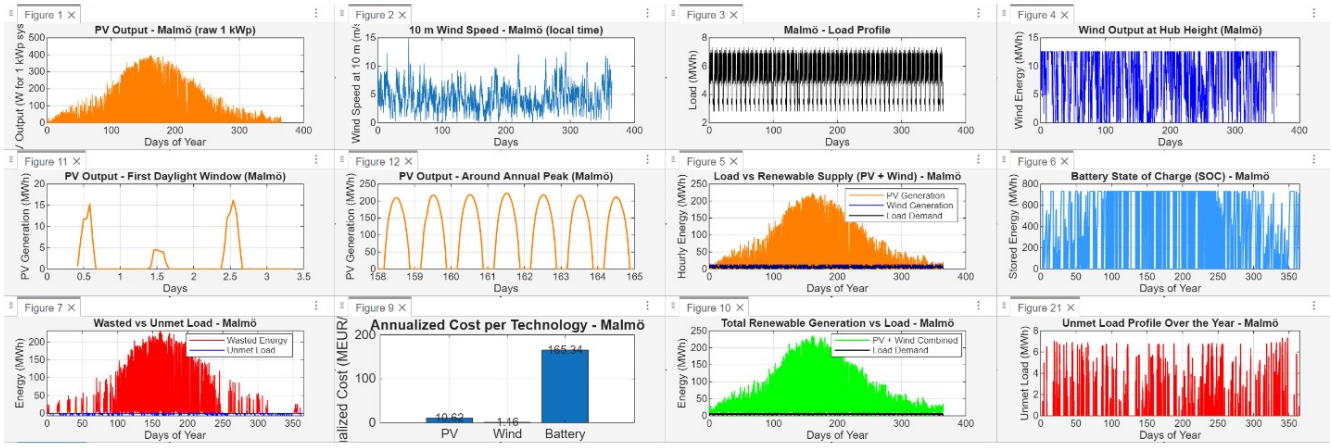


*Figure F 4. Comprehensive simulation results for the **Malmö** case study under Scenario 95–10.5, defined by a PV factor of 95 and a battery factor of 10.5, combined with **three wind turbines**. The figure presents the temporal behavior of solar PV output, wind speed and wind power generation, industrial electricity demand, battery state of charge, renewable electricity supply versus load, unmet load, and wasted energy over a full annual simulation period. In addition, the annualized cost distribution across PV, wind, and battery technologies is shown. Together, these results illustrate the interactions between generation, storage, demand, and cost at the city scale under southern Swedish climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*

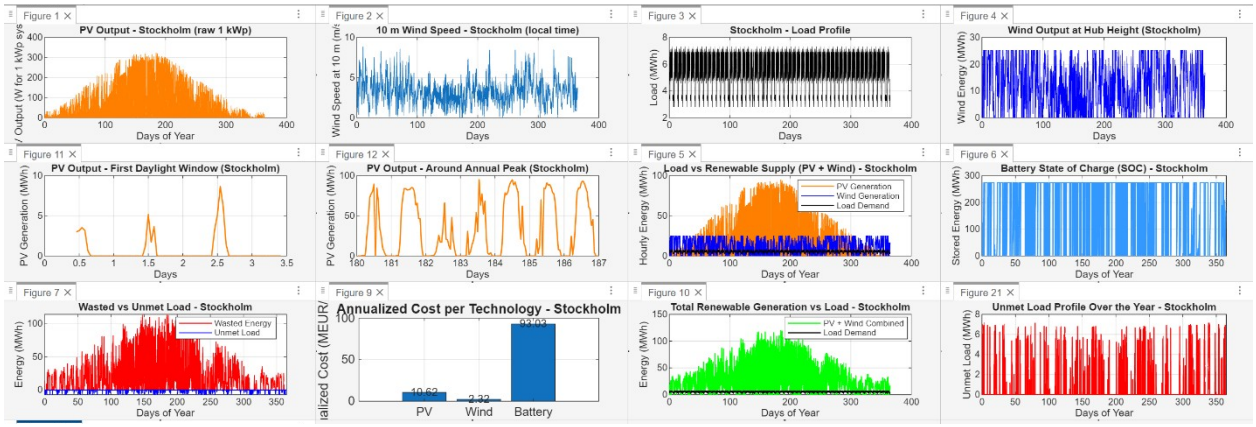


*Figure F 5. Comprehensive simulation results for the **Stockholm** case study under Scenario 50–7.5, defined by a PV factor of 50 and a battery factor of 7.5, combined with **six wind turbines**. The figure illustrates the temporal behavior of solar PV output, wind speed, and wind power generation, industrial electricity demand, battery state of charge, renewable electricity supply versus load, unmet load, and wasted energy over a full annual simulation period. In addition, the annualized cost distribution across PV, wind, and battery technologies is presented. Together, these results highlight the interactions between generation, storage, demand, and cost at the city scale under central Swedish climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*

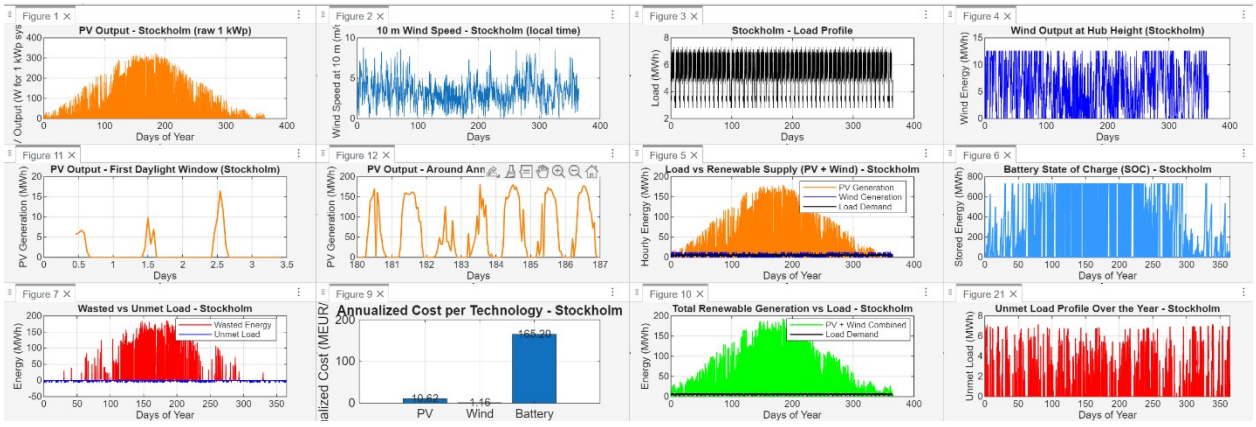


*Figure F 6. Comprehensive simulation results for the **Stockholm** case study under Scenario 95–10.5, defined by a PV factor of 95 and a battery factor of 10.5, combined with **three wind turbines**. The figure presents the temporal behavior of solar PV output, wind speed, and wind power generation, industrial electricity demand, battery state of charge, renewable electricity supply versus load, unmet load, and wasted energy over a full annual simulation period. In addition, the annualized cost distribution across PV, wind, and battery technologies is shown. Together, these results illustrate the interactions between generation, storage, demand, and cost at the city scale under central Swedish climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*

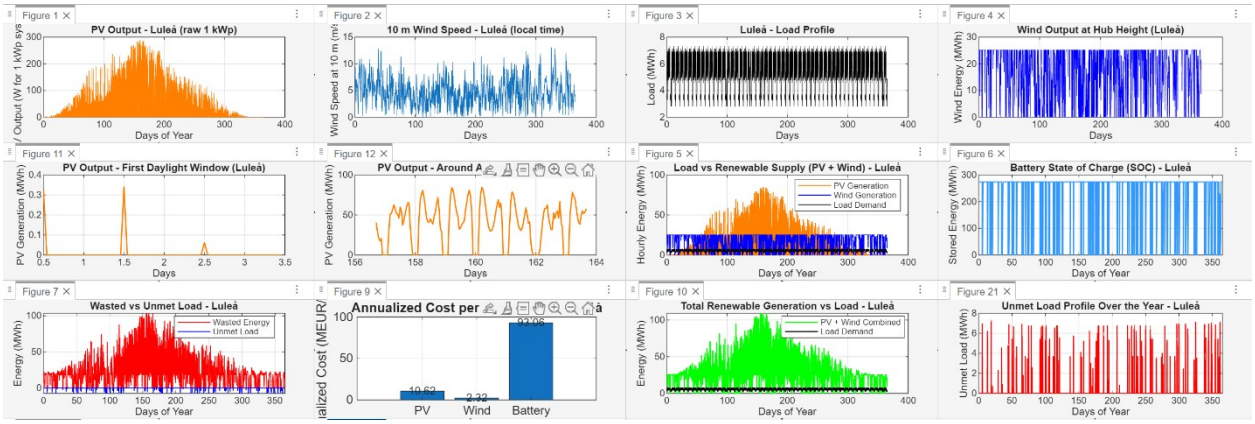


*Figure F 7. Comprehensive simulation results for the **Luleå** case study under Scenario 50–7.5, defined by a PV factor of 50 and a battery factor of 7.5, combined with **six wind turbines**. The figure illustrates the temporal behavior of solar PV output, wind speed, and wind power generation, industrial electricity demand, battery state of charge, renewable electricity supply versus load, unmet load, and wasted energy over a full annual simulation period. In addition, the annualized cost distribution across PV, wind, and battery technologies is presented. Together, these results highlight the interactions between generation, storage, demand, and cost at the city scale under northern Swedish climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*

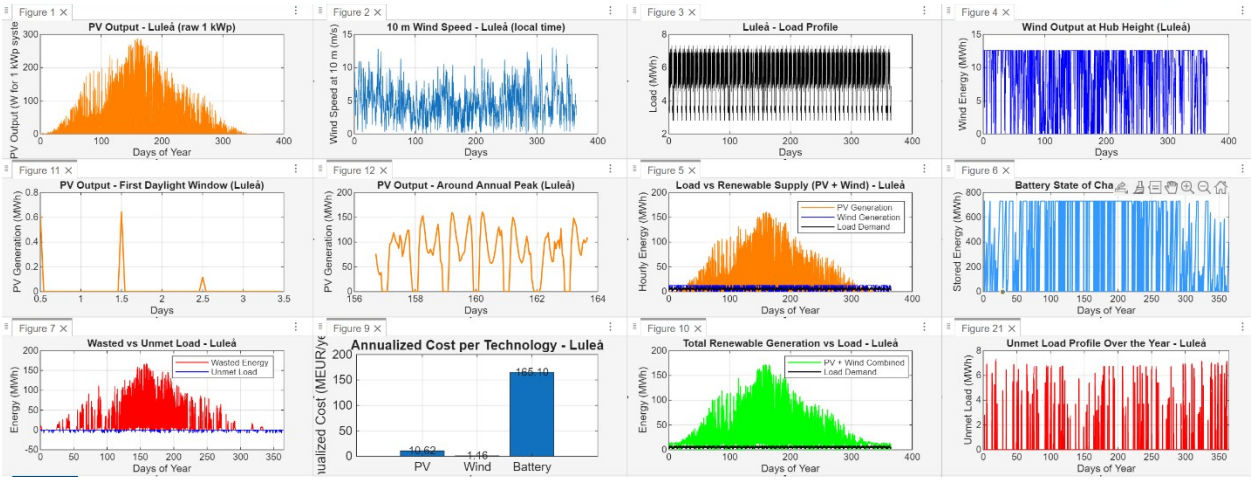


*Figure F 8. Comprehensive simulation results for the **Luleå** case study under Scenario 95–10.5, defined by a PV factor of 95 and a battery factor of 10.5, combined with **three wind turbines**. The figure presents the temporal behavior of solar PV output, wind speed, and wind power generation, industrial electricity demand, battery state of charge, renewable electricity supply versus load, unmet load, and wasted energy over a full annual simulation period. In addition, the annualized cost distribution across PV, wind, and battery technologies is shown. Together, these results illustrate the interactions between generation, storage, demand, and cost at the city scale under northern Swedish climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*

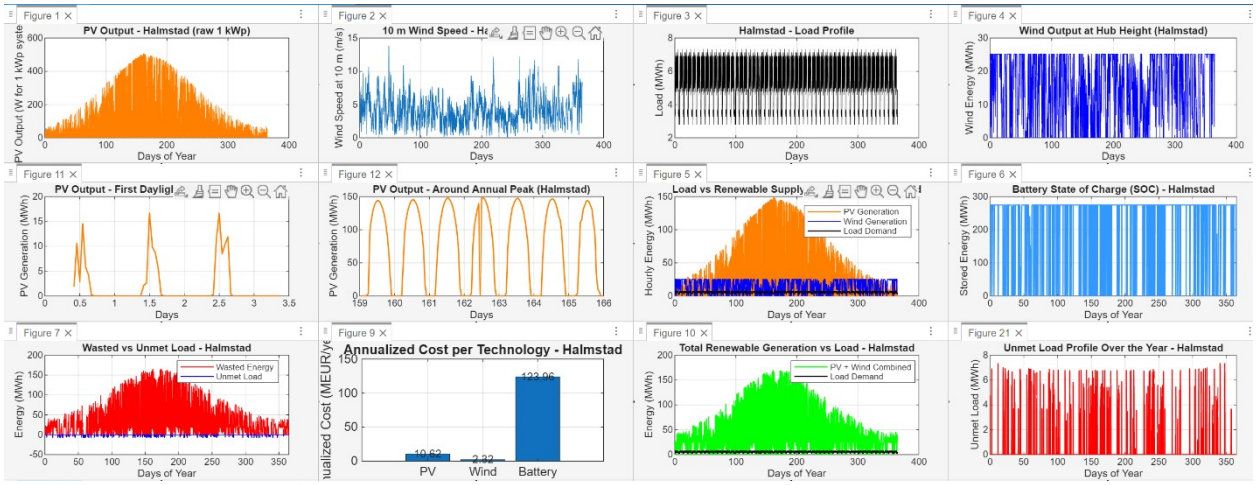


*Figure F 9. Comprehensive simulation results for the **Halmstad** case study under Scenario 50–7.5, defined by a PV factor of 50 and a battery factor of 7.5, combined with **six wind turbines**. The figure illustrates the temporal behavior of solar PV output, wind speed and wind power generation, industrial electricity demand, battery state of charge, renewable electricity supply versus load, unmet load, and wasted energy over a full annual simulation period. In addition, the annualized cost distribution across PV, wind, and battery technologies is presented. Together, these results highlight the interactions between generation, storage, demand, and cost at the city scale under southern Swedish climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*

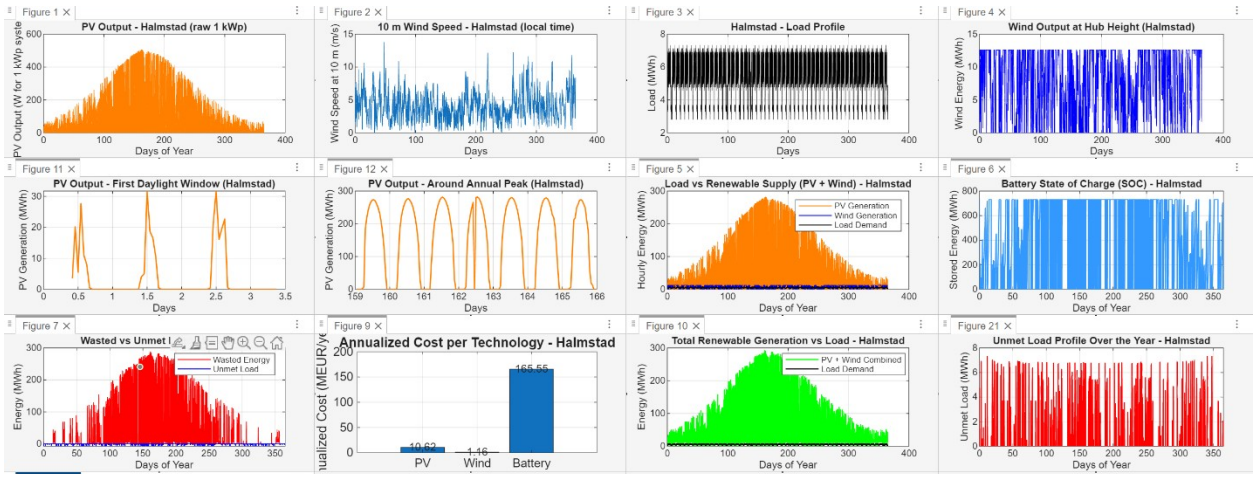


*Figure F 10. Comprehensive simulation results for the **Halmstad** case study under Scenario 95–10.5, defined by a PV factor of 95 and a battery factor of 10.5, combined with **three wind turbines**. The figure presents the temporal behavior of solar PV output, wind speed, and wind power generation, industrial electricity demand, battery state of charge, renewable electricity supply versus load, unmet load, and wasted energy over a full annual simulation period. In addition, the annualized cost distribution across PV, wind, and battery technologies is shown. Together, these results illustrate the interactions between generation, storage, demand, and cost at the city scale under southern Swedish climatic conditions. Source: Author's own simulation results based on measured load data (MATLAB).*